

Open-query games on ordinal spaces

A negative solution to the topological-sum problem,
with an exact Cantor–Bendixson classification

Research report

19 September 2026

For the convergent ordinal sequence $S = [0, \omega]$,

$$\text{sm}(S) = 1, \quad \text{sm}(S \sqcup S) = 2.$$

More generally, for every integer $k \geq 1$, the compact ordinal space

$$K_k = [0, \omega^{2^k - 1}]$$

satisfies $\text{sm}(K_k) = k$ and $\text{sm}(K_k \sqcup K_k) = k + 1$.

Complete ordinary mathematical proofs are included. The archive also contains an independent finite-poset verifier, exact data, and a machine-readable strategy certificate. The infinite-space results do not depend on numerical experiments. No claim of independent peer review or formal proof-assistant verification is made.

Part II, added on 29 September 2026 (batch 52), extends the theory from a membership bit to q output labels: sharp synchronization with overhead $\lceil \log_2 q \rceil$, the exact q -label invariant of convergent towers, and a polynomial/NP-complete dichotomy between two and three labels.

Part III, added on 29 September 2026 (batch 56), computes the q -label invariant exactly on every Hausdorff space with a finite nonempty derived set, through local filter capacities: three free-ultrafilter points need two questions at three labels, three convergent sequences three.

Part IV, added on 30 September 2026 (batch 65), characterizes the admissible layer words of every fixed coloring of every topological space by the reduced words surviving iterated color closures, classifies finite-depth behaviors, and gives the optimal height of a representative and an exact product law.

Abstract

We answer negatively Problem 1.3 of Chiozini, Csernák, and Soukup’s *Gamification of the T_0 -pseudoweight via cut-and-choose games on topological spaces*, in the version available as arXiv:2510.05754v3. The counterexample consists of two convergent sequences, so it lies entirely within compact metrizable ordinal spaces.

The main further result is an exact formula for every nonempty Hausdorff space of finite Cantor–Bendixson height. If $X^{(h)} \neq \emptyset$ and $X^{(h+1)} = \emptyset$, then

$$\text{sm}(X) = \begin{cases} \lceil \log_2(h+1) \rceil, & |X^{(h)}| = 1, \\ \lceil \log_2(h+2) \rceil, & |X^{(h)}| \geq 2. \end{cases}$$

The proof identifies finite open-query complexity with the logarithm of the minimum length of an alternating closed-layer decomposition. A canonical alternating-closure procedure computes that length and retains the orientation information lost by the scalar invariant sm .

Consequences include sharp one-move failures of the proposed sum identity at every positive finite level, exact finite-product formulae, a classification on countable scattered Hausdorff spaces, and a quadratic-time target algorithm for finite posets. Independent minimax computations verify the finite normal form on all naturally labelled posets through six elements and all their targets: 320,866 tests in total.

Addition of 29 September 2026 (batch 52). Part II, from a later manuscript, lets the Seeker determine one of $q \geq 2$ labels instead of a membership bit. Optimal depth is the ceiling of the binary logarithm of a minimum number of ordered open-suffix layers; on posets this number is a shortest common supersequence of chain words. Components with at most m layers need at most $(q-1)m+1$ jointly, so combining them can cost exactly $\lceil \log_2 q \rceil$ extra questions, sharply, already for fixed colorings of compact countable metrizable spaces. A space of last derivative rank h with s colors on its last derivative needs at most $(q-1)h+s$ layers, sharply; the worst case over q -colorings of t convergent towers of rank $m-1$ is $\lceil \log_2((q-1)(m-1) + \min(q, t)) \rceil$, which at $q=2$ agrees with the formula above for $h=m-1$. Exact optimization is polynomial for two labels and NP-complete for three, with a polynomial algorithm within $\lceil \log_2(q-1) \rceil$ questions of optimal.

Addition of 29 September 2026 (batch 56). Part III, from a further manuscript, answers Part II’s first research question for every q on Hausdorff spaces with a finite nonempty derived set. Every finite coloring has a profile digraph G_c on its q_c colors, and the least number of layers is $q_c + \tau(G_c)$ for the directed feedback vertex number τ . The worst case over q -colorings depends only on the partition capacities of the neighborhood filters at the nonisolated points; the bound $(q-1)h+s$ at $h=1$ is attained exactly when the points admit a bin partition with enough capacity. Hence three one-point free-ultrafilter spaces need at most two questions at three labels while three convergent sequences need three, although rank and derivative sizes agree; for sums of t ultrafilter spaces every alphabet-dependent threshold is computed by an exact directed extremal calculation.

Addition of 30 September 2026 (batch 65). Part IV, from a further manuscript, drops the separation and finiteness hypotheses for a fixed coloring. A word is admissible exactly when it contains, as subsequences, all reduced words v whose nested color closure F_v is nonempty, and a rejected word of length m is refuted by such a v of length at most m . At finite depth these words form a finite downset that the admissible words determine, and every nonempty finite downset arises, from finite posets and from compact countable metrizable spaces; this classifies finite-depth behaviors. The length of the longest such word is the least Cantor–Bendixson height of a Hausdorff representative, the formula $q_c + \tau(G_c)$ holds whenever the second derivative is empty, finite-poset obstructions need at most $mq(q-1)^{m-1}$ points, and independent products satisfy $\rho(X \times Y) = \rho(X) + \rho(Y) - 1$. Two colorings of one compact space that agree on every word of length at most three need three and four questions.

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1 The selected problem and the status of the answer

In the set-membership game, questions are restricted to open subsets of a space, and the aim is to decide membership in a fixed target subset. The invariant $\text{sm}(X)$ is the worst, over all targets, of the least ordinal number of questions needed. Chiozini, Csernák, and Soukup ask whether

$$\text{sm}\left(\bigsqcup_{i \in I} X_i\right) \stackrel{?}{=} \sup_{i \in I} \text{sm}(X_i). \quad (1.1)$$

This is their Problem 1.3; their Theorem 3.4 gives the upper bound consisting of the right-hand side followed by one additional move [1].

The answer to (1.1) is *no*. The proof in the next section is short and does not depend on any later theorem. In particular, the counterexample is not an artifact of non-Hausdorff topology, uncountable spaces, or an ambiguity about transfinite play. It uses a two-move game on a countable compact metrizable space. Repeating the same component countably many times also gives a counterexample when the index set is required to be infinite.

The more substantial development is the classification in theorem 6.2. Its mechanism explains exactly why the counterexample works. Different components may require opposite interpretations of the same open-set answer. A scalar question count forgets that orientation, while a pair of alternating-closure lengths preserves it.

Status and provenance. The arXiv record retrieved for this report lists version 3, dated 29 May 2026; the question remains explicitly printed there [1]. The report gives a negative solution to that stated question, not a solution to every open problem in the paper. A targeted literature search found no intervening resolution of this question. This is a search statement, not a proof of priority. The closed-layer machinery belongs to the classical setting of Hausdorff difference hierarchies; we do not claim that framework as new [2]. All lemmas needed for the results here are proved below.

[Added 29 September 2026, batch 52] Part II. Sections 1–13 and Appendices A–B are this report as of 19 September 2026 (Part I). Part II (section 14) is a later manuscript that extends the game from a membership bit to q output labels and reproduces Part I’s values at $q = 2$ (section 14.4). It does not change any statement of Part I.

2 A compact ordinal counterexample

Let $S = [0, \omega]$ have its order topology. Thus every natural number is isolated, and the neighborhoods of ω contain a final segment of $\mathbb{N}$. We use zero questions for a constant target: a subset equal to the whole space or to the empty set.

Lemma 2.1 (The one-question criterion). *For a subset A of any topological space X , membership in A can be decided with at most one open-set question if and only if A is open or closed.*

Proof. An open target is itself a legal question. A closed target can be tested by asking its open complement. Conversely, suppose the first question is U . The only two possible final states are U and $X \setminus U$. Both must be monochromatic for membership in A . If they have the same color, A is constant. Otherwise $A = U$ or $A = X \setminus U$. $\square$

Proposition 2.2. *For $S = [0, \omega]$,*

$$\text{sm}(S) = 1, \quad \text{sm}(S \sqcup S) = 2. \quad (2.1)$$

Proof. Every subset of S missing ω is open, since it consists entirely of isolated points. Every subset containing ω is closed, since its complement consists of isolated points. Hence every target takes at most one question. There are nonconstant targets, so $\text{sm}(S) = 1$.

Write S_0, S_1 for two disjoint copies, with limit points p_0, p_1 , and put

$$A = \{p_0\} \sqcup (S_1 \setminus \{p_1\}).$$

This set is not open: it contains p_0 but no isolated point of S_0 . It is not closed: it contains all isolated points of S_1 , but omits their limit p_1 . By lemma 2.1, one question cannot suffice.

Two questions suffice for *every* target B . First ask the clopen set S_0 , thereby determining which copy contains the remaining candidates. Inside that copy, B is open or closed, so one more open question decides membership. An open set in a component is open in the disjoint union. $\square$

The fact that spaces with at most one accumulation point have one-question membership complexity is also noted in [1, Section 7]; the failure of (1.1) comes from putting two such components together.

Corollary 2.3. *The proposed identity fails even when every component and the whole sum are compact, countable, metrizable, zero-dimensional, and homeomorphic to ordinal spaces. Specifically,*

$$S \sqcup S \cong [0, \omega \cdot 2], \quad \text{sm}([0, \omega \cdot 2]) = 2.$$

Proof. The two clopen intervals $[0, \omega]$ and $(\omega, \omega \cdot 2]$ partition $[0, \omega \cdot 2]$. Each has order type $\omega + 1$. The stated topological properties can also be seen directly from two disjoint convergent sequences. $\square$

Proposition 2.4 (An infinite index set). *For every cardinal $\kappa \geq 2$,*

$$\text{sm}\left(\bigsqcup_{i < \kappa} S\right) = 2.$$

Proof. The lower bound follows by restricting to two components. For the upper bound, fix a target A , and split the component indices into two classes so that A_i is open on the first class and closed on the second; when both apply, choose either class. One question asks the union of the components in the first class. A second question asks the open set whose trace is A_i on the first class and $S_i \setminus A_i$ on the second. The two answers determine the color. $\square$

3 The game, subspaces, and ordinal conventions

All ordinal intervals in this report carry their order topology. The notation $[0, \gamma]$ denotes a space whose underlying ordered set has order type $\gamma + 1$. A topological sum $X \sqcup Y$ is not being confused with an ordinal sum; when they are homeomorphic, this is stated separately.

Definition 3.1 (Open-query membership game). Fix $A \subseteq X$ and an ordinal β . At stage $\xi < \beta$, the Seeker chooses an open set U_ξ . The Hider answers $e_\xi \in \{0, 1\}$. Set

$$W_\eta = \bigcap_{\xi < \eta} \begin{cases} U_\xi, & e_\xi = 1, \\ X \setminus U_\xi, & e_\xi = 0. \end{cases}$$

The Seeker wins if $W_\beta \subseteq A$ or $W_\beta \cap A = \emptyset$. Write $\text{sm}(A, X)$ for the least winning length and $\text{sm}(X) = \sup_{A \subseteq X} \text{sm}(A, X)$, as in [1].

For ordinal-valued invariants we work with T_0 spaces, where a winning length exists: asking all open sets separates points. A statement of the form $\text{sm}(A, X) \leq d$ can also be read on an arbitrary topological space as the existence of a winning depth- d strategy, whether or not a least ordinal winning length exists. The finite normal form is valid in this broader sense.

An empty final state is winning. The Hider need not commit to a point in advance. For a finite decision tree, however, the game formulation and correctness on every fixed point are equivalent: a bad leaf contains two points of different colors, while an empty leaf is harmless. This equivalence will be used carefully in section 4.

Lemma 3.2 (Restriction to a subspace). *If $Z \subseteq X$ and $A \subseteq X$, then*

$$\text{sm}(A \cap Z, Z) \leq \text{sm}(A, X), \quad \text{sm}(Z) \leq \text{sm}(X).$$

Proof. Restrict every open question in a strategy on X to Z . It is open in the subspace, and every resulting state is the old state intersected with Z . For the second inequality, extend an arbitrary target of Z to a target of X . $\square$

Finite question counts are ordinary nonnegative integers. We use $\lceil \log_2 L \rceil$ only for positive integers L ; equivalently, it is the least integer d for which $L \leq 2^d$. This avoids floating-point or ordinal-logarithm conventions. In particular, $\lceil \log_2 1 \rceil = 0$.

4 Finite decision trees and alternating layers

Definition 4.1 (Alternating closed layers). For a nonempty space X and $A \subseteq X$, let $C_1 = A$ and $C_0 = X \setminus A$. A closed-layer decomposition of length $L \geq 1$ and outer color $c \in \{0, 1\}$ is a chain

$$X = F_0 \supseteq F_1 \supseteq \cdots \supseteq F_L = \emptyset \tag{4.1}$$

of closed sets such that

$$F_i \setminus F_{i+1} \subseteq C_{(c+i) \bmod 2} \quad (0 \leq i < L). \tag{4.2}$$

Empty layers are allowed. Let $L(A, X)$ be the least such length, allowing either outer color, and put $L(A, X) = \infty$ if no finite chain exists.

These are finite alternating differences of closed sets, equivalently finite Boolean combinations of open sets. This connects the game to the difference hierarchy, but the exact decision-depth statement below is proved directly.

Theorem 4.2 (Finite normal form). *For every nonempty topological space X , target A , and integer $d \geq 0$,*

$$\text{sm}(A, X) \leq d \iff L(A, X) \leq 2^d. \tag{4.3}$$

Consequently, whenever $L(A, X) < \infty$,

$$\text{sm}(A, X) = \lceil \log_2 L(A, X) \rceil. \tag{4.4}$$

4.1 From a query tree to an ordered collection of leaves

Lemma 4.3. *Complete a finite open-query tree to depth d , padding terminal branches with dummy questions. Number its leaves $0, 1, \dots, 2^d - 1$ in lexicographic order of answer strings, with $0 < 1$. If $\ell(x)$ is the leaf reached by the answers of x , then every upper level set*

$$\{x : \ell(x) \geq j\}$$

over the leaf order is open.

Proof. Proceed by induction on d . The case $d = 0$ is immediate. Let the root question be U , and put $M = 2^{d-1}$. Consider the two subtrees as query trees defined on all of X , regardless of which root answer would actually occur, and write ℓ_0, ℓ_1 for their leaf functions. If $j \leq M$, then

$$\{\ell \geq j\} = U \cup \{\ell_0 \geq j\};$$

for $j = M$, the second set is empty. If $j > M$, then

$$\{\ell \geq j\} = U \cap \{\ell_1 \geq j - M\}.$$

The induction hypothesis and closure of the topology under finite unions and intersections prove the claim. $\square$

Proof of the forward implication of theorem 4.2. A winning depth- d tree has a well-defined output color at each nonempty leaf. Assign an arbitrary color to each empty leaf. Read the 2^d leaf colors from the largest leaf down to the smallest, and group consecutive equal colors into runs. There are at most 2^d runs, and successive run colors alternate.

After deleting the largest i runs, the set of points still represented is a lower level set of ℓ . It is closed by lemma 4.3. These lower level sets, starting with X and ending with $\emptyset$, form the required chain. A run can represent no points, which is why empty layers were allowed. $\square$

4.2 Balanced search among closed layers

Proof of the reverse implication of theorem 4.2. Suppose (4.1)–(4.2) hold. Every point lies in a unique layer $F_i \setminus F_{i+1}$. The open question $X \setminus F_j$ tests whether $i < j$. Thus ordinary binary search can identify the layer index among L possibilities. Split the current interval of indices as evenly as possible at each question. Its largest remaining size after d questions is at most $\lceil L/2^d \rceil$. When $L \leq 2^d$, at most one layer remains, and that layer is monochromatic. The same reasoning applies to arbitrary Hider answers: either the state is empty or it is contained in a single layer. $\square$

Remark 4.4 (Why the logarithm is genuine). Sequentially peeling L layers would spend roughly L questions. Adaptivity permits balanced search through nested open thresholds, reducing the count to $\lceil \log_2 L \rceil$. Conversely, a depth- d tree has at most 2^d ordered leaf runs. Neither direction assumes that the space is ordered or Hausdorff.

[Added 29 September 2026, batch 52] q labels. Theorem 16.2 in Part II is the same normal form for a coloring by q labels: the least number of ordered layers with open suffixes replaces $L(A, X)$, and the depth is again its ceiling logarithm. At $q = 2$ it is theorem 4.2, proved by a second route (section 14.4).

5 Canonical alternating closures and orientation

The least layer length has a concrete canonical description. This is also the right quantity for understanding disjoint unions.

Definition 5.1 (Oriented closure sequence). For $c \in \{0, 1\}$ define

$$F_0^c = X, \quad F_{i+1}^c = \text{cl}_X(F_i^c \cap C_{(c+i+1) \bmod 2}). \quad (5.1)$$

Let $N_c(A, X)$ be the first integer L such that $F_L^c = \emptyset$, or ∞ when no such integer exists. The pair (N_0, N_1) is the *oriented closure profile* of the target.

Every F_i^c is closed and the sequence is decreasing. At step $i + 1$, one keeps the closure of points whose color would be wrong in the next layer. Therefore

$$F_i^c \setminus F_{i+1}^c \subseteq C_{(c+i) \bmod 2}.$$

Theorem 5.2 (Canonical minimality). *For each outer color c , $N_c(A, X)$ is the least possible length of a closed-layer decomposition with that color. In particular,*

$$L(A, X) = \min(N_0(A, X), N_1(A, X)). \quad (5.2)$$

Proof. If the canonical sequence terminates, its layers are a decomposition of the required length. Conversely, let $G_0, \dots, G_L$ be any closed-layer chain with outer color c . We show $F_i^c \subseteq G_i$ by induction. The assertion is true at $i = 0$. If it holds at i , then every wrong-colored point of F_i^c belongs to G_{i+1} , because $G_i \setminus G_{i+1}$ has the prescribed color. Taking closures preserves containment in the closed set G_{i+1} . Hence $F_{i+1}^c \subseteq G_{i+1}$. In particular $F_L^c = \emptyset$. $\square$

Lemma 5.3 (Two elementary profile rules). *Complementing the target swaps N_0 and N_1 . Also,*

$$N_c \leq N_{1-c} + 1$$

whenever the latter is finite.

Proof. The first assertion follows directly from (5.1). For the second, prepend an empty outer layer of color c to a decomposition whose outer color is $1 - c$; in the chain this means repeating X once at the beginning. Apply theorem 5.2. $\square$

Thus finite profile entries differ by at most one, but discarding the distinction between them can change a question count at a power-of-two threshold.

Proposition 5.4 (Exact finite profile of a sum). *Let $X = \bigsqcup_{i \in I} X_i$, $A_i = A \cap X_i$, and let $c \in \{0, 1\}$. For a nonempty family of nonempty components,*

$$N_c(A, X) = \sup_{i \in I} N_c(A_i, X_i), \quad (5.3)$$

where the supremum is taken in $\mathbb{N} \cup \{\infty\}$: an unbounded set of finite lengths has value ∞ . Hence, whenever the resulting minimum is finite,

$$\text{sm}(A, X) = \left\lceil \log_2 \left(\min_{c \in \{0, 1\}} \sup_i N_c(A_i, X_i) \right) \right\rceil. \quad (5.4)$$

Proof. Closure in a disjoint union is computed componentwise. Inductively, $F_n^c(A, X) \cap X_i = F_n^c(A_i, X_i)$ for every finite n . The global sequence is empty exactly when every component sequence is empty. Now use theorems 4.2 and 5.2. $\square$

The obstruction to (1.1) is already visible in the elementary inequality

$$\min_c \sup_i N_{i,c} \geq \sup_i \min_c N_{i,c}.$$

The two operations need not commute: each component may have its own preferred outer color, whereas one global closed-layer chain has only one outer color.

6 Exact classification at finite Cantor–Bendixson height

For a closed subspace F of a Hausdorff space, let F' be its set of nonisolated points. Set $X^{(0)} = X$, $X^{(n+1)} = (X^{(n)})'$; at a limit ordinal, use the intersection of the previous derivatives. Only finite derivatives are needed for the principal formula.

Definition 6.1. A nonempty Hausdorff space has *last derivative rank* h if

$$X^{(h)} \neq \emptyset, \quad X^{(h+1)} = \emptyset.$$

Its Cantor–Bendixson height is then $h + 1$. The terminal derivative $X^{(h)}$ is discrete; it need not be finite unless the space is compact.

Theorem 6.2 (Finite-height classification). *Let X be a nonempty Hausdorff space with last derivative rank $h < \omega$. Then*

$$\text{sm}(X) = \begin{cases} \lceil \log_2(h+1) \rceil, & |X^{(h)}| = 1, \\ \lceil \log_2(h+2) \rceil, & |X^{(h)}| \geq 2. \end{cases} \quad (6.1)$$

More precisely, the maximum of $L(A, X)$ over all targets is $h + 1$ in the first case and $h + 2$ in the second. The maxima are attained.

6.1 Why canonical closures lose one derivative at each step

Lemma 6.3. *If $F = \text{cl}_X B$ and x is isolated in F , then $x \in B$.*

Proof. An open neighborhood U with $U \cap F = \{x\}$ must meet B , because $x \in \text{cl}_X B$. Thus $x \in B$. $\square$

Lemma 6.4 (Derivative descent). *For every target and outer color, the canonical sequence satisfies*

$$F_{n+1}^c \subseteq (F_n^c)' \quad (n \geq 1), \quad F_n^c \subseteq X^{(n-1)} \quad (n \geq 1). \quad (6.2)$$

Proof. By lemma 6.3, every isolated point of F_n^c has color $(c + n) \bmod 2$. Therefore the opposite-colored points whose closure defines F_{n+1}^c all lie in $(F_n^c)'$. This derivative is closed in X , so their closure lies there too. Derivation is monotone on closed sets, giving the second assertion by induction from $F_1^c \subseteq X$. $\square$

Upper bounds in theorem 6.2. For every target and orientation, lemma 6.4 gives $F_{h+2}^c \subseteq X^{(h+1)} = \emptyset$. Hence $L(A, X) \leq h + 2$.

Now assume $X^{(h)} = \{p\}$, and write d for the target color of p . Choose c so that $(c + h) \bmod 2 = d$. We have $F_{h+1}^c \subseteq \{p\}$. If this set were nonempty, p would be isolated in it and, by lemma 6.3, would have color $(c + h + 1) \bmod 2 = 1 - d$, a contradiction. Thus $F_{h+1}^c = \emptyset$ and $L(A, X) \leq h + 1$. $\square$

6.2 Alternating rank layers give matching lower bounds

Write

$$D_j = X^{(j)} \setminus X^{(j+1)} \quad (0 \leq j \leq h).$$

These sets partition X . The set D_j is dense in $X^{(j)}$. Here are details that do not assume compactness: a finite-height space is scattered, because any nonempty subspace contains a point of least ambient rank, and that point is isolated in the subspace after restricting an isolating neighborhood at its rank. Every nonempty open subset of a scattered space contains a point isolated in that space. Apply this to $X^{(j)}$.

Lemma 6.5 (Parity target). *Color every point of D_j by $(b + j) \bmod 2$, where $b \in \{0, 1\}$, and let A_b be the color-1 set. Then*

$$N_b(A_b, X) = h + 1, \quad N_{1-b}(A_b, X) = h + 2. \quad (6.3)$$

Proof. For $h = 0$, the target is constant of color b , and the two lengths are 1 and 2, as asserted. Assume $h \geq 1$. With outer color b , the first closure is X' , because the opposite color contains D_1 , which is dense in X' , and is contained in X' . Repeating the argument shows $F_n^b = X^{(n)}$ for $0 \leq n \leq h + 1$. Thus the first empty term has index $h + 1$.

With outer color $1 - b$, the first closure is X , since it includes the dense rank-zero layer D_0 . Subsequent closures give $F_n^{1-b} = X^{(n-1)}$ for $1 \leq n \leq h + 2$. The first empty term now has index $h + 2$. The same computation covers $h = 0$. $\square$

Lemma 6.6 (Restriction of profiles). *For any subspace $Z \subseteq X$ and either color c ,*

$$N_c(A \cap Z, Z) \leq N_c(A, X).$$

Proof. The closure identity $\text{cl}_Z(B \cap Z) \subseteq Z \cap \text{cl}_X B$ gives, inductively, $F_n^c(A \cap Z, Z) \subseteq Z \cap F_n^c(A, X)$. $\square$

Lower bounds in theorem 6.2. The parity target of lemma 6.5 always has $L = h + 1$, proving the required lower bound when the terminal derivative is a singleton.

Suppose instead that $p \neq q$ lie in $X^{(h)}$. Take disjoint open neighborhoods U, V of p, q . Derivatives commute with restriction to an open subspace: $U^{(j)} = U \cap X^{(j)}$, and similarly for V . In particular both U and V have last derivative rank h .

On U , choose the parity target with bottom color 0. On V , choose the parity target with bottom color 1. Define the target arbitrarily outside $U \cup V$. By lemma 6.5, orientation 0 requires $h + 2$ layers on V , while orientation 1 requires $h + 2$ layers on U . By lemma 6.6, both global orientations require at least $h + 2$ layers. The upper bound already proved gives equality. Applying theorem 4.2 finishes the proof. $\square$

Remark 6.7 (What information survives). At finite height, the invariant sees the last derivative rank and whether the terminal derivative has one point or more than one. It sees neither the exact larger cardinality of that derivative nor the cardinalities of lower levels. Compactness and metrizability are not needed for [theorem 6.2](#).

7 Exact values on ordinal intervals

For $h, m \geq 1$ finite, put

$$X_{h,m} = [0, \omega^h \cdot m].$$

These are countable compact ordinal spaces. Standard background on ordinal spaces and Cantor–Bendixson characteristics is discussed in [\[3\]](#); the specific derivative calculation needed here is elementary.

Lemma 7.1. *For $1 \leq j \leq h$,*

$$X_{h,m}^{(j)} = \{\omega^j \eta : 0 < \eta \leq \omega^{h-j} \cdot m\}. \quad (7.1)$$

In particular,

$$X_{h,m}^{(h)} = \{\omega^h, \omega^h \cdot 2, \dots, \omega^h \cdot m\}, \quad X_{h,m}^{(h+1)} = \emptyset.$$

Proof. The nonisolated points of an ordinal interval are exactly its nonzero limit ordinals. Below $\omega^h \cdot m$, these are precisely the positive multiples of ω . This gives the formula for $j = 1$.

Within the ordered set of positive multiples of ω^j , a point with successor coefficient is isolated, including the first coefficient. A point with limit coefficient is a limit of smaller such multiples. The nonzero limit coefficients are the multiples of ω . Therefore the next derivative consists exactly of the positive multiples of ω^{j+1} in the indicated range. This proves the induction. At step h the remaining finite set is discrete, so the next derivative is empty. $\square$

Corollary 7.2 (Finite-exponent ordinal formula). *For positive finite h, m ,*

$$\text{sm}([0, \omega^h \cdot m]) = \begin{cases} \lceil \log_2(h+1) \rceil, & m = 1, \\ \lceil \log_2(h+2) \rceil, & m \geq 2. \end{cases} \quad (7.2)$$

Last rank h	One terminal point	At least two	Duplication increases sm
0	0	1	yes
1	1	2	yes
2	2	2	no
3	2	3	yes
4	3	3	no
5	3	3	no
6	3	3	no
7	3	4	yes
8	4	4	no
15	4	5	yes
31	5	6	yes
63	6	7	yes

The row $h = 0$ refers to arbitrary discrete spaces, not the notation $[0, \omega^0 \cdot m]$. A singleton has value 0; any larger discrete space has value 1.

Explicit hard targets. On $X_{h,1}$, color a point according to the parity of its Cantor–Bendixson rank. In Cantor normal form, the rank of a nonzero point is its last nonzero exponent; the point 0 has rank 0. On two copies, reverse all colors in one copy. The canonical closure lengths are then $h + 2$ in both orientations. This is an explicit witness, not merely an existence argument using the supremum over targets.

Why two copies form another ordinal space. For $h \geq 1$, the intervals

$$[0, \omega^h], \quad (\omega^h, \omega^h \cdot 2]$$

are clopen in $[0, \omega^h \cdot 2]$ and are each homeomorphic to $[0, \omega^h]$. The second interval has the same order type because removing the first point from an infinite indecomposable ordinal ω^h does not change that order type. Thus

$$X_{h,1} \sqcup X_{h,1} \cong X_{h,2}.$$

8 Sharp failures at every finite level, and sums more generally

Theorem 8.1 (Sharpness at every positive finite level). *For every integer $k \geq 1$, let*

$$K_k = [0, \omega^{2^k-1}].$$

Then

$$\text{sm}(K_k) = k, \quad \text{sm}\left(\bigsqcup_{i < \kappa} K_k\right) = k + 1 \quad \text{for every cardinal } \kappa \geq 2. \quad (8.1)$$

Thus the one-extra-move upper bound for sums is attained at every positive finite level, including by sums of two compact metrizable ordinal spaces.

Proof. Put $h = 2^k - 1$. Derivatives of a disjoint union are computed componentwise. Each component has last rank h and one terminal point; a sum of at least two has the same last rank and at least two terminal points. By theorem 6.2, the respective values are

$$\lceil \log_2(2^k) \rceil = k, \quad \lceil \log_2(2^k + 1) \rceil = k + 1.$$

□

Corollary 8.2 (Exact duplication threshold). *If X is Hausdorff of last finite derivative rank h and $|X^{(h)}| = 1$, then*

$$\text{sm}(X \sqcup X) > \text{sm}(X) \iff h + 1 \text{ is a power of 2.}$$

In that case the increase is exactly one.

Proof. The comparison is between $\lceil \log_2(h + 1) \rceil$ and $\lceil \log_2(h + 2) \rceil$. Consecutive positive integers cross a ceiling-logarithm threshold exactly when the smaller is a power of two. □

[Added 29 September 2026, batch 52] **More than two labels.** With q output labels the one-extra-move bound fails for fixed colorings: components of depth at most d can need exactly $d + \lceil \log_2 q \rceil$ questions jointly, for fixed colorings of finite sums of compact countable metrizable spaces (theorem 21.3); for $q = 3$ and $d = 1$ already of convergent sequences (section 21.1). For the supremum over colorings of finite sums of the towers $K_m \cong [0, \omega^{m-1}]$ ($m \geq 2$) the increase is still at most one (theorem 20.3 and the remark after it); in Part II's notation $K_k \cong K_{2^k}$.

8.1 Bounded finite heights

Let $\{X_i : i \in I\}$ be any nonempty family of nonempty Hausdorff spaces with finite last derivative ranks h_i . Suppose $H = \sup_i h_i < \omega$. Then H is attained and

$$\left(\bigsqcup_i X_i \right)^{(H)} = \bigsqcup_{h_i=H} X_i^{(H)}.$$

If there is a unique component of height $H + 1$ and that component has a singleton terminal derivative, the sum has value $\lceil \log_2(H + 1) \rceil$. In all other cases its value is $\lceil \log_2(H + 2) \rceil$. These two formulae can give the same numerical value.

8.2 Unbounded finite heights

The next upper bound is useful because the number of components need not be countable.

Proposition 8.3 (Parallel play with finite-time certification). *If every component X_i has finite $\text{sm}(X_i)$, then*

$$\text{sm}\left(\bigsqcup_i X_i\right) \leq \omega.$$

If the component values are unbounded among the integers, equality holds.

Proof. Fix a target A in the sum and, for each i , a finite winning strategy for $A_i = A \cap X_i$. We describe countably many rounds, each using three questions.

In round n , first ask the union of the next local questions in every component whose local strategy has not finished. Use a dummy open question in components that have finished. The single global answer advances all unfinished local simulations along the same answer history.

Next ask the union of the whole components whose current local simulation has finished with output color 0. Finally ask the analogous union for color 1. These are open sets because whole components are open. At either of these certification questions, a yes answer leaves a monochromatic state, and the Seeker has won. Empty local states may be assigned either output color.

Suppose all certification answers are no. Any point surviving the final intersection would lie in a particular component X_i . Its local strategy finishes after finitely many rounds. In that round or an earlier one, a certification question includes the whole component X_i , and a no answer removes the point. This is a contradiction. Thus the final state is empty.

The three questions per round have total order type $3 \cdot \omega = \omega$. For the lower bound when local values are unbounded, use subspace monotonicity. $\square$

In particular, for any family of finite-height Hausdorff spaces whose last ranks are unbounded in ω , the sum has membership number exactly ω . This conclusion is independent of the cardinality of the index set.

9 Finite products and the information lost by sm

Lemma 9.1 (Product derivatives). *Let X, Y be Hausdorff spaces of finite last derivative ranks h, k . For finite n ,*

$$(X \times Y)^{(n)} = \bigcup_{i+j=n} X^{(i)} \times Y^{(j)}. \quad (9.1)$$

Thus the last rank of the product is $h + k$, and its terminal derivative is $X^{(h)} \times Y^{(k)}$.

Proof. For closed subspaces $F \subseteq X$, $G \subseteq Y$, a point of $F \times G$ is isolated precisely when both coordinates are isolated in their respective subspaces. Therefore

$$(F \times G)' = (F' \times G) \cup (F \times G').$$

Derivation distributes over finite unions of closed sets: if a point is isolated or absent in every one of finitely many closed sets, intersect finitely many suitable neighborhoods to witness that it is not an accumulation point of their union. The reverse inclusion is immediate. Induction proves (9.1). At $n = h + k$ only the term with $i = h, j = k$ can be nonempty, and all terms vanish at the next stage. $\square$

Corollary 9.2 (Exact product formula). *For nonempty finite-height Hausdorff spaces $X_1, \dots, X_r$, put $H = \sum_j h_j$. Then*

$$\text{sm}\left(\prod_{j=1}^r X_j\right) = \begin{cases} \lceil \log_2(H+1) \rceil, & \text{every } X_j^{(h_j)} \text{ is a singleton,} \\ \lceil \log_2(H+2) \rceil, & \text{otherwise.} \end{cases}$$

In particular, for finite $r \geq 1$,

$$\text{sm}([0, \omega]^r) = \lceil \log_2(r+1) \rceil. \quad (9.2)$$

Knowing only the values of sm is not sufficient to compute product values. For example, let

$$P = [0, \omega^2], \quad Q = [0, \omega^3], \quad S = [0, \omega].$$

Then $\text{sm}(P) = \text{sm}(Q) = 2$, but

$$\text{sm}(P \times S) = 2, \quad \text{sm}(Q \times S) = 3.$$

The finer data consisting of the last rank and singleton-versus-multiple terminal derivative are compositional for finite products and bounded-height sums. The logarithmic compression to sm discards precisely the numerical information needed in this example.

10 Countable spaces and the transfinite-height boundary

The finite formula is not a transfinite ordinal-logarithm formula. On countable scattered spaces, all infinite Cantor–Bendixson heights lead to the same game value, namely ω .

The point-separating number $\text{ps}(X)$ is defined by the same game, but with winning condition $|W_\beta| \leq 1$ instead of monochromaticity for a target $[1]$.

Lemma 10.1 (A countable T_1 upper bound). *If X is countable and T_1 , then $\text{sm}(X) \leq \omega$. If X is also infinite, then $\text{ps}(X) = \omega$.*

Proof. Enumerate $X = \{x_0, x_1, \dots\}$. Ask the open complements $X \setminus \{x_n\}$ successively. A no answer leaves at most the singleton $\{x_n\}$. If every answer is yes, the final state is empty. This proves both upper bounds. A finite binary tree has finitely many leaves and cannot separate all points of an infinite space, proving the point-separation lower bound. $\square$

Theorem 10.2 (Countable scattered classification). *Let X be a nonempty countable scattered Hausdorff space. If its Cantor–Bendixson height is finite, theorem 6.2 applies. Otherwise*

$$\text{sm}(X) = \omega. \quad (10.1)$$

Proof. Only the lower bound in the infinite-height case remains. For every finite h , $X^{(h)}$ is nonempty and scattered, hence has an isolated point p . Choose an open neighborhood U in X with $U \cap X^{(h)} = \{p\}$. Derivatives commute with open restriction, so

$$U^{(h)} = \{p\}, \quad U^{(h+1)} = \emptyset.$$

By theorem 6.2 and subspace monotonicity, $\text{sm}(X) \geq \lceil \log_2(h+1) \rceil$. These integers are unbounded as h varies. Together with lemma 10.1, this yields (10.1). $\square$

Corollary 10.3 (All countable exponent intervals). *Let $0 < \alpha < \omega_1$ and $1 \leq m < \omega$. Then*

$$\text{sm}([0, \omega^\alpha \cdot m]) = \begin{cases} \lceil \log_2(h+1) \rceil, & \alpha = h < \omega, \ m = 1, \\ \lceil \log_2(h+2) \rceil, & \alpha = h < \omega, \ m \geq 2, \\ \omega, & \alpha \geq \omega. \end{cases}$$

Every one of these spaces has $\text{ps} = \omega$.

Proof. The finite-exponent cases were proved in corollary 7.2. When $\alpha \geq \omega$, the interval contains $[0, \omega^h]$ as a clopen subspace for every positive finite h . Their membership numbers are unbounded, while the whole space is countable and T_1 . Use lemma 10.1. $\square$

For example,

$$\text{sm}([0, \omega^\omega]) = \text{sm}([0, \omega^{\omega^2}]) = \text{sm}([0, \omega^{\omega^\omega}]) = \omega.$$

Thus extremely different countable order types can have the same membership number. This is a mathematical limitation of the invariant, not a limitation of the ordinal notation.

The countable scattered classification also covers every countable compact Hausdorff space. For completeness, the fact that such a space is scattered is proved in section A, rather than invoked as an unproved classification of countable compact spaces by ordinals.

11 A purely order-theoretic form for finite posets

Equip a finite poset P with the Alexandrov topology of upper sets. This is T_0 , and it is Hausdorff only when the order is an antichain. It is important not to mistake this topology for the order topology on the infinite ordinal spaces studied above.

Fix a target $A \subseteq P$ and color points by their membership in A . A strictly increasing chain is *alternating* if consecutive colors differ. Let a_b be the largest number of elements in an alternating chain whose greatest element has color b , with $a_b = 0$ if no point has that color.

Theorem 11.1 (Alternating-chain formula). *For every nonempty finite poset P and target A ,*

$$N_0(A, P) = 1 + a_1, \quad N_1(A, P) = 1 + a_0, \quad (11.1)$$

and therefore

$$\boxed{\text{sm}(A, P) = \lceil \log_2(1 + \min(a_0, a_1)) \rceil.} \quad (11.2)$$

Proof. Closure in the upper-set topology is downward closure. We claim that $F_n^c \neq \emptyset$ exactly when there is an alternating chain of length n ending in color $1 - c$.

More precisely, F_n^c is the downward closure of the least elements of all such chains. This is immediate for $n = 1$. In the next step, intersect with color $(c + n + 1) \bmod 2$ and take downward closure. A point of that color below the least element of an existing length- n alternating chain has the opposite color from that least element, so the inequality is strict and it can be prepended to the chain. Conversely, the first point of any length- $(n + 1)$ chain is produced this way. The induction follows.

The largest index of a nonempty F_n^c is consequently a_{1-c} , giving (11.1). Now apply theorems 4.2 and 5.2. $\square$

Corollary 11.2 (Quadratic-time computation). *Given the comparability relation of an n -element poset and a target, its optimal finite query depth is computable in $O(n^2)$ time after choosing a linear extension.*

Proof. Process points in a linear extension. Set

$$a(x) = 1 + \max\{a(y) : y < x, y \text{ has the opposite color to } x\},$$

where the maximum of the empty set is 0. Then $a_b = \max\{a(x) : x \text{ has color } b\}$. Each comparable pair is inspected at most once. Formula (11.2) completes the computation. $\square$

Example 11.3 (A finite orientation conflict). An alternating four-element chain has profile $(4, 5)$ or $(5, 4)$, depending on its orientation. Its target takes two questions. Two disjoint chains with opposite orientations have profile $(5, 5)$, so their combined target takes three questions. The archive contains an explicit optimal query tree for this eight-point example.

[Added 29 September 2026, batch 52] **Three labels.** Part II's six two-point chains 01, 02, 10, 12, 20, 21 (section 21.1) are the three-label analogue: each chain takes one question, their union three. On posets, the least layer count for q labels is the shortest common supersequence of the reduced chain words (theorem 17.1); its binary form (22.1) is (11.1) read from the other end of each chain, and for three labels the decision problem is NP-complete (theorem 22.4).

Remark 11.4. This order-theoretic formula is not obtained by applying the Hausdorff finite-height theorem to a non-Hausdorff space. Both results are separate consequences of the general finite normal form, and their hypotheses are kept distinct.

12 Algorithms, exact checks, and reproducibility

The proof of theorem 4.2 gives a strategy compiler. First construct the two canonical closure sequences; choose the shorter one; then binary-search its layer indices. At a node where the possible

indices are $a, \dots, b-1$, choose $j = \lfloor (a+b)/2 \rfloor$ and ask $X \setminus F_j$. A yes answer selects $a, \dots, j-1$, and a no answer selects $j, \dots, b-1$.

For a general infinite space, computing a closure is a mathematical operation, not necessarily an effective algorithm on a presentation of the space. The complexity claim in section 11 is specifically for finite posets whose comparability relation is supplied.

12.1 Independent verification paths

The program `code/verify.py` uses only the Python standard library. For each finite poset and every target, it compares three independently implemented calculations: exhaustive minimax over all upper-set questions, canonical downward-closure profiles, and the alternating-chain dynamic program. It also constructs a balanced strategy tree from the shorter closure chain and checks the color returned on every point.

The exhaustive minimax recurrence is

$$d_A(S) = \begin{cases} 0, & S \text{ is monochromatic,} \\ 1 + \min_U \max\{d_A(S \cap U), d_A(S \setminus U)\}, & \text{otherwise,} \end{cases}$$

where U ranges over upper sets splitting S nontrivially. States are encoded as bit masks, and the recursion is memoized. This recurrence does not use the closure formula being tested.

Number of points	Naturally labelled posets	Targets checked
1	1	2
2	2	8
3	7	56
4	40	640
5	357	11,424
6	4,824	308,736
Total	5,231	320,866

All checks passed. “Naturally labelled” means that every strict comparison $i <_P j$ satisfies $i < j$ as an integer. These are not counts of isomorphism classes. Every finite poset has at least one such labelling, so the search covers every isomorphism type through six points, with repetitions possible.

The included run took approximately 13 seconds in the execution environment used for this report. That observation is not a portable performance guarantee. The machine-readable result is in `data/verification.json`; the text log, count table, ordinal-value table, sharp-family table, and eight-point strategy certificate are provided alongside it.

12.2 Scope of the computational evidence

Finite upper-set topologies are not convergent sequences and do not model the Hausdorff accumulation behavior of ordinal intervals. Accordingly, the finite checks do *not* establish the infinite-space counterexample, the Cantor–Bendixson formula, or its transfinite consequences. Those results follow from the proofs in sections 2, 6 and 10. The computation instead checks the general finite game/closure/alternation mechanism in a large family where a direct independent minimax computation is possible.

To reproduce the included exhaustive run, execute

```
python3 code/verify.py --max-n 6
```

To rebuild the PDF, execute `make pdf`, or run `pdflatex` three times on `article.tex`. All inputs needed for compilation are local; no network access, external bibliography processor, or data regeneration is required.

[Added 29 September 2026, batch 52] Reruns. Without `--output`, `code/verify.py` writes into `data/` and replaces the recorded files there; pass `--output` with a scratch directory to keep them. Part II's verifier is `code/02-finite-output-verify.py` (section 25). Only Part II's verifier accepts `--brute-n`, and its `--max-n` stops at 6.

13 What is settled and what is not

The topological-sum identity (1.1) is false as stated. The counterexample can be expressed using the ordinary ordinal intervals $[0, \omega]$ and $[0, \omega \cdot 2]$, and the one-move defect is sharp at every positive finite level. The proof also yields exact computations for every Hausdorff space of finite Cantor–Bendixson height and every countable scattered Hausdorff space.

The two central distinctions are structural. First, the number of alternating layers is different from the number of adaptive questions: they are related by a ceiling logarithm. Second, the minimum over the two orientations is different from the pair of oriented lengths. Taking a disjoint union can force a common orientation and increase the minimum length by one; when that passes a power of two, it costs a full additional query.

No transfinite version of (4.4) is asserted. In particular, the symbol ∞ in the finite profile is not an ordinal rank and cannot simply be inserted into an ordinal logarithm. Nothing here determines the games at arbitrary uncountable lengths or resolves the other problems posed in [1]. Those are separate questions.

The research claim should be read at its demonstrated scale: an explicit negative solution to an explicitly stated question, together with a complete additional classification on a natural order-topological class. The proofs are intended to be independently checkable; their priority and presentation can be assessed separately from their mathematical content.

Part II. Finite-output open-query games

Sections 1–13 and Appendices A and B form the original report (Part I) of 19 September 2026, unchanged except for dated pointers to this addition, marked [Added 29 September 2026, batch 52]. Part II was added on 29 September 2026 in batch 52 of ProveIt’s incoming-report intake, from one manuscript written against the pinned state of Part I. It replaces the membership bit of Part I by a label from a finite alphabet of $q \geq 2$ letters. At $q = 2$ it reproduces Part I’s values (section 14.4).

14 Provenance, conventions and the binary case

14.1 One manuscript, one addition

Part II is batch-52 manuscript 12, *Finite-Output Open-Query Games: Sharp Synchronization and a Complexity Dichotomy*, an AI-assisted research article “prepared for Vladimir Reshetnikov” and dated 29 September 2026 (22 pages, US Letter, as delivered). It arrived as the archive `ProveIt_Finite_Output_Open_Query_Games.zip` (inner directory `finite_output_open_query_games/`) in ProveIt commit `11e1e9001` and was placed in `a701d9098`. It was written against ProveIt commit `5c695cfd` (in full `5c695cfd70a8b6c91bd5b2c1e99e3baecd5a864`), at which Part I’s `article.tex` had the Git blob `5e5ed2a806` and its `README.md` the blob `a8e024d1e3`. Both blobs are unchanged at the placement commit, so every statement the manuscript makes about “the ProveIt report” [4] refers to Part I as printed above.

Its code and data are shipped beside Part I’s files with the prefix `02-finite-output-` (Part I’s unprefix files count as 01): the verifier `code/02-finite-output-verify.py`, its Makefile, seven delivered records under `data/` (the verifier’s five outputs, its console log and the build log of the delivered PDF), and its source audit and research-status record in the report root. Its manuscript, README and PDF are not shipped and survive in the arrival commit. The report README lists every file and every name the delivered files still use.

The manuscript contributes, in the order printed here: a tree-layer normal form for q labels on arbitrary spaces (section 16); an exact shortest-common-supersequence representation for posets (section 17); compact countable metrizable realizations of every reduced word (section 18); the restricted-start universal-word lemma (section 19); the sharp finite Cantor–Bendixson bound $(q - 1)h + s$ and the exact q -label invariant of finite sums of convergent towers (section 20); sharp synchronization under topological sums, with exact extra cost $\lceil \log_2 q \rceil$ (section 21); a polynomial binary case against NP-complete ternary decision problems (section 22); a cyclic approximation within $\lceil \log_2(q - 1) \rceil$ questions and exact ideal search (section 23); finite obstructions (section 24); its finite checks (section 25); its proof-dependency audit, relation to the literature and a formalization plan (section 26); ten research questions (section 27); a conclusion; and certificate conventions (Appendix C). Every result, proof, example, remark, limitation and question of the manuscript is printed.

Where the merge had to choose.

- *Duplicated binary material.* The manuscript states that Part I’s binary counterexample, binary alternating-chain algorithm and binary Cantor–Bendixson results are not claimed anew. Its q -label normal form (theorem 16.2) contains Part I’s theorem 4.2 at $q = 2$, and its binary chain formula (proposition 22.1) is Part I’s theorem 11.1 in other notation. Both are printed in both

parts, because Part II states and proves them for q labels in its own notation: at $q = 2$ its proofs are second routes (Part II orders the nonempty leaves of an arbitrary tree, Part I pads the tree to full depth). The dictionary is section 14.4.

- *One renamed symbol.* The manuscript’s convergent tower K_m is printed as K_m throughout Part II, because Part I’s K_k is a different space (section 14.3). No other symbol is renamed, and no normalization is changed.
- *Front matter.* The manuscript’s title block, abstract and status paragraph are printed in section 14.2; its sections 1–14 are Sections 15–28 here, and its Appendix A is Appendix C. Its theorems keep their positions within each section: its Theorem $n.k$ is Theorem $(n + 14).k$ here. Its Research questions 1–10 keep their numbers.
- *Bibliography.* Both manuscripts cite Chiozini, Csernák and Soukup [1]; the two entries are merged, with both access dates. The entry [4] is Part I at the pin.
- *Shipped names.* Paths and commands in the manuscript’s section 25 and Appendix C are rewritten to the shipped names; see the note there. The rewritten text promises no file that is not shipped.

[Added 29 September 2026, batch 56] **Corrected count.** When Part II was written in, the two sentences above said “nine research questions” and “Research questions 1–9”. Part II prints ten (section 27), as did the delivered manuscript; the count is corrected here, and no question was added or removed. Part III’s research questions continue the same counter as Research questions 11–19.

Status (manuscript 12). The results of Part II are proved in ordinary mathematics. The modified-supersequence hardness theorem of Lagoutte and Tavenas (theorem 22.2) is an explicitly identified external input; the transfer to a power-of-two depth budget is proved here. Part II is not independently refereed and not proof-assistant verified; historical priority is not certified. It announces no named longstanding published open problem as solved and does not assert the word “breakthrough”. The literature on finite partitions, difference hierarchies, typed graph fusion and supersequences may contain related or equivalent formulations. The exact tower formula (theorem 20.3) concerns the supremum over all q -colorings of the specified spaces, whereas the compact sharp synchronization examples (theorems 21.2 and 21.3) concern particular colorings; the two must not be conflated. Finite computation does not prove the infinite-space statements, and the verifier’s six-point option was not run. Nothing in Part I or Part II is formalized in Lean or Rocq in ProveIt, and the Lean library that shares the `SetTheory/Cardinals` directory (large cardinals) contains nothing about open-query games.

14.2 The manuscript’s title block and abstract

Finite-Output Open-Query Games. Sharp synchronization and a complexity dichotomy. Topology / Order / Query complexity. Research article prepared for Vladimir Reshetnikov, 29 September 2026.

An open-query algorithm asks whether a hidden point belongs to an open set and must determine one of finitely many labels. We extend the binary membership framework explored in ProveIt to an arbitrary finite output alphabet. Every finite correct query tree has an ordered monochromatic layer representation, and the optimal worst-case depth is the ceiling of the binary logarithm of the minimum layer count. For upper-set topologies of posets, this layer count is exactly the shortest common supersequence length of the reduced color words of chains.

Our principal quantitative result is a sharp synchronization theorem. If each component admits at most m layers over $q \geq 2$ labels, their topological sum admits at most $(q - 1)m + 1$ layers, and

this bound is best possible. Consequently, components of depth at most d can require exactly $d + \lceil \log_2 q \rceil$ questions after being combined. Sharpness holds not only for finite posets but also for fixed colorings of compact countable metrizable spaces: an iterated convergent-sequence construction realizes every reduced word with exactly its supersequence profile.

We also prove the sharp finite Cantor–Bendixson bound $(q-1)h + s$, where h is the last derivative rank and s is its number of colors, and compute the worst-case q -label query invariant of finite sums of convergent towers exactly.

The same representation exposes an algorithmic threshold. Optimal depth is polynomial-time computable for two labels, whereas its decision version is NP-complete for three labels, even on a disjoint union of chains. The hardness uses the known three-letter modified-supersequence theorem; an exact common-prefix padding argument handles the power-of-two query budget. In contrast, a polynomial-time cyclic-layer algorithm is within $\lceil \log_2(q-1) \rceil$ questions of optimal, hence within one question for three labels. We also give finite obstruction bounds, exact algorithms, complete proofs, and reproducible checks on 102,331 colored finite posets.

Status. The results are proved here in ordinary mathematics. The modified-supersequence hardness theorem is an explicitly identified external input. This article is not independently refereed or proof-assistant verified; historical priority is not certified.

14.3 Notation of Part II

Part II fixes a finite alphabet Σ of q letters and studies a *coloring* $c : X \rightarrow \Sigma$; Part I’s target A is the case $\Sigma = \{0, 1\}$, $c = \mathbf{1}_A$. Several letters carry different meanings in the two parts. Each meaning below is fixed for its part; where a false reading is tempting, the table prints it beside the true one.

Symbol	Part I	Part II
K_k, K_m	$K_k = [0, \omega^{2^k-1}]$, with $\text{sm}(K_k) = k$ (theorem 8.1).	K_m (the manuscript’s K_m) is the m -th convergent tower of lemma 18.1, of last derivative rank $m-1$; $\mathsf{K}_m \cong [0, \omega^{m-1}]$ for $m \geq 2$, so $K_k \cong \mathsf{K}_{2^k}$. <i>Not</i> $\mathsf{K}_k = K_k$: its value is $\text{sm}_2(\mathsf{K}_k) = \lceil \log_2 k \rceil$.
c	An outer color $c \in \{0, 1\}$ of a closed-layer chain; C_c its color class.	The coloring $c : X \rightarrow \Sigma$. Part I’s outer color is the <i>last</i> letter w_m of a Part II word, not its first.
h	Last derivative rank.	Last derivative rank in section 20; but h_c in corollary 19.2 is the supremum of reduced chain-word lengths, and the column <code>h</code> of <code>data/02-finite-output-sharp_families.csv</code> is the word length m .
m	The multiplier in $X_{h,m} = [0, \omega^h \cdot m]$.	A number of layers, a word length, or the index of K_m .
q	In the lower-bound proof of theorem 6.2, a second point of $X^{(h)}$.	The alphabet size $ \Sigma $.
S, s	$S = [0, \omega]$.	$S \subseteq \Sigma$ a set of admissible first letters, $s = S $; in theorem 20.1, s is the number of colors on $X^{(h)}$.
D	$D_j = X^{(j)} \setminus X^{(j+1)}$; $d_A(S)$ the verifier’s minimax.	$D(X, c)$ the optimal fixed-coloring depth; $D(A)$ the verifier’s minimax on a mask A .
L	$L(A, X)$, the least closed-layer length.	$L(X, c)$, the least admissible word length; $L(X, \mathbf{1}_A) = L(A, X)$.
N	The oriented lengths N_0, N_1 .	N the total input length in the proof of theorem 22.4.

Symbol	Part I	Part II
$A, \mathcal{A}$	The target.	$\mathcal{A}_{q,m}(S)$ a family of reduced words; A a point mask (section 23) or a number (remark after theorem 20.3).
sm, sm_q	$\text{sm}(X)$, an ordinal, the supremum over targets of the least winning length.	$\text{sm}_q(X)$, the supremum over q -colorings of the finite depth D , or ∞ . At $q = 2$, $\text{sm}_2(X) = \text{sm}(X)$ whenever either is finite.
∞	“No finite decomposition”; not an ordinal.	“No finite bound works”; not an ordinal rank.
Layer order	$F_0 = X \supseteq F_1 \supseteq \dots$; layer 0 is the outer (open) one.	Positions $1, \dots, m$ from the bottom; suffixes are open. Part I’s layer i is position $m - i$, and $F_i = \{\lambda \leq m - i\}$.

Part II’s macros $\mathcal{W}, \mathcal{C}, \mathcal{A}, \text{scs}, D, L, \text{sm}_q$ and $\preccurlyeq$ are defined as in the manuscript; its leaf order (the whole no-subtree before the whole yes-subtree) is Part I’s lexicographic order with $0 < 1$.

14.4 The binary case: what Part II reproduces

Take $q = 2$, $\Sigma = \{0, 1\}$ and $c = \mathbf{1}_A$. The following checks were made when Part II was added; they are consequences of the two parts’ proofs, not new theorems.

1. *Normal form.* A layering λ with word $w_1 \dots w_m$ is the closed-layer chain $F_i = \{\lambda \leq m - i\}$ of outer color w_m , so $L(X, c) = L(A, X)$, and theorem 16.2 gives (4.4). For an alternating word ending in the letter e , the elimination (16.3) computes Part I’s canonical sequence (5.1) with outer color e term by term, so proposition 16.3 contains theorem 5.2.
2. *Posets.* Part I’s a_b reads chains from their greatest element, Part II’s r_a from their least; (22.1) and (11.1) both give $1 + \min(a_0, a_1)$ layers.
3. *Towers.* By lemma 18.1, $\bigsqcup_{i=1}^t K_m$ is Hausdorff with last derivative rank $h = m - 1$ and t terminal points, so theorem 6.2 gives $\lceil \log_2 m \rceil$ for $t = 1$ and $\lceil \log_2(m + 1) \rceil$ for $t \geq 2$: exactly (20.2) at $q = 2$. At $m = 2$, $t = 2$ this is the counterexample (2.1); at $m = 2^k$ it is (8.1) for finitely many copies. The formula was also compared with Part I’s recorded tables: all 129 rows of `data/ordinal_values.csv` ($h = 0, \dots, 128$) and all 16 rows of `data/sharp_family.csv` agree.
4. *The identification* $K_m \cong [0, \omega^{m-1}]$ ($m \geq 2$, used only to name Part I’s spaces). For $j \geq 1$, the space $[0, \omega^{j+1}]$ is the disjoint union of the clopen intervals $[0, \omega^j]$ and $(\omega^j n, \omega^j(n + 1)]$, $n \geq 1$, each homeomorphic to $[0, \omega^j]$ by the order-type argument of section 7, together with the point ω^{j+1} , whose neighborhoods contain all but finitely many of these intervals. A compact Hausdorff space is the one-point compactification of the complement of any of its points, so induction from $K_2 \cong [0, \omega]$ gives the claim.
5. *Sums.* At $q = 2$, theorems 21.2 and 21.3 give $m + 1$ layers and one extra question, as do lemma 5.3 and proposition 5.4; Part I’s theorem 8.1 attains the extra question for the supremum over targets, Part II’s families for fixed colorings. For $q \geq 3$ the exact overhead $\lceil \log_2 q \rceil$ exceeds one (section 21.1).
6. *Finite data.* The $q = 2$ rows of `data/02-finite-output-sharp_families.csv` with word lengths 2 and 4 are the chains 01, 10 (sum depth 2, a finite analogue of $S \sqcup S$) and 0101, 1010 (five layers, three questions), which is Part I’s eight-point example (profile (5, 5), `data/strategy_certificate.json`).

7. *The Cantor–Bendixson bound.* At $q = 2$, (20.1) reads $L \leq h + s$. Part I’s theorem 6.2 shows that both values of s that the terminal derivative allows are attained on *every* Hausdorff space of finite height, not only on towers; see the note after Research question 1.

15 The research question and its relationship to ProveIt

15.1 From a membership bit to a finite label

The natural object is a known labeling $c : X \rightarrow \Sigma$ of a topological space. A point $x \in X$ is hidden. An algorithm may ask any question of the form “is $x \in U$?”, where U is open. Questions may depend on previous answers. The algorithm must eventually report $c(x)$, not necessarily identify x . For two labels this is a set-membership problem. With more labels, different components may demand incompatible orders of resolving the labels.

Chiozini, Csernák, and Soukup introduced the topological cut-and-choose framework and asked whether the set-membership number of a topological sum always equals the supremum of its component values [1]. The ProveIt report *Open-query games on ordinal spaces* gives a binary counterexample and develops a finite alternating-layer normal form [4]. The present article starts from that direction but does not repeat its binary Cantor–Bendixson classification.

We address the following finite-output questions. How much extra information can synchronization across components require? Does the efficient binary optimization survive a third label? Can a finite non-Hausdorff obstruction be realized in familiar compact metrizable spaces? Which information, beyond a single optimal-depth number, composes correctly under topological sums?

The answers are the sharp bound $(q - 1)m + 1$, a binary/ternary complexity dichotomy, and the word profile introduced below. The compact realization result shows that the synchronization effect is not merely an artifact of finite specialization orders.

15.2 What is, and is not, being claimed

The repository baseline inspected here is the report in

```
SetTheory/Cardinals/docs/reports/
ordinals-and-order-types/games-on-ordinals/
open-query-membership-games/
```

at commit 5c695cfd70a8b6c91bd5b2c1e99e3baecd5a864. The README and the introductory part of its article were read, together with repository search results. This is a targeted comparison, not a proof audit of the whole ProveIt repository.

[Added 29 September 2026, batch 52] The baseline is Part I. That report is Part I above, and its `article.tex` and `README.md` are the same blobs at the pin and at the placement of this addition (section 14.1). The manuscript’s source audit (`02-finite-output-SOURCES.md`) says it read that README and lines 1–160 of the article.

The three-letter hardness of shortest common supersequences with no identical consecutive input letters is already known [5]. Related finite-partition and difference-hierarchy frameworks are also established subjects [9]. Thus neither multi-valued descriptive complexity nor the underlying supersequence hardness is announced as a discovery here. The mathematical contribution developed in this article is the explicit open-query representation, its sharp synchronization consequences,

the compact word realization, the exact depth reduction, and the additive query guarantee. Their historical novelty requires a broader independent literature review.

All games in this article have *finite worst-case budgets*. An infinite value means that no finite bound works. It is not an ordinal rank and makes no claim about transfinite determinacy, countable stopping times, or the full point-separating invariant of [1]. Except where expressly stated, we study a *fixed coloring*, not a supremum over all colorings of its space.

16 A normal form for arbitrary topological spaces

Fix a nonempty space X and a finite alphabet Σ , of size $q \geq 1$. No separation axiom is assumed.

Definition 16.1 (Depth, layers, and admissible words). For $c : X \rightarrow \Sigma$, let $D(X, c)$ be the minimum worst-case number of open questions in a finite correct deterministic decision tree, or ∞ if none exists. A word $w = w_1 \cdots w_m \in \Sigma^m$ is *admissible* if there is a map $\lambda : X \rightarrow \{1, \dots, m\}$ satisfying

$$c(x) = w_{\lambda(x)}, \quad U_j := \{x : \lambda(x) \geq j\} \text{ is open for every } j. \quad (16.1)$$

The fibers of λ are its ordered layers. Empty layers are allowed. Write $\mathcal{W}(X, c)$ for the set of admissible words, and set

$$L(X, c) = \min\{|w| : w \in \mathcal{W}(X, c)\},$$

with the minimum of the empty set interpreted as ∞ .

In a minimal representation there are no empty layers. Also, two consecutive layers carrying the same color can be merged: the remaining suffixes are still open. Thus a shortest admissible word is *reduced*, meaning that adjacent letters are distinct. This reduction is not deletion of all repeated letters; for example, 010 is reduced.

Theorem 16.2 (Tree-layer normal form). *For every (X, c) , the minimum number of nonempty leaves of a finite correct open-query tree equals $L(X, c)$. When this number is finite,*

$$D(X, c) = \lceil \log_2 L(X, c) \rceil. \quad (16.2)$$

In particular, a finite correct tree exists if and only if an admissible word exists.

Proof. Order the leaves of a query tree recursively, placing the entire no-subtree before the entire yes-subtree. We show that the union of any suffix of its ordered leaf cells is open. The assertion is proved by induction on the tree, with the current possible-point set given its subspace topology.

At the root, let the queried open set be U . If a suffix is contained entirely in the yes-subtree, induction expresses it as $U \cap V$ for an open set V of the current space. If the suffix includes part of the no-subtree, it contains the whole yes-subtree and has the form

$$U \cup ((X \setminus U) \cap V) = U \cup V,$$

again open. The same argument works in every current subspace. The one-leaf case is immediate. Removing empty leaf cells preserves the suffix property. Correctness makes each remaining leaf monochromatic. Its leaf index therefore defines a layering, so $L(X, c)$ is at most the number of nonempty leaves.

Conversely, given a layering with m nonempty layers, locate its index by balanced binary search. At an interval of candidate indices, split the interval as evenly as possible and query the open suffix U_j beginning at the split. This produces a correct tree with m leaves and depth at most $\lceil \log_2 m \rceil$.

A binary tree of depth d has at most 2^d leaves, so the first direction gives $L(X, c) \leq 2^d$ for any depth- d correct tree. The balanced construction supplies the opposite depth inequality and proves (16.2). $\square$

The theorem does not say that every optimal-depth tree has minimum leaf count. It says that minimum leaf count contains exactly the information needed to compute optimal worst-case depth. The class of all open sets is important: the proof uses closure under unions and finite intersections.

16.1 A fixed-word closure test

The following supplies a topological test without reference to a query tree. For a proposed word $w = w_1 \cdots w_m$, define closed sets by

$$G_0 = X, \quad G_{t+1} = \text{cl}_X(G_t \setminus c^{-1}(w_{m-t})) \quad (0 \leq t < m). \quad (16.3)$$

Thus we try to remove the top color, then the next color, retaining all limit points that cannot yet be discarded.

Proposition 16.3 (Greedy closed-set elimination). *The word w is admissible if and only if $G_m = \emptyset$.*

Proof. Suppose λ witnesses admissibility. Put $R_t = \{x : \lambda(x) \leq m - t\}$, so R_t is closed. Inductively $G_t \subseteq R_t$: after excluding color w_{m-t} , all remaining points of R_t lie in the closed set R_{t+1} . Taking closure therefore still stays in R_{t+1} . Consequently $G_m \subseteq R_m = \emptyset$.

Conversely, suppose $G_m = \emptyset$. Give a point removed at step $t + 1$ the index $m - t$. The removed set $G_t \setminus G_{t+1}$ has color w_{m-t} because G_{t+1} contains every point of G_t of another color. These removed sets partition X . The suffix beginning at index j is $X \setminus G_{m-j+1}$, which is open. This is the required layering. $\square$

The proposition is a finite colored elimination statement. It should not be confused with a classification of all transfinite partition hierarchies.

17 Posets and shortest common supersequences

Give a poset P its upper Alexandrov topology: U is open exactly when $x \in U$ and $x \leq y$ imply $y \in U$. A map $\lambda : P \rightarrow [m]$ has open suffixes exactly when it is order-preserving. This observation turns topology into a word-constrained monotone-map problem.

For a finite nonempty chain $x_1 < \cdots < x_r$, read the word $c(x_1) \cdots c(x_r)$ and collapse each maximal run of equal letters to one letter. Let $\mathcal{C}(P, c)$ denote the collection of all resulting reduced words. Write $u \preceq v$ when u is a subsequence of v , using a strictly increasing map of positions. For a word family $\mathcal{F}$, $\text{scs}(\mathcal{F})$ is the minimum length of a common supersequence, or ∞ when no finite common supersequence exists.

Theorem 17.1 (Exact chain-language representation). *For any poset, finite or infinite, and any finite word v ,*

$$v \in \mathcal{W}(P, c) \iff u \preceq v \text{ for every } u \in \mathcal{C}(P, c). \quad (17.1)$$

Consequently

$$L(P, c) = \text{scs}(\mathcal{C}(P, c)), \quad D(P, c) = \lceil \log_2 \text{scs}(\mathcal{C}(P, c)) \rceil \quad (17.2)$$

whenever the right-hand side is finite.

Proof. An admissible map is weakly increasing along a chain. At a color change its index must increase strictly, since one word position has only one color. Selecting one point per color run embeds the reduced chain word into v .

For the converse, greedily embed each reduced finite chain word into v . For $x \in P$, consider all finite chains ending at x , including the singleton chain. Their greedy last positions belong to the finite set $\{j : v_j = c(x)\}$. They are nonempty in number and all exist by hypothesis. Let $\lambda(x)$ be their maximum.

If $x < y$, take a chain ending at x whose greedy last position is $\lambda(x)$ and append y . If the last color repeats, reducing the word leaves the last greedy position unchanged; otherwise that position moves forward. The extended chain ends at y , so $\lambda(y) \geq \lambda(x)$. Also $v_{\lambda(x)} = c(x)$. Thus λ is a color-respecting order-preserving map, which proves admissibility. The formulas follow from Theorem 16.2. $\square$

17.1 A linear-time feasibility test on a finite DAG

Let the order be the reachability relation of an acyclic directed graph with n vertices and e edges, supplied in a topological order. The input may contain just cover edges, all strict comparisons, or any generating DAG. For a finite word v , write

$$\text{next}_a(b) = \min\{j \geq b : v_j = a\},$$

where the minimum is ∞ if the set is empty. Process the vertices in order and compute

$$b(x) = \max(\{1\} \cup \{f(y) : y \in \text{Pred}(x)\}), \quad f(x) = \text{next}_{c(x)}(b(x)). \quad (17.3)$$

Here $\text{Pred}(x)$ consists of incoming neighbors, not necessarily all predecessors. If a value is infinite the test rejects.

Proposition 17.2 (Least feasible map). *If the test succeeds, f is the pointwise least order-preserving map compatible with v and c . It succeeds exactly when v is admissible. With a next-position table, its running time is $O(q|v| + n + e)$.*

Proof. If g is any compatible monotone map, induction gives $f(y) \leq g(y)$ at already processed vertices. Hence $b(x) \leq g(x)$, and $g(x)$ is an occurrence of $c(x)$. The first possible such occurrence is at most $g(x)$. This proves minimality and shows that rejection precludes any compatible map. When all values are finite, monotonicity holds on every edge and hence on reachability. The recurrence uses one scan of the graph after constructing the next-position table backwards through the word. $\square$

The weak inequality $j \geq b(x)$ is essential. Comparable points of the same color may use the same layer. Replacing it by $j > b(x)$ would compute a different problem and would make even a constant-colored long chain unnecessarily hard.

Corollary 17.3 (Disjoint chains). *If P is a disjoint union of finite chains whose reduced color words are $w_1, \dots, w_r$, then*

$$L(P, c) = \text{scs}(w_1, \dots, w_r).$$

Proof. Every chain lies in a single component. Its reduced word is a subsequence of the component's reduced word. Thus the component words already impose all the constraints in Theorem 17.1. $\square$

18 Compact metrizable realizations of word profiles

Finite upper-set spaces are generally not Hausdorff. We now show that their word obstructions can survive in compact countable metrizable spaces. The result concerns the entire admissible-word profile, not only one numerical invariant.

Define K_1 to be a singleton. Recursively, K_m is the one-point compactification of the topological sum of countably many copies of K_{m-1} . Denote the new point by ∞_m . In particular, K_2 is a convergent sequence with its limit.

Lemma 18.1 (The convergent towers). *Each K_m is countable, compact, and metrizable. Its last nonempty Cantor–Bendixson derivative is $K_m^{(m-1)} = \{\infty_m\}$ for $m \geq 2$; $K_1^{(0)} = K_1$ is also a singleton. A neighborhood of ∞_m contains all but finitely many entire copies of K_{m-1} .*

Proof. A compact subset of a topological sum of compact components meets only finitely many components: the components themselves form an open cover. The assertion about neighborhoods follows directly from the one-point compactification. Countability and compactness follow by induction and this neighborhood property.

For metrizability, choose a metric $d_{m-1} \leq 1$ on each copy. Label copies by positive integers n . Within copy n , use $2^{-n}d_{m-1}$; give each of its points distance 2^{-n} from ∞_m ; and give points in distinct copies n, k distance $2^{-n} + 2^{-k}$. This is a metric inducing precisely the stated topology. Each copy is open and closed. Taking derivatives inside a copy agrees with its own derivatives, while ∞_m remains a limit point as long as infinitely many nonempty copies remain. Induction gives the derivative formula. $\square$

For a reduced word $w = a_1 \cdots a_m$, color K_m recursively: give ∞_m color a_1 , and give every copy of K_{m-1} the coloring for $a_2 \cdots a_m$. For $m = 1$, color its single point a_1 . Call this coloring c_w . Equivalently, a point of Cantor–Bendixson rank r has color a_{m-r} .

Theorem 18.2 (Exact compact word realization). *For every reduced word w and every finite word v ,*

$$v \in \mathcal{W}(K_{|w|}, c_w) \iff w \preceq v. \quad (18.1)$$

In particular, $L(K_m, c_w) = m$ and $D(K_m, c_w) = \lceil \log_2 m \rceil$.

Proof. The word w itself is admissible. Put the top point in layer 1, and use the recursive suffix layering, shifted by one, in every component. Suffixes starting after layer 1 are unions of open subsets of the open components; the first suffix is the whole space. Embedding these layer indices in any supersequence v preserves the openness of suffixes, so $w \preceq v$ implies admissibility.

For the converse, induct on m . The singleton case is immediate. Suppose λ witnesses admissibility of v on K_m , and set $j = \lambda(\infty_m)$, so $v_j = a_1$. The open suffix $\{x : \lambda(x) \geq j\}$ contains ∞_m . By Lemma 18.1, it contains an entire copy of K_{m-1} . On that copy the restricted layering uses only indices $j, \dots, |v|$. The induction hypothesis embeds $a_2 \cdots a_m$ in $v_j \cdots v_{|v|}$. Since $a_2 \neq a_1 = v_j$, the first position of this embedding is strictly greater than j . Prefixing position j embeds $a_1 \cdots a_m = w$ into v , as required. $\square$

Corollary 18.3 (Compact realization of any finite colored poset). *Every finite colored poset has the same admissible-word profile as some compact countable metrizable colored space of finite Cantor–Bendixson height.*

Proof. There are finitely many reduced chain words. Take the finite topological sum of one compact realization for each such word. A word is admissible on this sum exactly when it is admissible on every component: restriction proves necessity, and componentwise gluing proves sufficiency. Theorem 18.2 and Theorem 17.1 now give equality of profiles. A finite sum of compact countable metrizable spaces has the same properties. $\square$

This construction is an existence theorem and can be exponentially large if one lists all chain words of a poset. For a given list of chain words, however, it preserves that list directly as a succinct description of the compact instance.

19 The extremal supersequence lemma

The exact constants come from a small extremal word problem. Let $\mathcal{A}_{q,m}(S)$ consist of all reduced words of length $m \geq 1$ over Σ whose first letter lies in a fixed nonempty set $S \subseteq \Sigma$. Write $s = |S|$ and $\mathcal{A}_{q,m} = \mathcal{A}_{q,m}(\Sigma)$.

Theorem 19.1 (Restricted-start universal words). *For $q \geq 2$ and $1 \leq s \leq q$,*

$$\text{scs}(\mathcal{A}_{q,m}(S)) = s + (q - 1)(m - 1). \quad (19.1)$$

In particular,

$$\text{scs}(\mathcal{A}_{q,m}) = (q - 1)m + 1. \quad (19.2)$$

Proof. Order the alphabet cyclically, with the s letters of S first, and repeat this order indefinitely. Greedily embedding a word in $\mathcal{A}_{q,m}(S)$ puts its first letter at a position at most s . Every next letter differs from its predecessor and therefore occurs within the next $q - 1$ positions. The prefix of length $s + (q - 1)(m - 1)$ is thus a common supersequence.

For the lower bound, let v be any common supersequence. Choose a first letter in S whose first occurrence is latest. Its position is at least s . Having chosen a prefix ending with letter a at its greedy position j , consider the next occurrences after j of all $q - 1$ letters different from a . Every such occurrence exists while the chosen prefix has length less than m : otherwise that prefix followed by the missing letter could be extended to a word in $\mathcal{A}_{q,m}(S)$ that does not embed in v .

These $q - 1$ next occurrences occupy different positions. Choose the latest; its distance from j is at least $q - 1$. Continuing this adversarial choice gives a reduced word of length m whose greedy last position is at least $s + (q - 1)(m - 1)$. Its existence forces the same lower bound on $|v|$. $\square$

Corollary 19.2 (Bounded color alternation is exactly finite solvability). *For a nonempty colored poset, let h_c be the supremum of lengths of reduced finite chain words. A finite correct open-query tree exists if and only if $h_c < \infty$. In that case*

$$h_c \leq L(P, c) \leq (q - 1)h_c + 1 \quad (q \geq 2). \quad (19.3)$$

The upper bound is sharp for every $q \geq 2$ and $h_c \geq 1$.

Proof. Every chain word must embed into an admissible word, giving the lower bound. If all reduced chain words have length at most h_c , extend each shorter one to a reduced word of length h_c . The universal word from Theorem 19.1 embeds all of them, so Theorem 17.1 gives the upper bound. A disjoint union of chains carrying every word of $\mathcal{A}_{q,h_c}$ attains equality. Unbounded lengths preclude any finite word. $\square$

The lower bound in (19.3) can be supplemented by the number of colors actually present. Neither bound alone determines the answer: chain words of the same length may synchronize in very different ways.

20 Sharp Cantor–Bendixson bounds and exact tower invariants

For a space X , let $X^{(0)} = X$ and let $X^{(r+1)}$ be the nonisolated-point subspace of $X^{(r)}$. Suppose $X^{(h)} \neq \emptyset$ and $X^{(h+1)} = \emptyset$, where h is a nonnegative integer. The Cantor–Bendixson rank of x is the unique r with $x \in X^{(r)} \setminus X^{(r+1)}$.

Theorem 20.1 (A sharp finite-height layer bound). *Let X be a nonempty Hausdorff space of last derivative rank $h < \infty$. Let $c : X \rightarrow \Sigma$, where $q = |\Sigma| \geq 2$, and let $s = |c(X^{(h)})|$ be the number of colors appearing on its last nonempty derivative. Then*

$$L(X, c) \leq (q - 1)h + s, \quad D(X, c) \leq \lceil \log_2((q - 1)h + s) \rceil. \quad (20.1)$$

For every $q \geq 2$, $h \geq 0$, and $1 \leq s \leq q$, the layer bound is attained by a compact countable metrizable colored space having exactly s points in its last nonempty derivative, with distinct colors on those points.

Proof of the upper bound. Choose a cyclic order of the alphabet whose first s entries are the colors on $X^{(h)}$. Let v be the prefix of the repeated order of length $M = s + h(q - 1)$. Give a point of rank h the first occurrence of its color in v ; its index is at most s .

For a point of rank $r < h$, put

$$b_r = s + (h - 1 - r)(q - 1)$$

and assign it the first occurrence of its color at or after b_r . The interval $[b_r, b_r + q - 1]$ has q positions and contains the whole alphabet, so this assignment exists and lies in $[1, M]$. For two consecutive lower ranks, the maximum permitted index at the higher rank equals the minimum permitted index at the lower rank. The corresponding comparison also holds between rank h and rank $h - 1$: their boundary is s .

Every point x of rank r has an open neighborhood V_x satisfying $V_x \cap X^{(r)} = \{x\}$, because it is isolated in $X^{(r)}$. Every other point of V_x has smaller rank and hence an assigned index at least that of x . Thus $V_x \subseteq \{y : \lambda(y) \geq \lambda(x)\}$. Each suffix of λ is a union of such neighborhoods, and is open. This proves admissibility of v . For $h = 0$, the same argument simply orders the s colors of the discrete space. The depth estimate follows from Theorem 16.2. $\square$

To prove sharpness we need colorings richer than a single rank word. For $a \in \Sigma$ and $m \geq 2$, color the top point of K_m by a . List every reduced word u of length $m - 1$ whose first letter differs from a , and use infinitely many component copies with coloring c_u for each listed word. There are finitely many word types and countably many copies. Denote the resulting coloring by $b_{a,m}$. Put $b_{a,1} = a$ on K_1 .

Lemma 20.2 (All words with one prescribed initial color). *For every $m \geq 1$,*

$$\mathcal{W}(\mathbf{K}_m, b_{a,m}) = \{v : w \preceq v \text{ for every } w \in \mathcal{A}_{q,m}(\{a\})\}.$$

Proof. For $m = 1$ the assertion is the singleton case. For $m \geq 2$, suppose v is admissible and its layering gives the top point index j , with $v_j = a$. The open suffix beginning at j contains all but finitely many component copies. Since each word type u occurs infinitely often, one entire copy of each type lies in that suffix. Theorem 18.2 embeds u into $v_j \cdots v_{|v|}$, and its first letter differs from a , so this embedding starts after j . Prefixing j embeds every word au in the required family.

Conversely, suppose every au embeds into v . Let j be the first occurrence of a in v . In an embedding of any au , the initial a can be replaced by this earlier occurrence, so every u embeds after j . Use such an embedding and Theorem 18.2 to place the layers of every component after j ; place the top point at j . Suffixes beginning at most at j are the whole space, while later suffixes are unions of open subsets of components. This is an admissible layering. $\square$

Completion of the sharpness proof of Theorem 20.1. Let S be a set of s alphabet letters and take one copy of $(\mathbf{K}_{h+1}, b_{a,h+1})$ for each $a \in S$. For $h = 0$ these are s singletons. The last derivative consists of their s top points. By Lemma 20.2, the word profile of the sum requires exactly the words of $\mathcal{A}_{q,h+1}(S)$ as subsequences. Its minimum word length is $s + (q - 1)h$ by Theorem 19.1. $\square$

Now take a supremum over all colorings, in contrast to the fixed-coloring invariant $D(X, c)$:

$$\text{sm}_q(X) := \sup_{c: X \rightarrow \Sigma} D(X, c).$$

The alphabet has q available colors; a coloring need not use all of them.

Theorem 20.3 (Exact finite-output invariant of convergent towers). *For all integers $m, t \geq 1$ and $q \geq 2$,*

$$\boxed{\text{sm}_q\left(\bigsqcup_{i=1}^t \mathbf{K}_m\right) = \lceil \log_2((q-1)(m-1) + \min(q, t)) \rceil.} \quad (20.2)$$

In particular,

$$\text{sm}_q(\mathbf{K}_m) = \lceil \log_2(1 + (q-1)(m-1)) \rceil. \quad (20.3)$$

Proof. The last nonempty derivative is at rank $m - 1$ and has t points. Every coloring uses at most $\min(q, t)$ colors there. The upper bound is therefore Theorem 20.1. For the lower bound, choose $\min(q, t)$ distinct top colors and the corresponding colorings $b_{a,m}$. If extra copies remain, duplicate any one of those colored components. Duplicating its word constraint does not change the intersection of word profiles. The sharpness construction in Theorem 20.1 gives exactly the claimed layer count and hence depth. $\square$

For $q = 2$, (20.2) is $\lceil \log_2 m \rceil$ for one tower and $\lceil \log_2(m + 1) \rceil$ for two or more, consistent with the binary repository result [4]. The theorem above supplies the finite-alphabet extension for this compact family, including its worst-case colorings and explicit proofs.

[Added 29 September 2026, batch 52] Checked against Part I. The binary result is theorem 6.2 applied to $\bigsqcup_{i=1}^t \mathbf{K}_m$ (last derivative rank $m - 1$, t terminal points). At $m = 2$, $t = 2$ the value 2 is Part I's counterexample (2.1); since $\mathbf{K}_{2^k} \cong K_k$, the values k and $k + 1$ at $m = 2^k$ are (8.1). The formula agrees with all rows of Part I's recorded tables (section 14.4, items 3 and 4).

Remark 20.4 (A limitation and a useful distinction). Theorem 20.1 gives a sharp *uniform bound over a class of spaces*; it does not assert that every coloring attains it, or that every Hausdorff space of the specified height has the same multi-output invariant. Theorem 20.3 is an exact statement for the defined towers and their finite sums. When $m \geq 2$, its formula shows that summing these uncolored towers can increase sm_q by at most one, even though fixed colorings can exhibit the larger increase established next. Indeed, if $A = (q-1)(m-1)+1$, then $A \leq (q-1)(m-1)+\min(q, t) \leq A+q-1 \leq 2A$.

[Added 29 September 2026, batch 56] Beyond towers at $m = 2$. Part III computes sm_q for every Hausdorff space with a finite nonempty derived set (theorem 34.4). For $m = 2$ the value (20.2) holds for every such space whose nonisolated points are first countable or locally compact (corollary 36.2), and it fails for sums of free-ultrafilter spaces (theorem 38.1): at $q = t = 3$ the tower value is 3, the ultrafilter value 2 (35.1).

21 Sharp synchronization under topological sums

Proposition 21.1 (Profiles compose by intersection). *For any nonempty family of nonempty colored spaces, with the same alphabet,*

$$\mathcal{W}\left(\bigsqcup_{i \in I} X_i, \bigsqcup_{i \in I} c_i\right) = \bigcap_{i \in I} \mathcal{W}(X_i, c_i). \quad (21.1)$$

Proof. Restrict a layering of the sum to each component for one inclusion. For the other, take a layering of each component with the same word. Their maps glue, and every glued suffix is open because its restriction to each open component is open. One can choose component layerings by any such witnesses; alternatively Proposition 16.3 provides canonical layers for the prescribed word. $\square$

A scalar such as minimum depth or minimum length loses this compatibility information. Even the collection of shortest words alone need not be used as a substitute for the full admissibility profile: a longer locally admissible word may be useful for synchronization.

Theorem 21.2 (Sharp layer synchronization). *Let $q \geq 2$ and $m \geq 1$. If $L(X_i, c_i) \leq m$ for every i , then*

$$L\left(\bigsqcup_i X_i, \bigsqcup_i c_i\right) \leq (q-1)m + 1. \quad (21.2)$$

For every q, m the constant is best possible. Sharp examples can be chosen as finite disjoint unions of m -point chains, or as finite disjoint unions of copies of the compact countable metrizable space K_m with fixed rank-word colorings.

Proof. Each component admits a reduced word of length at most m . Every such word embeds into a reduced word of length m and hence into the single universal word of length $(q-1)m + 1$ in Theorem 19.1. Inserting empty layers according to this embedding makes the universal word admissible on every component. Proposition 21.1 gives the upper bound, with no restriction on the number of components.

For sharpness, take one component for each $w \in \mathcal{A}_{q,m}$, either the chain carrying w or the compact colored tower (K_m, c_w) . Each component has exactly m layers. By Corollary 17.3 or Theorem 18.2, the union's profile is exactly the common supersequence profile of $\mathcal{A}_{q,m}$, whose minimum length is $(q-1)m + 1$. There are $q(q-1)^{m-1}$ components. $\square$

Theorem 21.3 (Exact maximal extra query cost). *For $q \geq 2$ and $d \geq 0$, if every component has fixed-coloring depth at most d , then*

$$D\left(\bigsqcup_i X_i, \bigsqcup_i c_i\right) \leq d + \lceil \log_2 q \rceil. \quad (21.3)$$

Equality is attainable for every pair (q, d) , both by finite colored posets and by finite families of compact countable metrizable colored spaces.

Proof. Use Theorem 16.2 and Theorem 21.2 with $m = 2^d$. If $k = \lceil \log_2 q \rceil$, then $2^{k-1} \leq q - 1 < 2^k$, so

$$2^{d+k-1} < (q-1)2^d + 1 \leq 2^{d+k}.$$

Thus the ceiling of its logarithm is exactly $d + k$. The sharp examples of Theorem 21.2 have each component depth exactly d and union depth exactly $d + k$. $\square$

There is a simple independent intuition for the upper depth bound. Synchronize component trees for d rounds, taking at each round the union of the appropriate component queries. After that transcript, the surviving points in each component have one known color, but different components may have different colors. Queries that are unions of whole components resolve the remaining q possibilities in at most $\lceil \log_2 q \rceil$ rounds. The layer theorem does more: it finds the exact extremal number of layers and proves the depth loss sharp.

For the supremum-over-colorings invariant, the same upper estimate holds if $\text{sm}_q(X_i) \leq d$ for all i . It is sharp on finite upper-topology chains of 2^d points: their worst-case layer count is 2^d , attained by an alternating two-color word. The compact sharpness construction in Theorem 21.3 is a statement about its *specified colorings*. It does not assert that $\text{sm}_q(K_{2^d}) = d$ for $q > 2$; Equation (20.3) gives the different, correct value.

21.1 The first genuinely three-color example

Take six two-point chains with color words

$$01, \quad 02, \quad 10, \quad 12, \quad 20, \quad 21.$$

Each needs one question. The word 01201 is a common supersequence, with the following embeddings:

01	02	10	12	20	21
(1, 2)	(1, 3)	(2, 4)	(2, 3)	(3, 4)	(3, 5)

No word of length four contains all six ordered pairs. Such a word must use all three colors, and at least two of the colors occur only once; only one order of those two colors can then be a subsequence. Hence the sum has exactly five layers and exactly three questions. In particular, the binary “one extra question” bound is false for fixed three-color targets.

Replace each chain ab by a convergent sequence whose limit has color a and all isolated points have color b . Theorem 18.2 preserves every word constraint. This gives the same one-to-three increase on a compact countable metrizable space that is a finite sum of convergent sequences.

The six constraints are deletion-minimal for this three-query obstruction. More generally, if the ordered pair ba is deleted from the full q -color pair family, let R list all colors except a, b . The word $RabR$ has length $2q - 2$ and contains every remaining ordered pair. For $q = 3$ this is a four-layer, two-query solution after deleting any one component.

22 A binary/ternary complexity dichotomy

A finite input poset is given by a generating DAG or by its order relation, with an explicit color at every vertex. The decision problems ask whether $L(P, c) \leq k$ or whether $D(P, c) \leq d$, with an integer budget supplied as part of the input.

22.1 The binary case

For colors 0, 1, let r_a be the maximum length of a reduced chain word beginning with a , or zero if that color never occurs. Every reduced binary word alternates. An alternating master word starting with a embeds a reduced chain word beginning with a as soon as it has enough positions; embedding a word beginning with $1 - a$ costs one additional initial position.

Proposition 22.1 (Exact binary formula). *For a nonempty finite binary-colored poset,*

$$L(P, c) = \min\{\max(r_0, r_1 + 1), \max(r_1, r_0 + 1)\}. \quad (22.1)$$

Both L and D are polynomial-time computable.

Proof. A shortest master word is reduced and therefore alternating. The two possible initial colors give the two displayed lengths, by the preceding observation and Theorem 17.1. To compute the r_a , process the transitive order in a linear extension. For each vertex x and each starting color a , store the largest reduced-chain length beginning with a and ending at x . Extending a chain from $y < x$ increases the value by 1 precisely when $c(y) \neq c(x)$. This is a polynomial dynamic program; using an explicit order relation costs $O(n^2)$ after it is available. Finally apply Theorem 16.2. $\square$

This is the binary phenomenon already present in the repository report [4], now expressed in the chain-language notation. It is included to make the contrast with three colors precise, not as a new binary theorem.

[Added 29 September 2026, batch 52] **Where it is in Part I.** Theorem 11.1 and its quadratic-time corollary. Part I's a_b reads a chain from its greatest element, r_a here from its least; both formulas give $1 + \min(a_0, a_1)$ layers (section 14.4, item 2).

22.2 The imported hardness theorem

We use the following established result and no other external hardness assumption beyond ordinary NP-completeness terminology.

Theorem 22.2 (Lagoutte–Tavenas, modified SCS; external input). *Over three letters, deciding whether a finite family of words has a common supersequence of length at most k is NP-complete even when no input word has identical consecutive letters and no input word begins with a designated letter, which may be called 2.*

This is Corollary 6 of [5]; its strict-budget convention is equivalent by subtracting one from the budget. Its relationship to earlier typed-DAG fusion work of Darte is discussed there. We do not reprove that hardness theorem or claim that the reduced-word restriction is new.

Corollary 17.3 immediately transfers this result to the layer budget. The depth budget requires care: simply rounding $\log_2 k$ loses the exact threshold. The next lemma supplies the missing step.

Lemma 22.3 (Common-prefix cancellation). *For any finite word p and any nonempty finite family of finite words,*

$$\text{scs}(pw_1, \dots, pw_r) = |p| + \text{scs}(w_1, \dots, w_r). \quad (22.2)$$

Proof. Prefixing p to a shortest common supersequence proves the upper bound. For the reverse bound, let v contain every pw_i . Greedily embed p into v , and let its last position be j . For any embedding of pw_i , the prefix p finishes at a position at least j , so the embedded copy of w_i lies entirely after j . Consequently the suffix of v after j is a common supersequence of all w_i . Since $j \geq |p|$, this gives the lower bound. For the empty prefix the assertion is immediate. $\square$

Theorem 22.4 (Three colors suffice for NP-completeness). *For every fixed alphabet size $q \geq 3$, each of the following is NP-complete: deciding $L(P, c) \leq k$, and deciding $D(P, c) \leq d$. The result holds already for disjoint unions of chains.*

Proof. First consider membership in NP. Every n -point colored poset has a layering of at most n positions, obtained by listing its vertices in a linear extension. A word of length at most n together with a color-respecting monotone map to its positions is a polynomial certificate. Its correctness is checked on the generating edges. For the depth problem, test whether its length is at most 2^d . If $d \geq \lceil \log_2 n \rceil$, the answer is automatically yes; otherwise the integer 2^d is at most $2n$ and causes no encoding problem. The same certificate proves membership for a layer budget.

Layer hardness follows directly from Theorem 22.2 by making a chain for each input word. For depth hardness, take a modified-SCS instance $(w_1, \dots, w_r, k)$ over $\{0, 1, 2\}$ whose words do not begin with 2. Delete empty words. Trivial budgets are decidable directly and can be mapped to fixed yes- or no-instances, so assume $1 \leq k < N$, where $N = \sum_i |w_i|$.

Set

$$d = \lceil \log_2 k \rceil, \quad B = 2^d, \quad t = B - k.$$

If $t > 0$, choose the alternating word over $\{0, 2\}$ of length t ending in 2; call it p . If $t = 0$, let p be empty. Every word pw_i is reduced, including across the joining boundary, because no w_i starts with 2. Let P' be the disjoint union of chains carrying the words pw_i . By Corollary 17.3 and Lemma 22.3,

$$L(P', c') = t + \text{scs}(w_1, \dots, w_r).$$

Therefore

$$\begin{aligned} D(P', c') \leq d &\iff L(P', c') \leq B \\ &\iff \text{scs}(w_1, \dots, w_r) \leq B - t = k. \end{aligned}$$

The output has $N + rt$ vertices. Since $t < k \leq N$, its size is polynomial in the input size. This is an exact many-one reduction for the depth threshold. The three used colors can be viewed as a subset of any larger fixed alphabet. $\square$

[Added 29 September 2026, batch 56] **A second hardness route in Part III.** Part III's [proposition 39.1](#) is a second NP-completeness result, for profile-graph input with an unbounded alphabet, through directed feedback vertex set. For each fixed q that input model is solved by a finite search, which does not conflict with [theorem 22.4](#), whose instances are posets and tower lists with words of unbounded length.

22.3 The same hardness on succinct compact instances

The compact realization has an algorithmic consequence if the representation of an infinite space is specified rather than left implicit. Define a *tower-list instance* to be a finite list of reduced words $w_1, \dots, w_r$, denoting

$$\bigsqcup_{i=1}^r (K_{|w_i|}, c_{w_i}).$$

Its input size is the total word length, not the cardinality of the infinite underlying space. The allowed open sets are still all open sets of that space.

Corollary 22.5 (Compact rank-word instances). *For a fixed alphabet of at least three letters, deciding an exact finite layer budget or an exact finite query-depth budget on tower-list instances is NP-complete.*

Proof. Theorem 18.2 and Proposition 21.1 identify the entire admissible-word profile with the common supersequence profile of the input list. Thus the same certificates and the same padding reduction apply. $\square$

This is a precisely defined succinct-input result, not a claim about an unspecified computational presentation of arbitrary topological spaces.

23 Efficient approximation and exact algorithms

23.1 Cyclic master words

Fix a cyclic order of the q letters. For each of its q rotations, greedily map the input poset into the corresponding infinite periodic word using (17.3). Next occurrences are computed by modular arithmetic, so no infinite object is stored. Let M_r be the largest position used for rotation r . Return the shortest of the resulting prefixes, with its greedy rank map, and perform balanced suffix queries.

Theorem 23.1 (Additive approximation in query count). *For a finite nonempty q -colored poset, $q \geq 2$, the cyclic algorithm runs in $O(q(n + e))$ word operations and returns an admissible prefix of length at most*

$$M_{\text{cyc}} \leq 1 + (q - 1)(L(P, c) - 1). \quad (23.1)$$

Its query tree has depth at most

$$D(P, c) + \lceil \log_2(q - 1) \rceil. \quad (23.2)$$

For two colors it is exact. For three colors it is within one additional question of optimal.

Proof. Let w be a shortest admissible word of length L . It is reduced. Choose the cyclic rotation starting with its first letter. Greedily embed w into that periodic word. Its first position is 1, and every successive letter advances by at most $q - 1$, giving a last position at most $1 + (q - 1)(L - 1)$. Composing this word embedding with a layering for w produces a compatible monotone map into the periodic prefix. The greedy map is pointwise least by Proposition 17.2, so its last position is no larger. Minimizing across rotations proves (23.1).

Since $M_{\text{cyc}} \leq (q - 1)L$, taking logarithms and ceilings gives

$$\lceil \log_2 M_{\text{cyc}} \rceil \leq \lceil \log_2 L \rceil + \lceil \log_2(q - 1) \rceil.$$

The normal form converts this to the depth estimate. For $q = 2$, the length bound is $M_{\text{cyc}} \leq L$, forcing equality with the optimum.

Each rotation requires one graph scan. Every rank is at most qn , since at most n successive vertex assignments can increase a rank, each by at most q . The ranks therefore occupy $O(\log(qn))$ bits. This proves the stated word-operation bound; bit-operation costs include the corresponding integer arithmetic factor. $\square$

With alphabet $\{0, \dots, q-1\}$ and rotation starting with a , the next position of color c at or after a 1-based lower bound b is

$$b + (c - a - (b - 1) \bmod q),$$

where the remainder is in $\{0, \dots, q-1\}$. This is the implementation used in the verifier.

Empty positions may be deleted, and adjacent same-color occupied layers may then be merged, before building a query tree. These improvements only help; the theorem does not depend on applying them. The raw prefix-length bound is sharp for this particular algorithm: a chain following the reverse cyclic order has its first match at position 1 in the best rotation and each later match $q-1$ positions farther along. For example, the chain 2102 has optimum length 4 but uses positions 1, 3, 5, 7 in the best 012-cycle rotation. Deleting its unused positions recovers the optimum, so this example is *not* a lower bound against improved approximation algorithms.

23.2 Exact search on ideals

Let I be an order ideal, interpreted as the points already placed into early layers. Reading color a can process all points whose as-yet-unprocessed predecessors have that color. More precisely, define

$$T_a(I) = I \cup \{x \in P : \downarrow x \setminus I \subseteq c^{-1}(a)\}. \quad (23.3)$$

It is an ideal. It can be computed by a topological scan that adds a vertex of color a as soon as all its predecessors have been added.

Proposition 23.2 (Exact ideal automaton). *Starting at $\emptyset$, a word is admissible exactly when the successive transitions T_a reach P . Breadth-first search of this finite state graph therefore returns a shortest admissible word. If $J(P)$ is the number of order ideals, the running time is $O(qJ(P)(n+e))$, with at most $J(P) \leq 2^n$ states.*

Proof. After a word prefix has been read, process every vertex as early as its color and predecessors allow. Equation (23.3) is precisely the maximal possible extension using the next single color: a point can be included if and only if every still-unprocessed point below it can be included in this same monochromatic layer. Induction on the word length shows that this process contains the processed ideal of any other compatible partial layering. Thus it reaches all vertices if and only if some complete layering exists. Breadth-first search minimizes the number of transitions. Each of the q transitions per reached state takes one graph scan. $\square$

If a chain cover with chain lengths $n_1, \dots, n_r$ is supplied, every ideal intersects each chain in a prefix. Consequently

$$J(P) \leq \prod_{i=1}^r (n_i + 1).$$

This gives a polynomial algorithm for each fixed number of covering chains, including the familiar product-state algorithm for a fixed list of strings. It is an XP-type bound in that parameter, not an assertion of fixed-parameter tractability with an exponent independent of r .

The verifier uses exhaustive legal-question minimax as a separate algorithm. For a nonempty possible-point mask A that is not monochromatic,

$$D(A) = 1 + \min_U \max\{D(A \cap U), D(A \setminus U)\},$$

$$L_{\text{tree}}(A) = \min_U \{L_{\text{tree}}(A \cap U) + L_{\text{tree}}(A \setminus U)\},$$

where the minimum runs over open sets splitting A into two nonempty parts. A monochromatic nonempty mask has depth 0 and one leaf. The ideal automaton does not call these recurrences, making their agreement a useful independent finite check of the normal form.

24 Finite obstructions and bounded-budget decidability

Theorem 24.1 (Small finite witnesses of a failed layer budget). *For any poset P , finite or infinite, with q colors, if $L(P, c) > m$ for some $m \geq 1$, there is an induced finite colored subposet Q with*

$$|Q| \leq (m+1)q^m, \quad L(Q, c|_Q) > m. \quad (24.1)$$

Proof. For each word $v \in \Sigma^m$, Theorem 17.1 gives a reduced finite chain word not embedding into v . Stop its greedy embedding at the first failure. Each previous letter uses a strictly later position, so failure occurs within at most $m+1$ letters. Selecting the corresponding chain points gives an obstructing chain of at most that size.

Take the union of one such chain for each of the q^m words. If the induced subposet admitted a word of length at most m , append empty positions to obtain one of length exactly m . Its stored obstructing chain would then embed, a contradiction. This proves the bound. $\square$

In particular, a bounded finite query budget on an arbitrary Alexandrov poset is determined by its finite colored subposets. For a depth budget d , one may put $m = 2^d$ in (24.1). The numerical bound is an upper bound, not a claim of optimality or of a polynomial kernel.

On a finite DAG, enumerating the q^{2^d} words of length 2^d and applying Proposition 17.2 gives an exact algorithm with time

$$O\left(q^{2^d}(q2^d + n + e)\right).$$

Thus the problem is fixed-parameter tractable in the joint parameters q, d ; for fixed q , it is fixed-parameter tractable in the allowed query depth. This does not conflict with NP-completeness when the depth is part of an unbounded input.

Proposition 24.2 (Finite witnesses for sum incompatibility). *If every finite subfamily of a family of colored spaces has sum layer count at most m , then the entire sum has layer count at most m . If the bound fails, it fails already for a subfamily of at most q^m components.*

Proof. There are only q^m candidate words of length exactly m . If none is admissible on the whole sum, choose for each candidate a component on which it fails, using Proposition 21.1. Those at most q^m components already exclude every candidate. Padding an admissible shorter word shows that they also exclude every word of length at most m . $\square$

The two finite-witness statements have different domains. The induced-subposet statement is not a general finite-subspace theorem for Hausdorff spaces. Every finite subspace of a Hausdorff space is discrete, so it can have at most q necessary color layers, while the compact towers above have arbitrarily large layer complexity. Their obstructions lie in accumulation, not in finite Hausdorff subspaces.

25 Reproducible verification and certificates

The archive includes a standard-library Python verifier, not a proof-assistant formalization. Its full recorded run is, with the shipped name and an explicit output directory,

```
python3 code/02-finite-output-verify.py --max-n 5 --brute-n 4 --output <scratch>
```

A naturally labeled poset has every strict comparison directed from a smaller integer label to a larger one. Every finite poset admits such a labeling. The enumeration therefore covers every isomorphism type through the stated size, possibly many times; it is not an enumeration of unlabelled types.

For every binary and ternary coloring of every enumerated poset, the verifier compares exhaustive open-question minimax with shortest-word breadth-first search on ideals. It checks both minimum leaf count and the logarithmic depth formula, the binary formula when applicable, greedy feasibility, and the cyclic approximation bounds. The counts are:

Points	Naturally labeled posets	Binary colorings	Ternary colorings
1	1	2	3
2	2	8	18
3	7	56	189
4	40	640	3,240
5	357	11,424	86,751
Total	407	12,130	90,201

Thus there are **102,331 colored-poset tests**. Through four points, the program additionally enumerates all reduced chain words and solves their SCS problem by brute force. It checks **4,156 colored instances** in this way. For every candidate master word of lengths 1 through the point count in these instances, it compares chain subsequence containment, the greedy monotone map, and closed-set elimination. This gives **416,432 fixed-word checks**.

The restricted-start universal families are checked for $2 \leq q \leq 4$, $1 \leq m \leq 4$, and every $1 \leq s \leq q$: 36 parameter triples. For the 15 triples with $q \leq 3, m \leq 3$, the optimum is also found independently by exhaustive word enumeration. A further **1,176 tests** check exact common-prefix cancellation and the power-of-two reduction on pairs of small reduced ternary words.

The twelve-point example has a machine-readable certificate recording its predecessor masks, colors, master word, rank map, and balanced query tree. Every query is checked to be an upper set, and every point is followed to a correct leaf. The full outcomes are in `data/02-finite-output-verification.json`; tables and the strategy certificate are provided separately as CSV and JSON.

[Added 29 September 2026, batch 52] Shipped names. In the delivery the verifier was `code/verify.py`, its outputs lay in `results/`, and the command above read

```
python3 code/verify.py --max-n 5 --brute-n 4
```

Here `code/verify.py` is Part I's verifier, so the command above uses the shipped name. The recorded outputs are the five files and the console log

- `data/02-finite-output-verification.json`,
- `data/02-finite-output-strategy_certificate.json`,
- `data/02-finite-output-poset_counts.csv`,
- `data/02-finite-output-sharp_families.csv`,
- `data/02-finite-output-restricted_start_families.csv`,
- `data/02-finite-output-verification.log`.

The verifier writes unprefixed names; without `--output` it creates `results/` in the report directory. Its JSON files are written with the platform's line endings, so a rerun on Windows differs from the record by CRLF (and by the Python version and the elapsed time); its CSV files are CRLF on every platform, as delivered. The counts above agree with the recorded JSON and log.

These checks do not establish the infinite-space theorems, the general NP-hardness reduction, or historical novelty. Those claims rest respectively on the written proofs, the cited hardness input plus the written reduction, and a still-incomplete priority assessment. Finite Hausdorff subspaces are not used as substitutes for the compact tower arguments.

26 Proof dependencies, scope, and formalization

26.1 Proof dependency audit

The proofs form a short dependency chain. Tree flattening and balanced suffix search prove the normal form. The poset representation follows from monotone maps and finite greedy embeddings; compact realization follows from cofinite component neighborhoods. The adversarial next-occurrence argument proves the universal-word constant. These feed the synchronization theorem and the exact tower formula. Closed-set elimination supplies an independent topological normal-form test.

The only imported mathematical result needed for the NP-completeness claim is Theorem 22.2. The reduction to depth is proved here, including all boundary and input-size conditions. None of the topological or sharp-bound theorems depends on that complexity-theoretic input. No unproved conjecture is used as a theorem. The assertion that zero additive error cannot be guaranteed in polynomial time for three colors is, as usual, conditional on $P \neq NP$.

Indeed, Theorem 23.1 gives one extra query for three colors. A uniform guarantee of fewer than one extra query would, by integrality, be an exact algorithm and would contradict Theorem 22.4 if $P \neq NP$. In this precise integer-additive sense, the three-color approximation guarantee is optimal under that assumption. No analogous optimality claim is made for the bound $\lceil \log_2(q-1) \rceil$ when $q \geq 4$.

26.2 Relationship to the cited work

The binary repository layer framework is the direct starting point. Its binary counterexamples, binary alternating-chain algorithm, and binary Cantor–Bendixson results are not claimed anew. The finite-output exact tower formula, sharp synchronization family, and rounding-safe complexity transfer are the extensions developed here.

The SCS connection also has precursors in typed-graph fusion, as documented in [5]; a mere change of vocabulary would not be a priority claim. Likewise, Selivanov’s work places multi-valued partitions within a much broader descriptive-set-theoretic setting [9]. The present article is specifically about finite worst-case open-query budgets, explicit word certificates, and sharp constants. An independent comparison with the full partition-hierarchy and typed-fusion literature remains appropriate before submission as a novelty-certified paper.

26.3 A concrete proof-assistant plan

The finite part has a small trusted mathematical interface. A useful Lean formalization would proceed in the following order.

1. Define finite words, subsequence embeddings, reduced words, and color-respecting monotone maps to finite chains. Verify the greedy next-position recurrence and its least-map property.
2. Formalize finite binary query trees over a topology and the no-before-yes leaf order. Prove the suffix-open induction, then construct balanced suffix trees with the integer bound $m \leq 2^d$ rather than analytic logarithms.
3. Prove the chain-language equivalence, the universal-word upper bound, and the adversarial lower-bound witness. These yield synchronization with purely finite arithmetic at the final step.
4. Add a certified checker for a supplied word and rank map. A search program can remain untrusted: its positive certificates require only color agreement and monotonicity along input edges. Negative certificates can use an explicit failed chain for each candidate word or a verified ideal-state search, depending on the budget.
5. Formalize the convergent towers and their neighborhood property, then their rank-word profiles, the finite-height interval construction, and the exact tower invariant. The topological definitions should be kept separate from executable finite certificates.

The present archive contains no Lean file and makes no claim that any of these steps has been kernel-checked. The Python assertions are reproducible evidence about finite instances, not replacements for that formalization.

[Added 29 September 2026, batch 52] **Status in ProveIt.** None of these steps has been carried out in ProveIt, for Part II or for Part I: no Lean or Rocq file of the repository mentions open-query games, and the Lean library in the enclosing `SetTheory/Cardinals` directory is about large cardinals.

27 Further research questions and proposed directions

The following questions define a continuation program. They are not assertions that every formulation is a previously published open problem; their literature status should be checked individually before a priority claim.

Research question 1 (Exact invariants beyond convergent towers). For a general, possibly nonmetrizable or noncompact Hausdorff space of finite Cantor–Bendixson height, what local topological conditions make the uniform bound $(q - 1)h + s$ attainable? Can the invariant be characterized by the word profiles that accumulate at its nonisolated points?

The upper-bound proof uses only rank neighborhoods. Sharpness requires enough incompatible color patterns, and the tower construction supplies them by repeating each type cofinally. Separating those two ingredients is a concrete way to extend the classification without assuming that rank alone is always sufficient.

[Added 29 September 2026, batch 52] **Answered at $q = 2$; open for $q \geq 3$.** For two labels Part I answers both parts of this question. On every Hausdorff space of last derivative rank $h < \omega$ the bound $h + s$ is attained for $s = 1$ (the parity target of lemma 6.5) and, when $|X^{(h)}| \geq 2$, for $s = 2$ (the lower-bound proof of theorem 6.2); no condition beyond the Hausdorff property is needed, and $\text{sm}_2(X) = \text{sm}(X)$ depends only on h and on whether $X^{(h)}$ is a singleton, by (6.1). The question stands for $q \geq 3$.

[Added 29 September 2026, batch 56] **Answered for finite nonempty derived sets.** Part III answers both halves of this question for every q on Hausdorff spaces whose derived set is finite and nonempty, a proper subclass of $h = 1$. The accumulating profile of a coloring is a digraph G_c on its colors, whose arcs are the two-letter words accumulating at the nonisolated points, and $L(X, c) = q_c + \tau(G_c)$ with τ the directed feedback vertex number (theorems 33.1 and 33.4). The bound $(q - 1)h + s = q + s - 1$ is attained with exactly s colors on X' if and only if the nonisolated points admit a bin partition by their local partition capacities (theorem 34.5). Rank and the size of the last derivative do not determine sm_3 : three free-ultrafilter points give 2, three convergent sequences give 3 (35.1). If every nonisolated point is first countable or locally compact, some coloring attains the bound with $s = \min(q, t)$, where $t = |X'|$ (corollary 36.2). Heights $h \geq 2$ and infinite derived sets remain open (research questions 11, 12 and 14).

[Added 30 September 2026, batch 65] **Second half answered for fixed colorings on every space.** Part IV answers the second half of this question for every fixed coloring of every topological space, with no separation axiom. The relevant profile is the closure signature, the reduced words v whose nested color closure F_v is nonempty: a word is admissible exactly when it contains all of them as subsequences, and a rejected word of length m is refuted by one of length at most m (theorem 47.2). At finite depth the admissible words determine the signature (theorem 49.2), and finite-depth colored spaces are classified, up to equality of admissible words, by nonempty finite downsets of reduced words (theorem 50.5). On a Hausdorff space with $X'' = \emptyset$, whatever the size of X' , $L(X, c) = q_c + \tau(G_c)$ (theorem 51.3), extending Part III's formula. The signature is a global invariant of the coloring, built from nested closures, not a list of local capacities. The first half, which spaces attain $(q - 1)h + s$, and more generally which signatures the colorings of a given space can have, is not answered by Part IV and stays open for $q \geq 3$ beyond Part III's class (research questions 12 and 20).

Research question 2 (Small sharp synchronization families). What is the smallest number of reduced words of length at most m whose common supersequence length is $(q - 1)m + 1$? What is the smallest number needed merely to force the maximal depth $\lceil \log_2((q - 1)m + 1) \rceil$?

Our full family has $q(q - 1)^{m-1}$ words. For $m = 2$ the complete ordered-pair family is deletion-critical for its extremal layer count. For larger m , minimizing the obstruction family would substantially reduce both finite and compact examples.

Research question 3 (Optimal finite obstruction bounds). Determine the least function $B(q, m)$ such that every failure of an m -layer budget on a poset is witnessed by at most $B(q, m)$ points. How far can the bound $(m + 1)q^m$ be improved, and which colored posets are minimal obstructions?

The proof supplies explicit chains per candidate word but does not share their vertices efficiently. Critical SCS families are a natural first test class.

[Added 30 September 2026, batch 65] **Bound improved for finite posets.** For finite posets Part IV lowers the bound $(m + 1)q^m$ of theorem 24.1 to $mq(q - 1)^{m-1}$ (corollary 52.2): a rejected word of length m has a detected counterword of length at most m (theorem 47.2), and only the $q(q - 1)^{m-1}$ reduced candidates need witnesses. At $q = 3$, $m = 2$ this is 12 points instead of 27. The bound is not claimed optimal; the least $B(q, m)$ remains open, and Part IV asks it again as research question 21.

Research question 4 (Width and cover-graph parameters). For three colors, can the ideal-state algorithm be improved from an $(n + 1)^r$ -type bound for an r -chain cover to a fixed-parameter algorithm with parameter r ? Which cover-graph restrictions permit polynomial exact optimization when width is unbounded?

The finite-depth parameter is already fixed-parameter tractable by the word enumeration above. Width is different: long chains do not by themselves cause hardness, but synchronization among many chains does.

Research question 5 (One-extra-query algorithms with four or more colors). Does every fixed $q \geq 4$ admit a polynomial-time algorithm within one question of optimal, or is a larger additive gap necessary under an appropriate complexity assumption?

The cyclic construction gives $\lceil \log_2(q-1) \rceil$ extra questions. Exact three-color hardness only excludes zero error; it does not prove the cyclic bound optimal for larger alphabets. Improved SCS approximation or a direct query-tree method could give a better result.

Research question 6 (Compact profile representations). Can the full word profile of a finite colored DAG be realized by a compact metrizable tower description of polynomial size without listing every chain word? Which sharing operations on descriptions preserve the profile exactly?

The current construction is exact but can duplicate exponentially many chains. A compositional representation of shared suborders would connect the topology more directly to symbolic verification and succinct optimization.

[Added 30 September 2026, batch 65] Bears on this question. Part IV realizes every finite-depth word profile, of any space, by a finite sum of towers, one for each maximal word of its closure signature, of height at most its rank ρ ; that height is the least Cantor–Bendixson height of any Hausdorff representative (theorems 50.5 and 51.2). The number of towers can be as large as $q(q-1)^{\rho-1}$ (proposition 48.2), and no polynomial-size description is claimed; research question 22 asks for minimal representatives.

Research question 7 (Expected query cost and nonuniform costs). When points have a probability distribution, or queries have different costs, what replaces the layer-count logarithm? Can one optimize jointly over admissible layerings and ordered coding trees without enumerating all words?

Balancing by number of layers minimizes a worst-case bound, not expected cost. The flattening theorem still supplies an ordered structure, but the masses of the layers and the choice of word become essential data.

Research question 8 (Finite-output transfinite games). What analogue of the word profile describes well-founded, non-uniformly bounded query trees and games of ordinal length? Which synchronization laws survive at limit stages?

Neither $\lceil \log_2 L \rceil$ nor finite common-supersequence padding should be transferred mechanically to ordinal lengths. The treatment must specify the winning condition at limits and the kind of trees being ranked.

Research question 9 (Restricted query bases and noisy answers). How do the results change when only a designated basis of open sets may be queried, or when finitely many answers may be incorrect? What closure conditions on a query family retain a useful ordered-layer normal form?

The proof of Theorem 16.2 identifies the exact operations to investigate: finite intersections in a yes-branch and unions across a no-branch. Without these operations, the current equivalence need not persist.

Research question 10 (Certified integration into ProveIt). Can the finite greedy checker, the sharp-word adversary, and the compact rank-bound theorem be formalized as reusable interfaces rather than as isolated examples? Can proof-producing search return short negative certificates for practical three-color instances?

A successful integration would distinguish four outputs explicitly: a proved general theorem, a checked instance certificate, a computational experiment, and a literature-status claim. These are different forms of evidence.

28 Conclusion

The information needed to combine finite-output open-query algorithms is not just their depths. It is the set of color words in which their possible answers can be arranged with open suffixes. For posets this profile is an exact chain supersequence constraint; for compact convergent towers the same constraints are imposed by accumulation.

This representation yields three quantitative conclusions. First, the sharp uniform sum bound is $(q - 1)m + 1$ layers, with maximal depth increase $\lceil \log_2 q \rceil$. Second, finite Cantor–Bendixson height gives the sharp bound $(q - 1)h + s$, and the worst-case invariant of a finite sum of convergent towers is computed exactly by (20.2). Third, the introduction of a third label changes exact finite optimization from polynomial time to NP-completeness while still permitting a one-extra-query algorithm. All three conclusions have explicit witnesses and constructive proofs, and the finite part admits small, independently checkable certificates.

Part III. Filter profiles and open-query complexity

Sections 1–13 and Appendices A and B form Part I (19 September 2026); Sections 14–28 and Appendix C form Part II (batch 52). Both are unchanged except for dated pointers, marked [Added 29 September 2026, batch 52] or [Added 29 September 2026, batch 56], and the corrected count of Part II’s research questions (section 14.1). Part III was added on 29 September 2026 in batch 56 of ProveIt’s incoming-report intake, from one manuscript written against the pinned state of Parts I and II. It answers Part II’s research question 1 for every number q of labels on Hausdorff spaces with a finite nonempty derived set. At $q = 2$ it reproduces Part I’s values, and on sums of convergent sequences Part II’s tower formula (section 29.4).

29 Provenance, conventions and the relation to Parts I and II

29.1 One manuscript, one addition

Part III is batch-56 manuscript 05, *Filter Profiles and Open-Query Complexity: Exact finite-derivative classification, ultrafilter thresholds, and a three-color separation*, an AI-assisted research report “prepared for Vladimir Reshetnikov” and dated 29 September 2026 (20 pages, US Letter, as delivered). It arrived as the archive `ProveIt_Filter_Profiles_Open_Query_Complexity.zip` (inner directory `proveit_filter_games/`) in ProveIt commit `e47ed8547` and was placed in `ae718a441`. It was written against ProveIt commit `2d4919838` (in full `2d49198382cc068f46747d1b3d9067d47cee4f24`), which is later than Part II’s write (`f11edd3c5`). At that commit this report’s `article.tex` had the Git blob `75f0c4d7ba` (Parts I and II, including the batch-52 note under [research question 1](#)) and its `README.md` the blob `3357ea6820`; the manuscript records the first of these only. Both blobs are unchanged at the placement commit, so every statement the manuscript makes about “the repository report” refers to Parts I and II as printed above, and its “Part II, Section 27, Research question 1” is [research question 1](#) of section 27 here. Its source audit says that the README and the relevant article sections were read through the GitHub connector at that revision.

The manuscript cites that state of the report as its reference [ProveIt]: “V. Reshetnikov’s ProveIt repository, *Open-query games on ordinal spaces*, Part II, *Finite-Output Open-Query Games: Sharp Synchronization and a Complexity Dichotomy*, especially Section 27, Research questions 1–2. Repository revision `2d49198382cc068f46747d1b3d9067d47cee4f24`, article blob `75f0c4d7ba520326e6adf029163f672d1f764192`. Source inspected September 29, 2026.” It adds the repository URL and says that the exact relative path is recorded in its source audit, which is shipped. Inside this report the reference is internal, so its citations are replaced by cross-references (see below).

Its code and data are shipped beside those of Parts I and II, and their names carry the prefix `03-filter-profiles-`; Part I’s unprefixed files count as 01 and Part II’s as 02. They are the verifier `code/03-filter-profiles-verify.py`, its delivered Makefile, six recorded outputs under `data/` (three JSON files, two CSV tables and the console log), and its source audit `03-filter-profiles-SOURCES.md` in the report root. Its manuscript, PDF, delivery README and checksum ledger (`SHA256SUMS`, twelve entries under the delivery names, verified 12/12 at intake) are not shipped and survive in the arrival commit. The report README lists every file and every name the delivered files still use.

The manuscript contributes, in the order printed here: its statement of the question and of the main formulas (section 30); the ordered normal form, reproved (section 31); a clopen decomposition and a filter model for every Hausdorff space with a finite derived set (section 32); the exact admissible-word profile of every finite coloring as a directed graph, with layer count $q_c + \tau(G_c)$ for the directed feedback vertex number τ (section 33); local partition capacities, the capacity classification of sm_q and the bin-partition criterion for attaining the rank-one bound (section 34); the complete three-label classification and the separation of three ultrafilter points from three convergent sequences (section 35); the positive classification for first-countable or locally compact points (section 36); an exact directed extremal arc threshold with its equality graphs (section 37); every threshold for sums of free-ultrafilter spaces (section 38); an exact finite algorithm, its input qualification, NP-completeness for graph-profile input and a formalization route (section 39); its finite checks and dependency audit (section 40); nine research questions (section 41); a conclusion; and certificate conventions (Appendix D). Every result, proof, example, remark, limitation and question of the manuscript is printed.

Where the merge had to choose.

- *Front matter and numbering.* The manuscript's title block, abstract and status paragraph are printed in section 29.2; its table of contents is dropped. Its Sections 1–13 are Sections 30–42 here, and its Appendix A is Appendix D, after the appendices of Parts I and II. Its theorems keep their positions within each section: its Theorem $n.k$ is Theorem $(n + 29).k$ here. The manuscript numbers equations consecutively, (1)–(14); this report numbers them within sections, so its equation (2) is (30.2), and every equation keeps its label. Its Research questions 1–9 are Research questions 11–19 here, because the counter continues after Part II's ten.
- *Duplicated material.* The normal form theorem 31.3 is Part I's theorem 4.2 and Part II's theorem 16.2; the manuscript reproves it to be independent of the repository, and its proof, the same leaf-order and balanced-suffix argument as Part II's, is printed in full. The bound (34.3) is Part II's theorem 20.1 at $h = 1$, and corollary 36.2 contains Part II's theorem 20.3 at $m = 2$; the manuscript says so itself. The checks are collected in section 29.4.
- *Renamed macros.* The manuscript's `\smq` was `sm` with the subscript written after it (`smq`, `sm3`, `sm2`); Part II's `\smq` already includes the subscript q , so the manuscript's ten uses are set with Part I's `sm`. The printed symbol is unchanged. The manuscript's sans-serif layer count and depth $L(X, c)$, $d(X, c)$ are printed as Part II's $L(X, c)$ and $D(X, c)$, which denote the same numbers, so that Parts II and III share one notation. Its unused macros for a phrase, for support and for image are dropped, and its `\RaggedRight` table columns use `\raggedright`. No normalization is changed. The notation table is section 29.3.
- *The reference [ProveIt].* Its three uses are replaced by cross-references: to research question 1 in section 30, and to theorems 4.2 and 16.2 after theorem 31.3; the entry itself is quoted above.
- *Bibliography.* The manuscript's entries for Chiozini, Csernák and Soukup [1] and for Lagoutte and Tavenas [5] are merged with those of Parts I and II (the manuscript cites the first under the arXiv record's shorter title and the second as arXiv:1309.0422, 2013). Its entries [6, 7, 8] are added.
- *Shipped names.* Paths and commands in section 40 and Appendix D are rewritten to the shipped names, with an explicit output directory; the notes there quote the delivered text. The rewritten text promises no file that is not shipped.
- *Notes.* Five dated notes are added inside Part III: on the scope of the phrase “hence for $h = 1$ ” (section 30), on the relation of proposition 39.1 to Part II's theorem 22.4, on the status of formalization in ProveIt, and two on shipped names. In Part II, notes pointing to Part III follow

research question 1, the remark after theorem 20.3 and the proof of theorem 22.4, and a note in section 14.1 corrects its count of research questions; the title page and the abstract gain a paragraph each.

Status (manuscript 05). Part III is an AI-assisted, unrefereed research manuscript. Its arguments are conventional mathematical proofs, not Lean or Rocq formalizations, and nothing in it is formalized in ProveIt. The contribution it proposes as new is the filter-profile classification, that is, the identification of the finite-output invariant with local filter capacities, and its consequences: the capacity-attainment criterion, the three-color separation and the ultrafilter threshold laws. It makes no separate priority claim for the elementary supersequence/feedback-set reduction (lemma 33.3), the random-order bound (lemma 37.1) or the balanced-clique arc extremum (theorem 37.2); its literature search was targeted, not exhaustive, and does not certify priority. The classification concerns a finite nonempty derived set: it does not settle finite heights $h \geq 2$, infinite derived sets, transfinite play or expected-cost optimization, it makes no claim about arbitrary ordinal-length play or the point-separating invariant of [1], and it does not claim Part II’s Research question 2 on minimum synchronization families (its ultrafilter thresholds count local components while permitting every label at finitely many isolated exceptions, a specific input model). It does not refute Part II’s upper bound, which remains valid. Free ultrafilters are used in ZFC; their existence is not certified by the finite computations, which never construct or simulate an infinite ultrafilter. The only imported complexity-theoretic fact is the NP-completeness of directed feedback vertex set [8]; the reductions to layer count and to rounded depth are proved, for an explicit finite profile encoding only.

29.2 The manuscript’s title block and abstract

Filter Profiles and Open-Query Complexity. Exact finite-derivative classification, ultrafilter thresholds, and a three-color separation. Research report prepared for Vladimir Reshetnikov, September 29, 2026.

An open-query strategy identifies the label of a hidden point by adaptively asking membership in open sets. We continue the finite-output open-query program in ProveIt, concentrating on its question about the local hypotheses needed for sharp Cantor–Bendixson bounds. For every Hausdorff space with a finite nonempty derived set, we describe the complete word profile of every finite coloring by a finite directed graph. The minimum layer count is the number of used colors plus the graph’s minimum directed feedback vertex number; the minimum query depth is its binary logarithm rounded upward.

The worst-case invariant depends only on finite partition capacities of the neighborhood filters at the nonisolated points. A precise bin-partition criterion characterizes attainment of the rank-one layer bound. For three colors it yields a complete numerical classification. In particular, three one-point free-ultrafilter spaces require at most two questions, whereas three convergent sequences can require three, although their Cantor–Bendixson derivative cardinalities agree. For sums of arbitrary numbers of ultrafilter components, an exact directed extremal calculation gives every alphabet-dependent threshold, with a characterization of the graphs attaining the minimum arc count. Full proofs, constructive strategy certificates, finite exhaustive checks, and a further research program are included.

Status and scope. This is an AI-assisted, unrefereed research manuscript. The arguments below are conventional mathematical proofs, not Lean or Rocq formalizations. The new contribution proposed here is the filter-profile classification and its consequences for the repository’s finite-output question. We make no separate priority claim for the elementary supersequence/feedback-

set reduction or the extremal graph ingredients, and do not claim a solution of all finite-height or transfinite open-query problems. Free ultrafilters are used in ZFC; their existence is not certified by the finite computations.

29.3 Notation of Part III

Part III fixes a Hausdorff space X with a finite nonempty derived set $X' = \{p_1, \dots, p_t\}$ and a coloring $c : X \rightarrow [q] = \{1, \dots, q\}$, Part II's alphabet Σ of q letters. Several letters are used differently in Part III and in Parts I–II, and two carry two meanings inside Part III. Where a false reading is tempting, the table prints it beside the true one.

Symbol	Part III	Parts I–II
sm_q	$\text{sm}_q(X) = \sup_c D(X, c)$ over $c : X \rightarrow [q]$; colorings need not use every label. The manuscript's macro, printed identically.	Part II's $\text{sm}_q(X)$, the same invariant; Part I's $\text{sm}(X)$ is an ordinal, equal to $\text{sm}_2(X)$ when either is finite.
L, D	Least number of layers and least worst-case depth, printed L and d in the manuscript. In proposition 39.1, $D(G')$ is the depth of a space realizing the profile graph G' .	Part II's $L(X, c)$, $D(X, c)$; Part I's $L(A, X)$.
h , “rank one”	Part III settles the class $0 < X' < \infty$, where $h = 1$ and the rank-one bound is $(q-1) \cdot 1 + s = q + s - 1$. <i>False</i> : “Part III settles every space with $h = 1$ ”; $[0, \omega^2]$ has $h = 1$ and an infinite derived set (research question 14).	Last derivative rank.
t	$t = X' $, the number of nonisolated points.	Part II: the number of towers; $\bigsqcup_{i=1}^t K_2$ has $ X' = t$, so the meanings agree.
D_i, D, N_i, Z	$X = Z \sqcup N_1 \sqcup \dots \sqcup N_t$ with clopen $N_i \ni p_i$; $D_i = N_i \setminus \{p_i\}$ its isolated part; D an infinite set carrying a filter. <i>Not</i> the depth D.	Part I: $D_j = X^{(j)} \setminus X^{(j+1)}$ and the oriented lengths N_0, N_1 ; Part II: N an input length.
$\mathcal{F}, \mathcal{I}_{\mathcal{F}}, \mathcal{U}, S(\mathcal{F})$	A free filter, its dual ideal, a free ultrafilter, and the one-point space on $D \cup \{p\}$ whose neighborhoods of p are $\{p\} \cup A$, $A \in \mathcal{F}$. For the cofinite filter on a countable D , $S(\mathcal{F}) \cong [0, \omega] = \omega + 1$.	Part I: $S = [0, \omega]$; Part II: $S \subseteq \Sigma$ a set of first letters.
$b_i, b(\mathcal{F}), \mathbf{b}$	The partition capacity at p_i , in $\mathbb{N}_{>0} \cup \{\infty\}$, and the list of capacities. <i>Not</i> Part II's coloring $b_{a,m}$.	$b_{a,m}$ a coloring of K_m (section 20).
$B_i, B, \mathcal{B}$	The sets of colors whose cells are positive at p_i ; $\mathcal{B} = \mathcal{P}(D)/\mathcal{I}_{\mathcal{F}}$ a Boolean algebra.	—
s, S	In (34.3) and theorem 34.5: the number of colors on X' and their set, as in Part II's theorem 20.1. But in $T(q, r)$ and theorem 37.2, $q = ar + s$ with $0 \leq s < r$: s is a <i>remainder</i> . <i>False</i> : reading the s of $T(q, r)$ as a number of colors.	Part II: $s = S $, the number of colors on $X^{(h)}$.
a, a_i	$a_i = c(p_i)$, the center color at p_i ; but in $q = ar + s$, a is a quotient. <i>False</i> : the a of $T(q, r)$ is a color.	Part II: $a \in \Sigma$ a letter.
q_c, Σ_c	The number and the set of colors c uses.	—

Symbol	Part III	Parts I–II
$G_c, \tau, \alpha_{ac}, \ell$	The profile digraph on Σ_c ; the minimum directed feedback vertex number; the largest acyclic induced subgraph; the shortest word containing every vertex and arc as subsequences.	Part I: G_i open sets.
r, r_*, r_q, T	A bound on α_{ac} (the number of cliques of an extremal graph); $r_*(q, t)$ and r_q of (30.5) and corollary 38.2; $T(q, r)$ the least number of arcs with $\alpha_{ac} \leq r$.	Part II: r the number of words or chains.
k, P	$k = \lceil \log_2 q \rceil$ in corollary 38.2; a feedback budget in proposition 39.1 and the size of a feedback set in lemma 33.3; P a power of two in proposition 39.1.	Part I: K_k ; Part II: P a poset.
X_t	The sum of t one-point free-ultrafilter spaces (section 38).	Part I: $X_{h,m} = [0, \omega^h \cdot m]$.

29.4 Parts I and II inside Part III

The following checks were made when Part III was added; they are consequences of the three parts' proofs, not new theorems.

1. *Two labels.* For $q = 2$ every capacity list is realizable with both arcs between the two labels as soon as $t \geq 2$, and with at most one arc when $t = 1$; so theorem 34.4 gives $\text{sm}_2(X) = 1$ for $t = 1$ and 2 for $t \geq 2$, whatever the capacities. This is Part I's theorem 6.2 at $h = 1$ ($\lceil \log_2 2 \rceil = 1$, $\lceil \log_2 3 \rceil = 2$) for spaces with a finite nonempty derived set. The manuscript states it for ultrafilter sums in section 38.
2. *The upper bound.* When all q labels are used, (34.3) reads $L(X, c) \leq q + s - 1$, which is Part II's (20.1) at $h = 1$. Theorem 34.5 decides for every such space and every s whether it is attained.
3. *Towers.* Part II's K_2 is the convergent sequence $[0, \omega]$, which is compact and first countable. For t copies, corollary 36.2 gives $\lceil \log_2(q - 1 + \min(q, t)) \rceil$, which is (20.2) at $m = 2$. Part III extends that value to every space with a finite nonempty derived set whose nonisolated points are first countable or locally compact, and shows by theorem 38.1 that it fails for sums of free-ultrafilter spaces: at $q = t = 3$ the tower value is $\lceil \log_2 5 \rceil = 3$, the ultrafilter value 2 (35.1).
4. *No contradiction.* Part II's remark after theorem 20.3 states that theorem 20.1 is a uniform bound over a class and does not assert that every space of the given height has the same invariant; the README of this report says the same. The separation (35.1) shows that for $q = 3$ they do not. No statement of Part I or Part II is refuted, and none is retracted.
5. *Words.* By theorem 33.1, the words that accumulate at a point of a finite derived set are the two-letter words ab of the arcs of G_c , and $\ell(G)$ of lemma 33.3 is the shortest common supersequence of these words and of the single letters: the two-letter case of Part II's supersequence representation (theorem 17.1).
6. *Complexity.* Part II's theorem 22.4 (fixed $q \geq 3$, posets and tower lists) and proposition 39.1 (unbounded q , profile graphs) concern different input models; see the note after proposition 39.1.
7. *Independent checks.* A script written at intake and rerun at this write (not shipped) confirmed $\ell(G) = |V| + \tau(G)$ on all 4,165 loopless digraphs on one to four labelled vertices by a breadth-first search over words; $T(q, r)$ and the symmetry of every equality graph for $q \leq 4$; on 172

capacity lists over $\{1, 2, 3, \infty\}$ for $q = 2, 3, 4$, the ternary formula (30.3), the bin criterion against brute-force star realization, and, for lists of infinite capacities, (20.2) at $m = 2$; the layer formula of theorem 38.1 for $q \leq 4$; and all 63 rows of `data/03-filter-profiles-query_jump_thresholds.csv` against a direct scan.

30 The repository question and the main conclusions

30.1 A specific source and a specific question

The starting point is ProveIt’s report *Open-query games on ordinal spaces*, particularly Part II, section 27, Research question 1 (Parts I and II above). The inspected revision is

2d49198382cc068f46747d1b3d9067d47cee4f24.

The report asks which local conditions make the finite-output upper bound $(q - 1)h + s$ attainable, and whether accumulating word profiles determine the invariant. Here h is the index of the last nonempty Cantor–Bendixson derivative and s is the number of colors on that last derivative. The report explicitly distinguishes this question for $q \geq 3$ from its binary classification.

We solve the profile problem for all Hausdorff spaces whose derived set is finite and nonempty, hence for $h = 1$. We also determine exactly when the bound is attained in that class. This is a resolution of a substantive special case of the repository question and a negative answer to the possible stronger assertion that rank and last-derivative size alone always suffice for finite-output games. It is not a refutation of the report’s upper bound: that bound remains valid.

[Added 29 September 2026, batch 56] **Scope: “hence for $h = 1$ ”.** The class settled here, Hausdorff spaces whose derived set X' is finite and nonempty, lies inside the class $h = 1$ but is a proper subclass of it: the ordinal space $[0, \omega^2)$ has the infinite discrete derived set $\{\omega \cdot n : n \geq 1\}$ and empty second derivative, so $h = 1$ there too. Such spaces are the subject of research question 14. Part II’s research question 1 is therefore answered for finite nonempty derived sets, not for all of $h = 1$ (see the note after that question).

The broader cut-and-choose setting comes from Chiozini, Csernák and Soukup [1]. Our invariant is finite-output and finite-depth; no claim below concerns arbitrary ordinal-length play or the point-separating invariant of that paper.

30.2 Overview of the results

Fix a nonempty Hausdorff space X with

$$X' = \{p_1, \dots, p_t\}, \quad 1 \leq t < \infty.$$

For a coloring c , write $q_c = |c[X]|$. There is a canonically determined local profile at each p_i , after choosing disjoint clopen neighborhoods. It consists of the color $a_i = c(p_i)$ and the colors B_i whose cells accumulate at p_i . Form a loopless digraph G_c on the used colors with arcs

$$a_i \longrightarrow b \quad (b \in B_i, b \neq a_i). \tag{30.1}$$

If $\tau(G)$ is the minimum number of vertices whose deletion destroys all directed cycles, our first exact formula is

$$\boxed{L(X, c) = q_c + \tau(G_c), \quad D(X, c) = \lceil \log_2(q_c + \tau(G_c)) \rceil.} \tag{30.2}$$

The full admissible-word profile, not merely these two minima, is described in theorem 33.1.

A second invariant $b_i \in \mathbb{N}_{>0} \cup \{\infty\}$ measures how many disjoint color cells can simultaneously accumulate at p_i . These capacities determine the worst case over all q -colorings. For prescribed s colors on X' , the maximum possible layer count equals $q + s - 1$ if and only if the indices $1, \dots, t$ can be partitioned into s nonempty classes, each with total capacity at least $s - 1$ (theorem 34.5).

For three colors, put

$$u = |\{i : b_i = 1\}|, \quad v = |\{i : b_i \geq 2\}|.$$

Then the entire worst-case query invariant has the simple form

$$\text{sm}_3(X) = \begin{cases} 3, & v + \lfloor u/2 \rfloor \geq 3, \\ 2, & v + \lfloor u/2 \rfloor < 3. \end{cases} \quad (30.3)$$

Here $b_i = 1$ means exactly that the local neighborhood filter is an ultrafilter. Thus even the first nontrivial finite-output case detects local information absent from the derivative cardinalities.

Finally, for a sum X_t of t free-ultrafilter components, define

$$T(q, r) = ra(a - 1) + 2as, \quad q = ar + s, \quad 0 \leq s < r, \quad (30.4)$$

and

$$r_*(q, t) = \min\{r \in \{1, \dots, q\} : T(q, r) \leq t\}.$$

The exact worst-case formulas are

$$\max_{c: X_t \rightarrow [q]} L(X_t, c) = 2q - r_*(q, t), \quad \text{sm}_q(X_t) = \lceil \log_2(2q - r_*(q, t)) \rceil. \quad (30.5)$$

All displayed formulas are proved below, including their boundary cases.

31 Open questions, layers, and admissible words

Definition 31.1. Let $c : X \rightarrow \Sigma$ be a coloring into a finite set. An open-query strategy is a finite binary decision tree whose internal nodes are labeled by open subsets of X . A point x follows the yes branch at a node labeled U precisely when $x \in U$. A strategy is correct if the label at the reached leaf is $c(x)$ for every $x \in X$. The minimum worst-case depth is $D(X, c)$. We set

$$\text{sm}_q(X) = \sup\{D(X, c) : c : X \rightarrow \{1, \dots, q\}\}.$$

Colorings need not be continuous. Queries have unit cost, and the objective is worst-case rather than expected depth.

Definition 31.2. A layer representation of c is a finite ordered partition $X = E_1 \sqcup \dots \sqcup E_m$ into nonempty monochromatic sets such that every suffix

$$U_j = \bigcup_{k \geq j} E_k$$

is open. Its word is the sequence of colors of its layers. The minimum number of layers is $L(X, c)$. A word is called admissible if it supports such a representation with empty positions permitted.

Allowing empty positions makes the profile upward closed under taking supersequences. It does not change the minimum length: delete empty positions, and merge consecutive positions of the same color when useful.

Theorem 31.3 (Ordered normal form). *Whenever a finite layer representation exists,*

$$D(X, c) = \lceil \log_2 L(X, c) \rceil.$$

Moreover, the least possible number of nonempty leaves of a correct strategy equals $L(X, c)$.

Proof. Order the leaves of a decision tree recursively by putting every no-subtree before the corresponding yes-subtree. We prove that every suffix of this leaf order has an open union of its point sets. At the root let the open question be U . A suffix beginning inside the yes-subtree is relatively open in U , by induction, and is therefore open in X . A suffix beginning in the no-subtree has the form

$$U \cup ((X \setminus U) \cap V) = U \cup V$$

for some open $V \subseteq X$, again by induction using the relative topology on $X \setminus U$. Thus the ordered nonempty leaves give a layer representation. A depth- d tree has at most 2^d leaves, so $L(X, c) \leq 2^d$.

Conversely, given m ordered layers, ask membership in a suffix whose boundary divides the list into two parts of sizes differing by at most one. Continue recursively on the appropriate interval of positions. Every question is one of the original globally open suffixes. Balanced binary search has depth $\lceil \log_2 m \rceil$ and exactly m leaves. Taking the minimum proves both assertions. $\square$

This is the normal-form mechanism used in the repository report (Part I, theorem 4.2; Part II, theorem 16.2); it is reproduced to make the present proof independent of any formalization or unverified implementation in that repository.

32 Every finite derived set has a filter model

32.1 The topological decomposition

Lemma 32.1. *If X is Hausdorff and $X' = \{p_1, \dots, p_t\}$ is finite, there are pairwise disjoint clopen neighborhoods N_i of the p_i such that*

$$X = Z \sqcup N_1 \sqcup \dots \sqcup N_t,$$

where Z is discrete and clopen and $N'_i = \{p_i\}$.

Proof. For each pair of distinct p_i, p_j , choose disjoint open neighborhoods. Intersect the finitely many neighborhoods chosen at each point to obtain pairwise disjoint open sets N_i , with no other p_j in N_i . The complement of N_i is open: every isolated point in it is an open singleton, and each $p_j, j \neq i$, has the disjoint neighborhood N_j . Hence N_i is clopen. Their finite union is clopen; its complement Z contains no nonisolated point. Finally, an open neighborhood of a nonisolated point cannot make that point isolated, so $N'_i = \{p_i\}$. $\square$

Write $D_i = N_i \setminus \{p_i\}$. Every point of D_i is isolated, and D_i is infinite. Define

$$\mathcal{F}_i = \{A \subseteq D_i : \{p_i\} \cup A \text{ is open in } N_i\}.$$

This is a proper filter containing all cofinite subsets of D_i . We use “free filter” in precisely this sense. Conversely, for an infinite set D and such a filter $\mathcal{F}$, let $S(\mathcal{F}) = D \cup \{p\}$, with all subsets of D open and the open sets containing p exactly $\{p\} \cup A$, $A \in \mathcal{F}$. This is a zero-dimensional Hausdorff space with derivative $\{p\}$. Indeed, cofinite neighborhoods separate p from isolated points, and every basic neighborhood of p is clopen.

32.2 Finite color quotients of a filter

The dual ideal of $\mathcal{F}$ is

$$\mathcal{I}_{\mathcal{F}} = \{A \subseteq D : D \setminus A \in \mathcal{F}\}.$$

A set is $\mathcal{F}$ -positive if it does not belong to $\mathcal{I}_{\mathcal{F}}$. Equivalently, it meets every member of $\mathcal{F}$. For subsets of the isolated part, this says exactly that p belongs to their closure. The ideal contains all finite subsets of D .

For a finite coloring of D , let C_b be its color cells and put

$$B = \{b : C_b \text{ is } \mathcal{F}\text{-positive}\}. \quad (32.1)$$

The set B is nonempty: a finite union of members of a proper ideal cannot be all of D .

Lemma 32.2 (Positive-cell criterion). *For every set J of colors,*

$$\bigcup_{b \in J} C_b \in \mathcal{F} \iff B \subseteq J.$$

Proof. If $B \subseteq J$, the complement is a finite union of nonpositive cells and belongs to $\mathcal{I}_{\mathcal{F}}$. Conversely, if J omits a positive cell C_b , its union cannot belong to $\mathcal{F}$, since it is disjoint from a set meeting every filter member. $\square$

This elementary finite quotient is the key distinction between cofinite and ultrafilter convergence. For a free ultrafilter, exactly one color cell is positive. For the cofinite filter on an infinite set, every infinite color cell is positive.

33 The complete profile is a directed graph

Let $c : X \rightarrow \Sigma$ use exactly the q_c colors in $\Sigma_c = c[X]$. In the clopen decomposition of lemma 32.1, set

$$a_i = c(p_i), \quad B_i = \{b : c^{-1}(b) \cap D_i \text{ is } \mathcal{F}_i\text{-positive}\}.$$

Let G_c be the simple loopless digraph on Σ_c whose arcs are $a_i \rightarrow b$ for $b \in B_i \setminus \{a_i\}$. Repeated arcs are recorded once. An arc $a \rightarrow b$ represents the two-letter subsequence requirement ab , not a requirement of adjacency in a word.

Theorem 33.1 (Exact graph profile). *A finite word w on Σ_c is admissible for (X, c) if and only if it contains every color of Σ_c and, for every arc $a \rightarrow b$ of G_c , it contains ab as a subsequence.*

Proof. Suppose first that a layer representation has word w , and p_i is in position k . The suffix from k is an open neighborhood of p_i . Its intersection with D_i belongs to $\mathcal{F}_i$, so it meets every positive color cell. Thus each $b \in B_i$ occurs at a position $j \geq k$. If $b \neq a_i$, then $j > k$, which gives $a_i b$ as a subsequence. All used colors must occur somewhere.

Conversely, suppose the word has all these properties. Assign p_i to the *first* occurrence of a_i in w , and assign every isolated point of color b , including points in Z , to the *last* occurrence of b . For each $b \in B_i$, the assigned position of its isolated points is at least the assigned position of p_i . This is immediate if $b = a_i$; otherwise it follows from the subsequence condition.

Consider any suffix of the resulting assignment. If it does not contain p_i , its intersection with N_i is a subset of the discrete part, so is open. If it contains p_i , it contains all positive color cells of D_i ; lemma 32.2 says that its isolated part belongs to $\mathcal{F}_i$, so it is again open. The intersection with Z is open. Since the decomposition is clopen, the suffix is globally open. The assignment is color-respecting and proves admissibility. $\square$

Remark 33.2 (Why arbitrary words do not occur here). At a point of this first nontrivial derivative level, a finite coloring creates only two-letter constraints. Higher Cantor–Bendixson levels can create longer word constraints, which are not encoded by a single digraph on the colors. The reduction is complete for a finite derived set, not a purported classification of all scattered spaces.

33.1 Two-letter constraints and directed feedback sets

For a loopless digraph G on a finite vertex set V , let $\alpha_{\text{ac}}(G)$ be the largest size of a subset inducing an acyclic digraph. Then $\tau(G) = |V| - \alpha_{\text{ac}}(G)$. Let $\ell(G)$ be the shortest length of a word containing every vertex and every directed edge as a subsequence.

Lemma 33.3. *For every nonempty finite loopless digraph,*

$$\ell(G) = |V| + \tau(G).$$

An optimal word can be constructed from any minimum feedback vertex set and a topological ordering of its complement.

Proof. Let w satisfy the requirements, and let R be the set of vertices appearing at least twice. Every vertex outside R appears once. Their order in w respects all arcs among them, so $G[V \setminus R]$ is acyclic. Consequently R is a feedback vertex set and

$$|w| \geq |V| + |R| \geq |V| + \tau(G).$$

Conversely, write a minimum feedback set as $R = (r_1, \dots, r_k)$, and let T be a topological ordering of $G[V \setminus R]$. The word

$$r_1 \cdots r_k T r_k \cdots r_1 \tag{33.1}$$

contains every required pair. Arcs inside T are respected by its topological order. Arcs from R to T use the first block, and arcs from T to R use the last block. Every pair of distinct vertices of R occurs in both orders across its two blocks. The word has length $|V| + k$. A minimum feedback set has a nonempty complement, so no degenerate empty middle block is needed. $\square$

The shortest-common-supersequence viewpoint has a substantial independent literature; see, for example, Lagoutte and Tavenas [5]. We supply lemma 33.3 as an elementary bridge, not as a claim to have introduced either supersequences or feedback vertex sets.

Theorem 33.4 (Exact coloring complexity). *For every finite coloring of a Hausdorff space with a finite nonempty derived set, equation (30.2) holds. Both minima are finite, and equation (33.1) followed by balanced suffix search constructs an optimal strategy from a minimum feedback vertex set.*

Proof. Combine theorems 31.3 and 33.1 and lemma 33.3. $\square$

Corollary 33.5 (Positive label weights). *Assign each used color a positive weight w_b , and charge this weight once for each occurrence of the color in a layer word. The minimum weighted layer cost is*

$$\sum_{b \in \Sigma_c} w_b + \min_{R: G_c - R \text{ acyclic}} \sum_{b \in R} w_b.$$

Proof. The repeated vertices of any admissible word form a feedback set, giving the lower bound. The doubled-feedback-set construction attains it for a minimum-weight feedback set. $\square$

This is a statement about weighted layer occurrences. It is not a formula for expected query cost or for arbitrary unequal costs of open questions.

34 Local partition capacity and exact attainability

34.1 A local Boolean invariant

Definition 34.1. The partition capacity $b(\mathcal{F})$ of a free filter is the supremum of the numbers of pairwise disjoint $\mathcal{F}$ -positive subsets of its underlying set. Its value lies in $\mathbb{N}_{>0} \cup \{\infty\}$. Here ∞ means that such families exist at every finite size; no particular infinite family is asserted by the notation.

Disjoint positive sets can be enlarged to a partition by adding the unassigned remainder to one of them. Thus $b(\mathcal{F})$ is also the maximum possible number of positive color cells, or the statement that this number is unbounded over finite colorings.

Proposition 34.2. *The following statements hold.*

1. *Restricting a filter to one of its members does not change its partition capacity.*
2. *$b(\mathcal{F}) = 1$ if and only if $\mathcal{F}$ is an ultrafilter.*
3. *If $b(\mathcal{F}) = b < \infty$, then $\mathcal{F}$ is the intersection of exactly b distinct free ultrafilters, uniquely determined as an unordered family. Conversely, such an intersection has capacity b .*

Proof. For the first assertion, if $A \in \mathcal{F}$, its complement belongs to the dual ideal. Intersecting a positive set with A preserves positivity. This transfers disjoint positive families in both directions.

For the remaining assertions, consider the Boolean algebra $\mathcal{B} = \mathcal{P}(D)/\mathcal{I}_{\mathcal{F}}$. Positive sets represent its nonzero elements. Suppose the largest size of a disjoint nonzero family is the finite number b . A family of maximum size covers the unit: otherwise its complement could be appended. Each member is an atom: otherwise splitting it would give a larger family. Every Boolean-algebra element is a union of some of these atoms, because intersecting it with an atom gives either that atom or zero. Thus $\mathcal{B}$ is the Boolean algebra of subsets of a b -element set. Its b atom evaluations lift to distinct ultrafilters on D . An element of $\mathcal{P}(D)$ lies in all of them exactly when its quotient class is the unit, which is equivalent to membership in $\mathcal{F}$.

Conversely, finitely many distinct ultrafilters can be separated by finitely many sets. The Boolean algebra generated by these sets has a cell belonging to each ultrafilter and to no other. It follows that the map to the b simultaneous truth values is onto $\{0, 1\}^b$, with the intersection filter as the inverse image of the all-one vector. Its dual quotient therefore has exactly b atoms. The case $b = 1$ gives the second assertion and uniqueness follows from the quotient atoms. $\square$

Consequently $b_i = b(\mathcal{F}_i)$ is independent of the clopen decomposition: two chosen neighborhoods have a common neighborhood obtained by intersection, and restriction does not alter the capacity.

34.2 Realizable stars and a complete finite optimization

Lemma 34.3. *At a point with capacity b_i , any prescribed center color a_i and any prescribed set A_i of at most $\min(b_i, q - 1)$ other colors can be realized as exactly its outgoing neighbors in the profile graph. All q colors may be made to occur globally without changing any prescribed local graph.*

Proof. If A_i is nonempty, partition D_i into $|A_i|$ positive sets and give them the colors in A_i . If A_i is empty, color all of D_i by a_i . Give p_i its prescribed center color. Any missing labels can be inserted at finitely many isolated points in one component, avoiding one reserved representative of every label already used. Finite sets belong to the dual ideal, and removing finitely many points from a positive set preserves positivity. Therefore these changes introduce no positive color cell and destroy none. $\square$

For a list of capacities $\mathbf{b} = (b_1, \dots, b_t)$, let $\mathcal{G}_q(\mathbf{b})$ consist of all loopless digraphs on $\{1, \dots, q\}$ expressible as the union of stars

$$\{a_i \rightarrow x : x \in A_i\}, \quad |A_i| \leq \min(b_i, q - 1), \quad a_i \notin A_i.$$

Centers may repeat and stars may be empty.

Theorem 34.4 (Capacity classification). *For every Hausdorff X with finite nonempty derived set and local capacity list $\mathbf{b}$,*

$$\max_{c: X \rightarrow [q]} L(X, c) = q + \max_{G \in \mathcal{G}_q(\mathbf{b})} \tau(G), \quad (34.1)$$

$$\text{sm}_q(X) = \left\lceil \log_2 \left(q + \max_{G \in \mathcal{G}_q(\mathbf{b})} \tau(G) \right) \right\rceil. \quad (34.2)$$

In particular, $\text{sm}_q(X)$ depends only on the truncated capacities $\min(b_i, q - 1)$, not on the underlying infinite cardinalities or the particular ultrafilters.

Proof. Every coloring using all q labels gives a graph in the displayed class by the definition of capacity. Conversely, every graph in the class is realized by lemma 34.3, using all q labels. A coloring using fewer labels can be enlarged to use all labels by finite changes at isolated points, preserving its graph and adding isolated graph vertices. By theorem 33.4, this cannot lower its layer count. The maxima are over finite graph classes, and the logarithm is monotone, so theorem 33.4 gives both equalities. $\square$

34.3 Exactly when the rank-one bound is sharp

Let $S = \{c(p_i) : 1 \leq i \leq t\}$ and $s = |S|$. All arcs of G_c originate in S , so every directed cycle lies in S . Consequently

$$\tau(G_c) = s - \alpha_{\text{ac}}(G_c[S]), \quad L(X, c) = q_c + s - \alpha_{\text{ac}}(G_c[S]) \leq q_c + s - 1. \quad (34.3)$$

This recovers the repository bound $(q - 1)h + s$ at $h = 1$, when all q colors occur.

Theorem 34.5 (Sharp local attainment criterion). *Fix $q \geq 1$ and $1 \leq s \leq \min(q, t)$. There is a coloring using all q labels, exactly s labels on X' , and having $L(X, c) = q + s - 1$, if and only if there is a partition*

$$\{1, \dots, t\} = I_1 \sqcup \dots \sqcup I_s, \quad I_j \neq \emptyset, \quad \sum_{i \in I_j} b_i \geq s - 1 \quad (1 \leq j \leq s). \quad (34.4)$$

An infinite capacity is interpreted as exceeding every finite threshold.

Proof. By equation (34.3), equality holds precisely when $\alpha_{ac}(G_c[S]) = 1$. A loopless digraph on s vertices has acyclic number one exactly when every pair of distinct vertices is joined by both arcs: if one arc is missing, that pair is acyclic; conversely every pair in a complete bidirected graph is a directed two-cycle.

Group the nonisolated points according to their center colors. In a group I_j , the outgoing stars must cover all $s - 1$ other center colors. Each star covers at most b_i of them, so equation (34.4) is necessary.

Conversely, under equation (34.4), assign center color j to every p_i with $i \in I_j$. Distribute the $s - 1$ other center colors among those indices, assigning at most b_i distinct targets to index i . This is possible because the total capacity is sufficient; some indices may receive no targets. Realize these stars by lemma 34.3. The induced graph on the center colors is complete bidirected. Add any missing labels by finite isolated exceptions. Then equation (34.3) gives exactly $q + s - 1$ layers. $\square$

This is an exact criterion for *layer sharpness*. Because a ceiling logarithm can identify different layer counts, failure of the criterion does not automatically imply a smaller worst-case query depth. The next section isolates a case where the depth really changes.

35 Three colors: a complete classification and a separation

Theorem 35.1 (All finite-derived-set spaces, three colors). *Let X be Hausdorff with a finite nonempty derived set. Put*

$$u = |\{i : b_i = 1\}|, \quad v = |\{i : b_i \geq 2\}|.$$

Then equation (30.3) holds.

Proof. Every such space is infinite, so some coloring uses all three labels; at least two queries are then necessary. By theorem 33.4, at most five layers, hence three queries, suffice. Three queries are necessary for a coloring precisely when it needs five layers, which is equivalent to its graph being complete bidirected on three colors.

Apply theorem 34.5 with $q = s = 3$. We need three nonempty groups, each of total capacity at least two. Every capacity at least two can supply one such group by itself; every two capacities equal to one can supply one group. Thus three groups exist if $v + \lfloor u/2 \rfloor \geq 3$. Conversely, in any three satisfactory groups, at most v groups contain a capacity at least two. Each remaining group consumes at least two of the u unit capacities. Hence the same inequality is necessary. Extra indices can be added to any group. $\square$

35.1 Two spaces with the same derivative cardinalities

Choose a free ultrafilter $\mathcal{U}$ on $\mathbb{N}$, whose existence is a standard consequence of ZFC, and let

$$X = S(\mathcal{U}) \sqcup S(\mathcal{U}) \sqcup S(\mathcal{U}).$$

Its capacity list is $(1, 1, 1)$. On the other hand, let

$$Y = (\omega + 1) \sqcup (\omega + 1) \sqcup (\omega + 1)$$

with each summand a convergent sequence and its limit. Its local filters are cofinite and its capacity list is (∞, ∞, ∞) . Both spaces are countably infinite and zero-dimensional Hausdorff, and

$$|X'| = |Y'| = 3, \quad X'' = Y'' = \emptyset.$$

Nevertheless theorem 35.1 gives

$$\boxed{\text{sm}_3(X) = 2 \quad \text{and} \quad \text{sm}_3(Y) = 3.} \quad (35.1)$$

Thus rank, cardinality, and the cardinality of the last derivative do not determine this invariant, even in this very small derivative-height class. The example changes local topology, not the number of nonisolated points.

For an explicit hard coloring of Y , color the three limit points 0, 1, 2, respectively. In the component with limit color a , partition the isolated sequence into two infinite sets with the two other colors. The resulting profile contains all six directed pairs on three colors. It requires five layers; 01210 is an optimal word.

For X , each component contributes at most one arc, so a coloring has at most three distinct arcs. It cannot contain all six directed pairs. Therefore four layers always suffice. The coloring with component profiles 01, 12, 20 gives the cycle $0 \rightarrow 1 \rightarrow 2 \rightarrow 0$, and the optimal word 0120 has four layers. Writing the nonisolated points as p_0, p_1, p_2 and their discrete parts as D_0, D_1, D_2 , those layers can be chosen as

$$\{p_0\}, \quad D_0 \sqcup \{p_1\}, \quad D_1 \sqcup \{p_2\}, \quad D_2.$$

Balanced suffix search uses two questions. The supplied certificate file contains both this tree and a five-layer tree for the convergent example.

35.2 Why the counterexample is not locally compact

Proposition 35.2. *A one-point free-ultrafilter space on $\mathbb{N}$ is not locally compact at its nonisolated point and admits no injective sequence converging to it.*

Proof. Let $A \in \mathcal{U}$. Split A into two infinite parts. Exactly one part, say B , belongs to $\mathcal{U}$. The cover of $\{p\} \cup A$ consisting of $\{p\} \cup B$ and the singleton sets from $A \setminus B$ has no finite subcover. Basic neighborhoods are closed, so a compact neighborhood containing one would make that basic neighborhood compact, a contradiction.

If an injective sequence converged to p , split its range into the even- and odd-indexed subsequences. At least one of these two sets is not in $\mathcal{U}$; its complement belongs to $\mathcal{U}$ and gives a neighborhood of p omitting infinitely many terms. This contradicts convergence. $\square$

36 Positive classification under ordinary local hypotheses

Proposition 36.1. *If a nonisolated point of a Hausdorff space with finite derived set is first countable, its local partition capacity is infinite. The same conclusion holds if the space is locally compact at that point.*

Proof. Pass to its one-point clopen neighborhood. In the first-countable case, choose a decreasing neighborhood base and recursively choose a new isolated point from each base member. This is possible because a neighborhood of a nonisolated point in a T_1 space is infinite. The resulting injective sequence converges to the point. Splitting it into any prescribed finite number of infinite subsequences gives disjoint positive sets.

In the locally compact case, take a compact neighborhood contained in the one-point clopen neighborhood. Within that compact subspace, every open neighborhood of the nonisolated point has finite complement: the complement is closed, compact, and discrete. Its isolated part is infinite. Partition that part into arbitrarily many infinite sets. Each accumulates at the nonisolated point in the compact subspace and hence in the ambient space. $\square$

Corollary 36.2. *Suppose X is Hausdorff, $0 < |X'| = t < \infty$, and every nonisolated point is either first countable or has a compact neighborhood. Then for every $q \geq 1$,*

$$\begin{aligned} \max_{c: X \rightarrow [q]} L(X, c) &= q - 1 + \min(q, t), \\ \text{sm}_q(X) &= \lceil \log_2(q - 1 + \min(q, t)) \rceil. \end{aligned}$$

Proof. For any coloring, equation (34.3) bounds the layers by $q - 1 + \min(q, t)$, also after adjoining unused labels. Every capacity is infinite by proposition 36.1; therefore the partition criterion is satisfied with $s = \min(q, t)$. It supplies equality. Apply theorem 31.3. $\square$

This extends the rank-one convergent-tower formula to all spaces in the stated classes. Compactness and first countability are sufficient, not necessary: the exact hypothesis for layer attainment is the capacity partition criterion.

Example 36.3 (Mixed local behavior). For three labels, capacity lists $(2, 2, 1)$ and $(2, 2, 1, 1)$ have worst-case depths two and three, respectively. In the second list the three required groups are the two single entries of capacity two and the pair of unit entries. An arbitrarily large capacity at just one point cannot be divided between several different center colors: each nonisolated point has only one color. This is why summing all capacities without the partition requirement would be incorrect.

37 An exact directed extremal calculation

To obtain a closed formula for unit capacities, we need the minimum number of arcs in a digraph on q vertices whose acyclic number is at most r . Antiparallel arcs are allowed, but loops and multiple arcs are not. The random-order method below belongs to the Caro–Wei family of arguments; stronger directed bounds appear in [6, 7]. Only the elementary one-sided version is needed here, and we prove it directly.

Lemma 37.1 (Random-order bound). *For every finite loopless digraph G ,*

$$\alpha_{\text{ac}}(G) \geq \sum_{v \in V(G)} \frac{1}{d^+(v) + 1}.$$

Proof. Choose a uniformly random ordering of the vertices. Select a vertex if it appears before all its out-neighbors. Within the selected set, every arc goes forward in the ordering, so the selected set is acyclic. A vertex of out-degree d is selected with probability $1/(d+1)$. Taking expectations proves the bound. $\square$

Theorem 37.2 (Exact arc threshold and equality). *Let $1 \leq r \leq q$, and write $q = ar + s$ with $0 \leq s < r$. The minimum number of arcs in a loopless digraph on q vertices with $\alpha_{\text{ac}}(G) \leq r$ is*

$$T(q, r) = ra(a-1) + 2as.$$

The graphs attaining this minimum under the stated acyclic-number constraint are exactly the disjoint unions of r complete bidirected graphs whose sizes differ by at most one: s have size $a+1$, and $r-s$ have size a .

Proof. For every nonnegative integer d ,

$$\frac{1}{d+1} \geq \frac{2a-d}{a(a+1)}. \quad (37.1)$$

Indeed, the difference is

$$\frac{(d-a)(d-a+1)}{(d+1)a(a+1)} \geq 0,$$

since the numerator is the product of consecutive integers. Equality holds precisely for $d = a-1$ or $d = a$.

If m is the number of arcs, summing equation (37.1) and using lemma 37.1 gives

$$r \geq \alpha_{\text{ac}}(G) \geq \sum_v \frac{1}{d^+(v)+1} \geq \frac{2aq-m}{a(a+1)}.$$

Thus

$$m \geq 2aq - ra(a+1) = ra(a-1) + 2as = T(q, r).$$

The union of balanced complete bidirected graphs described in the statement has acyclic number exactly r and has exactly that many arcs, proving sharpness.

It remains to prove the equality classification. If $m = T(q, r)$ and $\alpha_{\text{ac}}(G) \leq r$, all inequalities above are equalities. In particular, the random selected set in lemma 37.1 has expectation r and always has size at most r . It must therefore have size exactly r for every ordering.

There can be no asymmetric arc $u \rightarrow v$ with $v \not\rightarrow u$. Compare an ordering beginning u, v with the ordering beginning v, u , leaving all other vertices in the same later order. In the first ordering both u and v are selected. In the second, only v is selected. The selection status of every other vertex is unchanged, since both u, v remain before it. This changes the total by one, a contradiction. Thus all arcs occur in opposite pairs.

Regard the resulting symmetric graph as undirected. It cannot contain an induced path $u-v-w$ with no edge uw . Put this triple first in the ordering. The order u, w, v selects its two endpoints; the order v, u, w selects only its middle vertex. All later selection statuses again remain unchanged. This is impossible. An undirected graph with no induced three-vertex path is a disjoint union of cliques: otherwise a shortest path between nonadjacent vertices in a connected component begins with such an induced path.

Finally, equality in equation (37.1) requires every out-degree to be $a-1$ or a , so each clique has size a or $a+1$. Its acyclic number is its number of components, which must be r . The sum of component sizes gives the claimed multiplicities. $\square$

Remark 37.3 (What is and is not being claimed as new). The theorem is an elementary directed extremal calculation with familiar balanced-clique extremizers. Its proof is included in full so the open-query thresholds do not rely on a possibly mismatched graph citation. We do not assert historical priority for this standalone graph theorem. The source search located stronger degree-based acyclic-set bounds, not a priority certification for this exact reformulation.

38 All ultrafilter-sum thresholds

Let

$$X_t = \bigsqcup_{i=1}^t S(\mathcal{U}_i), \quad t \geq 1,$$

where the $\mathcal{U}_i$ are arbitrary free ultrafilters on infinite sets. The sets and ultrafilters need not be isomorphic. An additional discrete summand makes no difference to the following formulas.

Theorem 38.1. *For $q \geq 1$, define $r_*(q, t)$ as in equation (30.5). Then equation (30.5) holds.*

Proof. Each component has capacity one and contributes at most one arc. Conversely, any digraph on q vertices with at most t arcs can be realized: use one component for each arc, give unused components an empty star, and introduce unused labels by finite exceptions. Therefore

$$\max_c L(X_t, c) = q + \max\{\tau(G) : |V(G)| = q, |E(G)| \leq t\}.$$

Since $\tau(G) = q - \alpha_{ac}(G)$, theorem 37.2 says that the minimum possible acyclic number under this arc budget is exactly $r_*(q, t)$. This proves the layer formula. Apply theorem 31.3 for depth. The set defining r_* is nonempty because $T(q, q) = 0$. For $q = 1$, the formulas give one layer and zero questions, as required. $\square$

38.1 Exact threshold for the extra question

Corollary 38.2. *Let $q \geq 2$ and $k = \lceil \log_2 q \rceil$. Put*

$$r_q = 2q - 2^k - 1, \quad t_q = T(q, r_q).$$

Then $1 \leq r_q < q$, and

$$\text{sm}_q(X_t) = \begin{cases} k, & 1 \leq t < t_q, \\ k + 1, & t \geq t_q. \end{cases} \quad (38.1)$$

Proof. Using all q colors forces at least k questions. Since $\tau(G) \leq q - 1$, at most $2q - 1 < 2^{k+1}$ layers are needed, so at most $k + 1$ questions suffice. The larger depth occurs exactly when some graph with at most t arcs satisfies

$$q + \tau(G) > 2^k \iff \alpha_{ac}(G) \leq 2q - 2^k - 1 = r_q.$$

Apply theorem 37.2. The inequalities for r_q follow directly from $2^{k-1} < q \leq 2^k$, including the case that q is a power of two. $\square$

The threshold varies substantially with the alphabet size. If q is a power of two, $t_q = 2$. If $q = 2^{k-1} + 1$, then $r_q = 1$, so $t_q = q(q-1)$. Thus the number of components needed for one additional question ranges from two to a quadratic function of the number of labels. This variation comes from the exact rounding boundary of a binary tree, not from a change in derivative height.

Colors q	Baseline depth k	r_q	First t requiring $k+1$
2	1	1	2
3	2	1	6
4	2	3	2
5	3	1	20
6	3	3	6
7	3	5	4
8	3	7	2
9	4	1	72
10	4	3	24
11	4	5	14
12	4	7	10
13	4	9	8
14	4	11	6
15	4	13	4
16	4	15	2

The archive contains the exact table through $q = 64$ (shipped as `data/03-filter-profiles-query_jump_thresholds.csv`).

38.2 Binary and ternary checks

For two colors, $r_*(2, 1) = 2$, while $r_*(2, t) = 1$ for $t \geq 2$. Thus

$$\text{sm}_2(X_1) = 1, \quad \text{sm}_2(X_t) = 2 \quad (t \geq 2),$$

which agrees with the repository's binary distinction between a singleton and a nonsingleton last derivative.

For three colors,

$$\max_c L(X_t, c) = \begin{cases} 3, & t = 1, \\ 4, & 2 \leq t \leq 5, \\ 5, & t \geq 6. \end{cases}$$

The query depth is consequently two for $1 \leq t \leq 5$, and three for $t \geq 6$. In particular, three limit points do not suffice to force the extra query in the ultrafilter case, whereas they do in the cofinite case.

Corollary 38.3 (Layer saturation with prescribed top colors). *For unit capacities and $2 \leq s \leq \min(q, t)$, the layer value $q + s - 1$ is attainable with exactly s colors on the derived set if and only if $t \geq s(s-1)$. For $s = 1$, it is always attainable.*

Proof. Each of the s nonempty bins in equation (34.4) needs at least $s-1$ unit entries, and this condition is also sufficient. $\square$

39 Algorithms, input models, and formalization boundaries

39.1 An exact finite algorithm

For a supplied profile graph on q used labels, enumerate subsets R of its vertex set in increasing cardinality. Test whether $G - R$ is acyclic by a topological sort. The first successful subset is a minimum feedback vertex set. Construct the word equation (33.1), assign nonisolated points to first occurrences and isolated points to last occurrences as in theorem 33.1, and balance suffix queries.

A direct implementation uses

$$O(2^q(q + |E|))$$

time for the graph optimization. This bound is independent of the sizes of the infinite discrete pieces once their finite profiles have been supplied. Directed feedback vertex set has more sophisticated parameterized algorithms; [8] describes that line of work. Those algorithms are optional and are not needed for any theorem here.

Essential input qualification. We do not claim an algorithm can determine ultrafilter membership for an arbitrary presented infinite color cell. The finite algorithm takes the positive-cell sets B_i , or equivalently the graph, as input. In the explicit extremal constructions the coloring is monochromatic on each ultrafilter component apart from finitely many exceptions, so these profiles are known without an oracle. For arbitrary colorings, obtaining the profile is a separate semantic or effective problem.

39.2 Why exact optimization can still be difficult

Proposition 39.1. *For finite graph-profile input with an unbounded alphabet, deciding whether $L \leq M$ is NP-complete. Deciding whether $D \leq d$ is also NP-complete. These statements already hold for profiles realizable by finite sums of one fixed free-ultrafilter space, using colorings constant on each discrete component outside finitely many exceptions.*

Proof. A feedback vertex set and a topological ordering of its complement are polynomially checkable certificates. By theorem 33.4, the layer question with $M = q + k$ is exactly the directed feedback vertex set question $\tau(G) \leq k$. The latter is a classical NP-complete problem; see the discussion and references in [8]. Every graph profile is realizable by one ultrafilter component per arc.

For depth, it is necessary to account for rounding rather than merely quote the layer reduction. Given G on q vertices and a nontrivial bound $0 \leq k \leq q - 1$, let P be the least power of two satisfying $P \geq q + k$, and adjoin

$$z = P - q - k$$

isolated graph vertices. If G' is the resulting graph, then $\tau(G') = \tau(G)$, and

$$D(G') \leq \log_2 P \iff q + z + \tau(G) \leq P \iff \tau(G) \leq k.$$

The number z is $O(q)$, so the construction is polynomial. Unused colors are realized at finite isolated exceptions. A dummy empty-star component may be included when needed. Membership in NP follows from the same certificate, with the integer inequality $q + |R| \leq 2^d$; when d is unnecessarily large the instance is immediately positive by the universal $2q - 1$ layer bound. $\square$

Unlike the general reduced-word supersequence setting studied in [5], our profiles have only two-letter constraints. This is complexity for an explicit finite profile encoding, not for an unspecified encoding of arbitrary topological spaces or infinite sets. For every fixed q , the graph search has a fixed finite bound.

[Added 29 September 2026, batch 56] Relation to Part II’s hardness theorem. The last sentence does not conflict with Part II’s theorem 22.4, by which exact optimization is NP-complete for every fixed $q \geq 3$. The input models differ: there the instances are finite posets (disjoint chains already suffice) or succinct tower lists, whose words have unbounded length, and the difficulty is the shortest common supersequence of those words; here the input is a profile graph on the colors, which for a space with finite nonempty derived set has at most $q(q-1)$ arcs, so for fixed q only finitely many inputs exist. Proposition 39.1 is a second NP-completeness route, through directed feedback vertex set with an unbounded alphabet.

39.3 A concrete formalization route

The argument separates naturally into four interfaces. First formalize ordered words, subsequences, open suffixes, and the tree-to-layer normal form. Second formalize digraphs, topological orders, and the repeated-letter proof of $\ell(G) = q + \tau(G)$. Third formalize the finite partition lemma for filters and its graph-profile bridge. Finally add the finite-derived-set decomposition, the Boolean-capacity argument, and the random-order extremal theorem.

The graph portions admit purely finite certificates. A positive certificate consists of a feedback set, a topological order, an admissible word, and a balanced tree. A negative certificate can list a directed cycle in every induced subgraph that would contradict the proposed feedback lower bound, or use an independently verified exhaustive subset search. The filter existence and topology must remain separate mathematical obligations. The present archive contains no `.lean` or `.v` file and makes no kernel-verification claim.

[Added 29 September 2026, batch 56] Status in ProveIt. None of these interfaces has been formalized in ProveIt: no Lean or Rocq file of the repository mentions open-query games, filter capacities or directed feedback vertex sets, and the Lean library in the enclosing `SetTheory/Cardinals` directory is about large cardinals.

40 Reproducible checks and proof audit

40.1 Independent finite computations

Run the standard-library Python program, with the shipped name and an explicit output directory, as

```
python3 code/03-filter-profiles-verify.py --output <scratch>
```

It enumerates all loopless directed graphs on up to four labeled vertices, allowing antiparallel arcs. Independently, it enumerates all reduced words through length $2q - 1$, records their covered ordered pairs, and performs a supermask-minimum transform. That computes the shortest admissible word for every target graph without calling the feedback-set routine. The two answers are then compared.

Vertices q	Digraphs checked	Reduced words enumerated
1	1	1
2	4	6
3	64	93
4	4,096	4,372
Total	4,165	4,472

For every graph, the verifier also constructs a doubled-feedback-set word and checks all required subsequences directly. It checks the extremal arc threshold for every $1 \leq r \leq q \leq 4$, and verifies that every equality graph has the balanced complete-bidirected-component structure.

There are also **11,515 positive-cell word checks**: direct existence of a common permitted position for the nonisolated point is compared with the pairwise subsequence requirements. A further **98 capacity cases** compare exhaustive star-union realizations with an independent bin-assignment test for theorem 34.5. Finally, exact threshold tables and balanced extremal constructions are generated through 64 colors. The recorded run passed every assertion.

These numbers describe finite evidence, not the scope of the theorems. In particular, no finite experiment can instantiate a free ultrafilter on $\mathbb{N}$, prove the clopen decomposition for arbitrary Hausdorff spaces, or establish historical novelty.

[Added 29 September 2026, batch 56] Shipped names. The delivered command was `python3 code/verify.py`, without an output option. Here `code/verify.py` is Part I's verifier, so the command above uses the shipped name. Without `--output` the verifier writes `results/` beside `code/`, with unprefixed names. The recorded outputs are shipped as

- `data/03-filter-profiles-verification.json`,
- `data/03-filter-profiles-verification.log` (the console output, which is the same JSON),
- `data/03-filter-profiles-capacity_checks.json`,
- `data/03-filter-profiles-strategy_certificates.json`,
- `data/03-filter-profiles-extremal_thresholds.csv`,
- `data/03-filter-profiles-query_jump_thresholds.csv`.

The counts above agree with the recorded JSON, which also records ten extremal threshold cases (one for each $1 \leq r \leq q \leq 4$). A rerun on a copy (29 September 2026, Python 3.14.4, about four seconds) reproduced both CSV files byte for byte (they are CRLF on every platform, as delivered) and the JSON files and console log after CRLF $\rightarrow$ LF (the verifier writes JSON with the platform's line endings).

40.2 Dependency and status audit

Claim	Mathematical inputs	Status here
Normal form	Finite binary trees and open suffixes	Reproved; repository input
Graph profile	Clopen decomposition and finite filter quotient	Full proof
Layer/FVS formula	Repeated letters and topological ordering	Full elementary proof
Capacity criterion	Finite Boolean quotient and star realization	Full proof
Three-color separation	Capacity criterion and a free ultrafilter	Full ZFC proof
Ultrafilter thresholds	Random-order acyclic bound and integer support line	Full proof with equality cases
Complexity assertions	Classical directed-FVS NP-completeness	Established input cited; reduction proved
Finite checks	Exhaustive Python enumeration	Executed; not proof-assistant verification

The substantive novelty proposed for further independent review is the identification of the finite-output topological invariant with local filter capacities and the resulting exact separation and threshold laws. The literature check is not exhaustive. The mathematical statements do not depend on a claim of priority.

41 Further research questions and topics

The first two directions continue the original repository question beyond the class settled here. The remaining directions arise from the exact classification above; they are proposed problems, not assertions that each formulation is a previously published open problem.

Research question 11 (The next derivative level). For Hausdorff spaces with finite nonempty second derivative and empty third derivative, what finite local data replace the positive-color stars? Can an iterated filter quotient encode all admissible words without exponentially listing every lower-level profile?

The obstruction is correlation: a neighborhood may contain whole lower-level profiles, rather than independent color cells. Merely attaching a scalar capacity to each point is unlikely to record all such interactions; a proof or a counterexample would sharply delimit this extension.

[Added 30 September 2026, batch 65] **Bears on this question.** For a fixed coloring of a Hausdorff space with $X''' = \emptyset$, the closure-signature words of length at most three are complete data for admissibility (lemma 51.1 and theorem 47.2), and proposition 54.2 at $r = 3$ gives two three-colorings of one compact space with $X''' = \emptyset$ that agree on every word of length at most two but need 7 and 5 layers, so the words of length three cannot be dropped. Part IV gives no capacity or local-star form of these data, and does not treat the worst case over colorings; the question stays open.

Research question 12 (Compact and first-countable finite-height spaces). Does the repository bound $(q-1)h+s$ attain its full layer value under natural compactness or first-countability hypotheses at every finite height? The present paper proves this at $h = 1$, not at larger heights.

A promising approach is to construct enough incompatible colored profiles inside disjoint neighborhoods while controlling their accumulation at the next derivative. The proof must not assume metrizability or a countable local base when neither is available.

Research question 13 (Intermediate-capacity extremal formulas). Find a closed formula for

$$\max_{G \in \mathcal{G}_q(b_1, \dots, b_t)} \tau(G)$$

when some capacities lie strictly between one and $q - 1$. Determine whether suitable unions of bidirected cliques always optimize any important uniform-capacity subcase.

For a common finite capacity $b \geq 1$, a graph is realizable exactly when $\sum_x \lceil d^+(x)/b \rceil \leq t$, with zero out-degree contributing zero. This follows by splitting each outgoing neighborhood into at most b -element stars. It converts the question into a discrete degree-cost extremal problem, but the linear-arc proof of theorem 37.2 does not by itself solve the nonlinear cost problem.

Research question 14 (Infinite derived sets). Which Hausdorff spaces with an infinite discrete derived set admit a useful filter-component decomposition for finite-output games? When such a global decomposition fails, can local profiles still determine the finite-depth invariant?

The finite proof uses pairwise disjoint neighborhoods chosen for finitely many nonisolated points. Replacing “finite” by “infinite” without a separation argument would be invalid.

[Added 30 September 2026, batch 65] **Second half answered for fixed colorings when $X'' = \emptyset$.** Part IV, which does not cite this question, answers its second half for every fixed coloring of a Hausdorff space whose derived set is discrete, that is, $X'' = \emptyset$: then $L(X, c) = q_c + \tau(G_c)$, where G_c has an arc $a \rightarrow b$ exactly when some point of color a lies in the closure of the color class of b (theorem 51.3). No filter-component decomposition, compactness or first countability is used. Which spaces admit a useful decomposition, and the worst case sm_q over colorings of such spaces, are not addressed.

Research question 15 (Expected query cost). For a probability distribution on the hidden point, characterize the minimum expected depth over all admissible layer assignments and ordered binary trees. How does the optimum depend on masses of positive and null color cells in the filter quotient?

The weighted-layer identity in corollary 33.5 is only a preliminary algebraic analogue. Expected depth depends on leaf masses, and redistributing a color among repeated positions can change those masses.

Research question 16 (Effective presentation of local profiles). For computably presented filters and color cells, when can the positive-color sets B_i be computed or approximated? Which oracle strength is necessary for exact strategy synthesis in natural presentations?

This isolates the semantic task deliberately excluded from the finite feedback-set algorithm. Even a tiny profile graph need not be effectively recoverable from arbitrary infinite-set data.

Research question 17 (Near-extremal stability). If $\alpha_{\text{ac}}(G) \leq r$ and $|E(G)|$ is close to $T(q, r)$, must G be close to a balanced union of complete bidirected graphs, in a quantitative sense useful for almost-sharp query families?

The equality proof supplies two concrete defects: asymmetric arcs and induced three-vertex paths. A stability theorem would need to control how many such defects can coexist with small slack in the random-order bound.

Research question 18 (Recovering topology from the alphabet spectrum). How much of a finite capacity list can be recovered from the sequence $(\text{sm}_q(X))_{q \geq 1}$? Does recording maximum layer counts instead of rounded query depths make the capacity multiset identifiable?

The ceilings erase information: at three labels, one ultrafilter component has maximum layer count three, whereas two through five components have maximum layer count four, yet all have depth two. This suggests studying the unrounded layer spectrum first.

Research question 19 (Choice strength and definability). What is the weakest set-theoretic assumption supporting a countable three-color separation of the kind in equation (35.1)? Can comparable separations be obtained from explicitly definable neighborhood filters under additional descriptive restrictions?

The example here uses a free ultrafilter and is an existence construction in ZFC, not an explicitly computable list of open neighborhoods. Nothing in the finite certificate files resolves this foundational issue.

42 Conclusion

At the first nontrivial derivative level, the relevant local information is not merely the existence of an accumulation point. It is the family of color cells that may accumulate there simultaneously. A finite coloring turns that information into directed pair constraints; the cost of making those constraints compatible is exactly a feedback vertex number.

This gives a complete fixed-color profile theorem for every Hausdorff space with a finite nonempty derived set, an exact local criterion for attainment of the rank-one bound, a full three-label classification, and all alphabet-dependent thresholds for ultrafilter sums. The resulting three-color separation shows concretely why the binary Cantor–Bendixson classification does not extend unchanged to finite output alphabets. Higher derivative levels, infinite derived sets, and general cost models remain distinct problems.

Part IV. Closure-word duality

Sections 1–13 and Appendices A and B form Part I (19 September 2026); Sections 14–28 and Appendix C form Part II (batch 52); Sections 29–42 and Appendix D form Part III (batch 56). They are unchanged except for dated pointers, marked [Added 29 September 2026, batch 52], [Added 29 September 2026, batch 56] or [Added 30 September 2026, batch 65]. Part IV was added on 30 September 2026 in batch 65 of ProveIt’s incoming-report intake, from one manuscript written against the pinned state of Parts I–III. For every fixed coloring of every topological space it characterizes the admissible layer words by the reduced words that survive iterated color closures. This answers the fixed-coloring half of Part II’s research question 1 on every space, extends Part III’s feedback-vertex formula to every Hausdorff space with empty second derivative, and improves Part II’s finite obstruction bound (section 43.4).

43 Provenance, conventions and the relation to Parts I–III

43.1 One manuscript, one addition

Part IV is batch-65 manuscript 05, *Closure-Word Duality for Open-Query Games: Universal finite models, optimal topological compression, finite-budget certificates, and an exact product law*, an AI-assisted research manuscript “prepared for Vladimir Reshetnikov” and dated 30 September 2026 (25 pages, US Letter, as delivered). It arrived as the archive `ProveIt_Closure_Word_Duality.zip` (inner directory `ProveIt_Closure_Word_Duality/`, main file `article.tex`) in ProveIt commit 529a082b9 and was placed in 3fd85006b. It was written against ProveIt commit a4268e78e (in full a4268e78ebd0f6bf71ba609c07f4b3415748f532), which is later than Part III’s write (f8a20bcb1). At that commit this report’s `article.tex` had the Git blob d1e50fd006 (Parts I–III, including the notes of batches 52 and 56), which the manuscript records, and its `README.md` the blob 465cb1e062; both blobs are unchanged at the placement commit. Every statement the manuscript makes about “the repository report” therefore refers to Parts I–III as printed above: its “Part II, Theorem 16.2 and Proposition 16.3” are theorem 16.2 and proposition 16.3, its “Research question 1 in Part II” is research question 1, and its “Part II, Research question 3” is Part II’s Research question 3 (section 27). Its source audit says that the `README` and the article’s line ranges 500–640, 1150–1360, 1480–1690, 2100–2340, 2640–2800 and 3780–3970 at that revision were read through the connected GitHub tool.

The manuscript cites that state of the report as its reference [Repo]: “ProveIt research collection, *Open-query games on ordinal spaces*, Parts I–III, 19 and 29 September 2026, AI-assisted research report. Pinned repository commit a4268e78ebd0f6bf71ba609c07f4b3415748f532. Article: `SetTheory/Cardinals/docs/reports/ordinals-and-order-types/games-on-ordinals/open-query-membership-games/article.tex`. Pinned source on GitHub. Accessed 30 September 2026. The supplied source audit gives the exact inspected ranges and identifies which results are reused.” This is not the entry [4] of Part II, which is Part I at an earlier commit. Inside this report the reference is internal, so its two citations are replaced by cross-references (see below).

Its code and data are shipped beside those of Parts I–III, and their names carry the prefix 04-closure-words-; Part I’s unprefix files count as 01, Part II’s as 02 and Part III’s as 03. They are

- `code/04-closure-words-verify.py`, the verifier;

- `code/04-closure-words-Makefile`, its delivered Makefile;
- `data/04-closure-words-verification.json`, the recorded run;
- `data/04-closure-words-validation.json`, the delivered build and package record;
- `04-closure-words-SOURCES.md`, the source audit, in the report root.

The manuscript, its PDF and its delivery README are not shipped and survive in the arrival commit; the archive had no checksum ledger. The report README lists every file and every name the delivered files still use.

The manuscript contributes, in the order printed here: its statement of the problem, of the repository frontier and of its contribution (section 44); the tree-layer normal form and greedy elimination, reproved (section 45); the closure signature and its operator calculus (section 46); closure-word duality with short failure certificates (section 47); the finite-rank criterion, size bounds and the rank bounds on arbitrary spaces (section 48); recovery of the signature from the admissible words (section 49); finite and compact realizations and the classification of finite-depth behaviors (section 50); the optimal Hausdorff realization height and the feedback-vertex formula when $X'' = \emptyset$ (section 51); finite-budget replicas and the improved obstruction bound (section 52); the product signature and rank law (section 53); two colorings of one compact space separated only by words of length four, and a hierarchy of such examples (section 54); algorithms and the effective boundary (section 55); its finite checks (section 56); nine research questions (section 57); a conclusion; a proof dependency and claim audit (Appendix E); and small worked certificates (Appendix F). Every result, proof, example, remark, limitation and question of the manuscript is printed.

Where the merge had to choose.

- *Front matter and numbering.* The manuscript's title block, abstract and status paragraph are printed in section 43.2; its table of contents is dropped. Its Sections 1–15 are Sections 44–58 here, and its Appendices A and B are Appendices E and F, after the appendices of Parts I–III, with “(Part IV)” added to their titles. Its theorems and equations keep their positions within each section: its Theorem $n.k$ is Theorem $(n + 43).k$ here and its equation $(n.k)$ is $(n + 43.k)$. Its Research questions 1–9 are Research questions 20–28 here, because the counter continues after Part III's nineteen.
- *Duplicated material.* The manuscript reproves, and credits, what it takes from Parts II and III, so that its argument does not depend on an unformalized external lemma; the proofs are printed in full. The normal form theorem 45.2 is theorems 4.2, 16.2 and 31.3; the greedy elimination proposition 45.3 is proposition 16.3; the universal constant lemma 48.3 is (19.2); the tower lemma lemma 50.4 contains lemma 18.1 and theorem 18.2; the admissible-word half of proposition 50.2 is proposition 21.1; and the feedback-set argument in the proof of theorem 51.3 is that of lemma 33.3. The checks are collected in section 43.4.
- *Renamed macros and symbols.* The manuscript's one-argument closure macro `\cl` (an overline) clashes with Part I's `\cl` (cl, no argument); its seventeen uses are set with a new `\cwcl`, and the printed overline is unchanged. Its `\rk` clashes with Part I's `rk` and is written ρ , as it printed. Its `\depth`, printed depth, is set with Part II's `D`; its admissible-word set $\mathcal{W}$ is Part II's $\mathcal{W}$, its word length Part II's $|\cdot|$, and its italic layer count $L(X, c)$ is printed as Part II's $L(X, c)$ (the same number). Its signature $\mathcal{S}$ is set with `\cwSig`, printed unchanged as $\mathcal{S}$. Its basis $\mathcal{B}(X, c)$ is printed $\mathcal{B}^{\text{bas}}(X, c)$, because Part III's $\mathcal{B}$ is a Boolean algebra. Its tower K_k is printed K_k , as in Part II (section 14.3), since Part I's K_k is a different space. Its unused height macro and its macro for the pinned commit are dropped. No normalization is changed. The notation table is section 43.3.

- *The reference [Repo]*. Its two uses are replaced by “(printed above)” in section 44 and by theorem 16.2 and proposition 16.3 after proposition 45.3; the entry itself is quoted above. The phrase “Research question 1 in Part II” is printed as a cross-reference to research question 1, with the same words.
- *Bibliography*. The manuscript’s entries for Chiozini, Csernák and Soukup [1], Lagoutte and Tavenas [5] and Selivanov [9] are the works already cited by Parts I–III, and are merged with those entries; its annotations are added to them.
- *Shipped names*. The source-audit file, the verifier, the recorded output and the rerun command in sections 44 and 56 are printed with the shipped names and an explicit output path; the note in section 56 quotes the delivered text. The rewritten text promises no file that is not shipped.
- *Notes*. Four dated notes are added inside Part IV: on the scope of “in full generality” (section 44), on Part I’s product law (section 53), on shipped names (section 56) and on the relation of the research questions to those of Parts II and III (section 57). In Parts II and III, notes pointing to Part IV follow research question 1 and Part II’s Research questions 3 and 6, and research questions 11 and 14; the title page and the abstract gain a paragraph each, and the entry [1] a dated sentence on its two titles.

Status (manuscript 05). Part IV is an AI-assisted, unrefereed research manuscript. Its new claims are ordinary mathematical theorems proved in it, not Lean- or Rocq-checked results; it ships executable finite checks but no proof-assistant files, and nothing in it is formalized in ProveIt. It credits as prior material the tree-layer normal form, greedy elimination, the chain interpretation, the compact word construction and the universal-word constant of Parts II and III, and claims as its contribution their extension through closure-word duality and its consequences. “New here” means new relative to the inspected report, not certified historical priority; its literature checks were targeted, not exhaustive. It does not claim to settle the uniform-coloring or transfinite questions of the cited literature: it does not determine which signatures the colorings of a prescribed space can have, so the attainability half of research question 1 stays open. It does not identify word equivalence with Wadge equivalence or with any partition hierarchy, and does not claim the shortest-common-supersequence problem or its three-letter hardness [5]. Its algorithms are relative to stated closure and emptiness operations, not polynomial-time algorithms for arbitrary topological input; its product law concerns the specified pair coloring, not the supremum over colorings of a product; its obstruction bound is an upper bound, not claimed optimal. The finite checks test finite instances only; every infinite-space statement rests on the written proofs, and the recorded run time of 4.282 seconds is specific to the delivering environment.

43.2 The manuscript’s title block and abstract

Closure-Word Duality for Open-Query Games. Universal finite models, optimal topological compression, finite-budget certificates, and an exact product law. ProveIt research series. AI-assisted research manuscript prepared for Vladimir Reshetnikov, 30 September 2026.

A hidden point of a topological space is to be assigned one of finitely many known labels, using adaptive questions about open sets. We associate to the labeling a language of reduced words detected by iterated color closures. We prove, without separation assumptions, that a proposed layer word is admissible exactly when it contains every detected word as a subsequence. Moreover, failure of an m -letter proposal has a detected counterword of length at most m . This converts every finite-budget problem, on an arbitrary space, into a finite shortest-common-supersequence

problem.

For finite-depth games the detected language is a finite downset, uniquely recoverable from the entire admissibility profile. Every such downset is realized both by a finite union of colored chains and by a compact countable metrizable space. Its longest word has length equal to the least possible Cantor–Bendixson height of a Hausdorff realization. We obtain an exact feedback-vertex-set formula when the second derivative is empty, including spaces with infinitely many nonisolated points; an improved finite-poset obstruction bound; and the product law $\rho(X \times Y, c \times e) = \rho(X, c) + \rho(Y, e) - 1$. A pair of three-colorings of the same compact space agree on all patterns of length at most three and on every derivative’s color cardinalities, yet need respectively three and four queries. Full proofs, an explicit algorithm, a research agenda, and independent exhaustive finite checks accompany the article.

Status and scope. The new claims are ordinary mathematical theorems proved in this manuscript, not Lean- or Rocq-checked results. The repository’s tree-layer normal form, greedy elimination test, chain interpretation, compact word construction, and universal-word constant are credited as prior material. The contribution is their extension through closure-word duality and its consequences. Historical priority has not been independently certified. This is not a claim to settle all uniform-coloring or transfinite questions in the cited literature.

43.3 Notation of Part IV

Part IV fixes a nonempty space X , with no separation axiom unless stated, a finite alphabet Σ of q letters and a coloring $c : X \rightarrow \Sigma$, as Part II does; $A_a = c^{-1}(a)$. Several letters are used differently in Part IV and in Parts I–III. Where a false reading is tempting, the table prints it beside the true one.

Symbol	Part IV	Parts I–III
$\mathcal{S}(X, c),$ $\rho(X, c)$	The closure signature, the nonempty reduced words v with $F_v \neq \emptyset$, and its rank, the supremum of their lengths (section 46). On a finite poset $\mathcal{S}(P, c) = \mathcal{C}(P, c)$ (lemma 50.1).	Part II: $\mathcal{C}(P, c)$ the reduced chain words of a poset; h_c in corollary 19.2 is $\rho(P, c)$.
T_a, R_a, F_v	$T_a(F) = \overline{A_a \cap F}$, the color closure; $R_a(F) = \overline{F \setminus A_a}$; F_v the nested closure along v . <i>False:</i> T_a is Part II’s automaton transition; F_v is a layer or a filter.	Part II: $T_a(I)$, a transition on order ideals (23.3); Part I: F_i closed layers; Part III: $\mathcal{F}$ a filter, $T(q, r)$ an arc count.
$\overline{B}$	The closure of B (the manuscript’s one-argument $\backslash \text{cl}$).	Part I: cl ; Part II: cl_X .
A_a	The color class $c^{-1}(a)$.	Part I: A the target; Part II: $\mathcal{A}_{q,m}(S)$ a word family, A a mask.
$\mathcal{W}, L, D$	Admissible words, least layer count, least worst-case depth: the manuscript’s $\mathcal{W}, L$ and depth.	Part II’s $\mathcal{W}, L, D$ (same notions); Part I’s $L(A, X)$.
$\mathcal{B}^{\text{bas}}(X, c), B$	The maximal antichain of a finite signature (the manuscript’s $\mathcal{B}$); B also a finite antichain or basis in theorems 50.3 and 50.5 and section 55. <i>False:</i> Part III’s $\mathcal{B}$.	Part III: $\mathcal{B} = \mathcal{P}(D)/\mathcal{I}_{\mathcal{F}}$ a Boolean algebra; B_i, B sets of colors.
$B(q, m),$ $B_*(q, m)$	$B(q, m)$ the least obstruction size, as in Part II’s Research question 3; $B_*(q, m) = mq(q-1)^{m-1}$ the bound (52.1).	Part II: the bound $(m+1)q^m$ (theorem 24.1).

Symbol	Part IV	Parts I–III
K_k, K_b	The tower of height k (the manuscript's K_k), and the tower colored by a rank word b , reduced or not. Its new point has neighborhoods containing all but finitely many copies, so it is Part II's K_k . <i>Not</i> Part I's K_k .	Part I: $K_k = [0, \omega^{2^k-1}]$; Part II: K_m, c_w .
height, h	The Cantor–Bendixson height, the least k with $X^{(k)} = \emptyset$; it is $h+1$ for the last derivative rank h . The space Z of height four has $h=3$. <i>False</i> : “height” = h .	h the last derivative rank; Appendix B: the height is $h+1$.
D, D_m	A finite downset of reduced words; D_m the signature truncated at length m ; in the proof of theorem 51.3, a topological ordering. <i>Not</i> the depth D .	Part I: D_j ; Part II: D ; Part III: D_i , D isolated parts.
$U(q, r), U_i, U_j$	$U(q, r) = (q-1)r+1$, Part II's $\text{scs}(\mathcal{A}_{q,r})$ (19.2); blocks U_i in lemma 49.1; open suffixes U_j (55.3).	Part II: U_j open suffixes.
$N(q, r)$	The number of nonempty reduced words of length at most r .	Part I: N_0, N_1 ; Part II: N an input length.
G_c, q_c, τ	As in Part III: arcs $a \rightarrow b$ for $A_a \cap \overline{A_b} \neq \emptyset$; for a finite derived set this is Part III's digraph. G_t is the elimination chain, as in Part II.	Part III: G_c, q_c, τ ; Part II: G_t .
r, s, e, Γ	r a rank; in theorem 53.1, r, s the factor ranks and $e: Y \rightarrow \Gamma$ the second coloring. <i>False</i> : s is a number of colors.	Part II: s colors on $X^{(h)}$; Part III: r a bound on α_{ac} , s a remainder.
t, ℓ_i	The number and the lengths of basis words in section 55.	Part III: $t = X' $, $\ell(G)$.
$Z, c_{\text{hard}}, c_{\text{soft}}$	The sum of 24 towers of theorem 54.1 and its two colorings.	Part III: Z the clopen discrete part.
$\varepsilon, \preceq$	The empty word; the subsequence relation.	Part II writes $\preceq$ for the same relation.

43.4 Parts I–III inside Part IV

The following checks were made when Part IV was added; they are consequences of the four parts' proofs, not new theorems.

1. *Normal form and elimination.* Theorem 45.2 and proposition 45.3 are theorem 16.2 and proposition 16.3; the chain (45.3) is (16.3), and at $q=2$ both are Part I's theorem 4.2.
2. *Posets.* On a finite poset, lemma 50.1 identifies the signature with Part II's reduced chain words $\mathcal{C}(P, c)$, so theorem 47.2 there is theorem 17.1, and (48.4) is (19.3) with $h_c = \rho(P, c)$. Part IV extends both to every topological space, and adds the truncation to constraints of length at most $|w|$.
3. *Finite height.* On a Hausdorff space of last derivative rank h , lemma 51.1 gives $\rho \leq h+1$, so (48.4) gives $L \leq (q-1)(h+1)+1 = (q-1)h+q$. This is weaker than Part II's sharp theorem 20.1, $(q-1)h+s$ with $s \leq q$, as the manuscript says after corollary 48.4.
4. *Rank one.* For a Hausdorff space with a finite nonempty derived set, the arcs (51.2) are those of Part III's G_c (a point p_i of color a in the closure of A_b), so (51.3) is Part III's theorem 33.4. Theorem 51.3 assumes only $X'' = \emptyset$: the derived set may be infinite, and no compactness, first

countability or filter decomposition is used. It concerns a fixed coloring; Part III's capacity results on sm_q are not extended.

5. *Obstructions.* For finite posets corollary 52.2 improves the bound $(m+1)q^m$ of theorem 24.1 to $mq(q-1)^{m-1}$; for example 12 instead of 27 points at $q=3$, $m=2$. Theorem 24.1 is stated for every poset, corollary 52.2 for finite ones.
6. *Products.* The rank law (53.2) is the coloring analogue of Part I's lemma 9.1; see the note after theorem 53.1. The binary phenomenon of Appendix F (chains 01 and 10) is the finite-poset form of the clash of orientations behind Part I's counterexample (section 2).
7. *Separations.* Theorem 54.1 and Part III's (35.1) separate different things: Part III compares two spaces through sm_3 ; Part IV compares two colorings of one space whose signatures agree up to length three. Neither contradicts the other, and no statement of Parts I–III is refuted or retracted.
8. *Independent checks.* A script written at intake and rerun at this write (not shipped) enumerated all 389 labelled topologies on at most four points and their three-colorings and reproduced the counts of section 56 (29,577 colorings, 21,639 with a finite tree, 109 distinct finite signatures). It compared duality and the short-certificate test with a brute-force search for layerings, which checks suffix openness directly and does not use elimination, on every coloring with at most three points and a random 2% of those with four (73,452 comparisons in the rerun, 69,612 at intake; all agree). It confirmed $U(q, r) = (q-1)r + 1$ by exact shortest-supersequence search for $(q, r) = (2, 3), (2, 5), (3, 2), (3, 3), (3, 4), (4, 2), (4, 3)$; the layer counts 9 and 7 and the basis sizes 24 and 12 of theorem 54.1; the product signature and the rank law on 300 random pairs of two-colored preorders with at most three points; and $L = q + \tau(G)$ on all 4,165 loopless digraphs on at most four labelled vertices. The worked certificates of Appendix F were checked by hand.

44 The problem, the repository frontier, and the contribution

Suppose a point x of a known space X is hidden. One may ask whether x belongs to any chosen open subset of X , and later questions may depend on earlier answers. The objective is not necessarily to identify x : it is to output its known color $c(x)$ in a finite alphabet Σ . The central issue is how topology constrains the order in which colors can be separated. Ordinary cardinalities of color classes do not capture that constraint.

Chiozini, Csernák and Soukup study a broader, transfinite cut-and-choose framework and associated point-separating and set-membership invariants [1]. Their arXiv record was checked at version 3, dated 29 May 2026. We work only with finite worst-case budgets and with a specified coloring. These restrictions are substantive: a supremum over all colorings and a transfinite game need additional arguments.

44.1 The inspected baseline

The direct baseline is the ProveIt report *Open-query games on ordinal spaces*, Parts I–III (printed above), at commit

a4268e78ebd0f6bf71ba609c07f4b3415748f532.

Its article is the Git blob `d1e50fd006a8ba1fe645b41b4fb37ee35f7043fe`. Part II identifies open-query trees with ordered monochromatic layers, represents finite colored posets by chain-word

constraints, and realizes individual words on compact convergent towers. Part III treats spaces with finite nonempty derived set by local filters and a directed graph on colors.

Research question 1 in Part II asks for exact invariants beyond convergent towers and for a characterization through accumulating word profiles. Its Part III annotation explicitly leaves higher finite heights and infinite derived sets beyond the finite-derived-set treatment. The present work answers the *fixed-coloring profile* part in full generality. It does not determine which profiles can be produced by all colorings of each prescribed space. That remaining realizability problem includes the uniform local-capacity question raised in the same research program.

[Added 30 September 2026, batch 65] **Scope: “in full generality”.** The phrase refers to the second half of research question 1, for one coloring at a time: on every space, the admissible words of a fixed coloring are characterized by its closure signature (theorems 47.2, 49.2 and 50.5). The first half, attainability of $(q - 1)h + s$ over all colorings of a given space, stays open beyond Part III’s class, as the next sentence says; the question is therefore answered in part, and the note after research question 1 records the new scope.

The comparison is targeted. We inspected the report’s definitions, normal-form and elimination arguments, synchronization and complexity discussion, research questions, and finite-derived-set results. We did not audit every mathematical assertion in ProveIt. The supplied source audit, shipped here as `04-closure-words-SOURCES.md`, records paths, the pin, and the inspected ranges.

44.2 A map of the results

For a closed set $F \subseteq X$ and color a , put

$$T_a(F) = \overline{F \cap c^{-1}(a)}.$$

For a word $v = v_1 \cdots v_k$, let $F_v = T_{v_1} \cdots T_{v_k}(X)$, with operators acting from right to left. The *closure signature* consists of the nonempty reduced words v for which $F_v \neq \emptyset$. Reduced means that adjacent letters are different; letters may recur later.

The main results, with the precise hypotheses given below, are these.

1. **Universal duality and short certificates.** A word w is admissible exactly when every word of the closure signature is a subsequence of w . If w has $m \geq 1$ letters, only signature words of length at most m need be tested (theorem 47.2).
2. **Complete finite-depth classification.** Finite-depth games, identified when they admit exactly the same words, are classified by nonempty finite downsets of reduced words. Their unique maximal antichains give exact finite SCS instances. They have finite-poset and compact countable metrizable realizations; the maximum signature length is the optimal Hausdorff realization height (theorems 50.5 and 51.2).
3. **Consequences beyond the previous local graph model.** The graph formula extends to every Hausdorff space with $X'' = \emptyset$. Finite budgets have uniformly bounded finite replicas and improved actual subposet obstructions. Independent products have additive signature rank, minus one. Higher-order closures cannot be replaced by pairwise or triple summaries (theorems 51.3, 52.1, 53.1 and 54.1).

The finite-topology checks compare independent minimax optimization with the word and closure calculations. They include all labeled topologies on at most four points and all their three-colorings. They are tests of finite instances, not a substitute for any infinite-space proof.

44.3 Novelty boundary and related frameworks

The repository already supplies the normal form and the single-word closure elimination test used in section 45. Those are not claimed as new here. Neither is the SCS problem or its hardness for reduced words over three letters: the latter is established by Lagoutte and Tavenas [5]. Selivanov's work on difference and fine hierarchies of finite partitions provides a broader descriptive-set-theoretic context [9]. We do not identify our equivalence relation with Wadge equivalence or claim that existing partition hierarchies were previously unavailable.

The demonstrated extension is an explicit, all-space duality for the entire admissibility language, its finite-budget truncation, a recovery theorem, optimal-height realization, and the stated consequences. These are the mathematical claims to assess. A literature-wide priority determination would require a more extensive comparison than the source review performed here.

45 Open-query trees and ordered layers

Throughout, X is nonempty and Σ is a finite nonempty alphabet. The coloring $c : X \rightarrow \Sigma$ need not be continuous or surjective. Unless otherwise specified, there is no separation assumption. Write $A_a = c^{-1}(a)$, and let $q = |\Sigma|$. Unused colors may be discarded when improving a numerical bound.

Definition 45.1 (Admissibility). For $m \geq 1$, a word $w = w_1 \cdots w_m$ is *admissible* for (X, c) if there exists $\lambda : X \rightarrow \{1, \dots, m\}$ such that

$$c(x) = w_{\lambda(x)}, \quad \{x : \lambda(x) \geq j\} \text{ is open for every } j. \quad (45.1)$$

Empty fibers are allowed. Let $\mathcal{W}(X, c)$ be the set of admissible words and

$$L(X, c) = \min\{|w| : w \in \mathcal{W}(X, c)\},$$

with value ∞ when the set is empty. Let $D(X, c)$ be the minimum worst-case depth of a correct finite binary open-query tree, or ∞ .

Inserting empty layers preserves admissibility. Adjacent layers of the same color can be merged, because the suffix sets at the surviving boundaries remain open. Thus a shortest admissible word has no equal adjacent letters. We use $v \preceq w$ for the ordinary scattered-subsequence relation.

Theorem 45.2 (Tree-layer normal form; repository baseline). *The minimum number of nonempty leaves of a correct finite open-query tree is $L(X, c)$. If finite, then*

$$D(X, c) = \lceil \log_2 L(X, c) \rceil. \quad (45.2)$$

Proof. Order a tree's leaves recursively, with the no-subtree before the yes-subtree. Every suffix of its leaf cells is open. To see this, use induction in the current possible-point subspace. If the current query is U , a suffix wholly in the yes-subtree is $U \cap V$ for a relatively open V . A suffix meeting the no-subtree contains all of the yes-subtree and is

$$U \cup ((X \setminus U) \cap V) = U \cup V.$$

The subspace version of the same identity proves the induction. Remove empty leaves. Correctness makes each remaining cell monochromatic, so its position is a layering. Conversely, a layering with m nonempty fibers permits balanced binary search of its index using the open suffixes in (45.1). This gives m leaves and depth at most $\lceil \log_2 m \rceil$. A depth- d binary tree has at most 2^d leaves, which proves the converse inequality and the formula. $\square$

Proposition 45.3 (Greedy elimination; repository baseline). *For closed F put $R_a(F) = \overline{F \setminus A_a}$. Given $w = w_1 \cdots w_m$, define*

$$G_0 = X, \quad G_{t+1} = R_{w_{m-t}}(G_t) \quad (0 \leq t < m). \quad (45.3)$$

Then $w \in \mathcal{W}(X, c)$ if and only if $G_m = \emptyset$.

Proof. Suppose λ is a layering for w . Its prefixes $H_t = \{\lambda \leq m - t\}$ are closed. Induction gives $G_t \subseteq H_t$: a point of H_t with color different from w_{m-t} must belong to H_{t+1} , and closure preserves containment in this closed set. Thus $G_m \subseteq H_m = \emptyset$.

Conversely, the G_t decrease and are closed. If $G_m = \emptyset$, give $G_t \setminus G_{t+1}$ the index $m - t$. Such a point has color w_{m-t} , since $G_t \setminus A_{w_{m-t}} \subseteq G_{t+1}$. The index- $\geq j$ suffix is $X \setminus G_{m-j+1}$, which is open. These layers partition X and prove admissibility. $\square$

Both statements are reproduced to make the new argument independent of an unformalized external lemma. Their prior appearance is Part II, theorem 16.2 and proposition 16.3 of the pinned report.

46 The closure signature

Definition 46.1 (Color closures and detected words). For a closed $F \subseteq X$, define

$$T_a(F) = \overline{A_a \cap F}, \quad F_{a_1 \cdots a_k} = T_{a_1}(T_{a_2}(\cdots T_{a_k}(X) \cdots)), \quad F_\varepsilon = X. \quad (46.1)$$

Let $\text{red}(v)$ be obtained from v by replacing each nonempty constant run by one letter. The *closure signature* is

$$\mathcal{S}(X, c) = \{v \neq \varepsilon : v = \text{red}(v), F_v \neq \emptyset\}. \quad (46.2)$$

Its rank is

$$\rho(X, c) = \sup\{|v| : v \in \mathcal{S}(X, c)\} \in \mathbb{N} \cup \{\infty\}.$$

Since $X \neq \emptyset$, its signature is nonempty and its rank is at least one.

The expression F_{abc} records a genuinely nested condition:

$$F_{abc} = \overline{A_a \cap \overline{A_b \cap \overline{A_c}}}.$$

It is not the intersection of the three individual color closures. A point need not witness all stages by a chain of actual specialization comparisons; in Hausdorff spaces nontrivial stages arise from accumulation instead.

Lemma 46.2 (Operator calculus). *On the lattice of closed subsets, each T_a is monotone, contractive, idempotent, and preserves finite unions. Consequently*

$$F_v = F_{\text{red}(v)}, \quad v \preceq u \implies F_u \subseteq F_v. \quad (46.3)$$

The signature is a downset among nonempty reduced words ordered by subsequence.

Proof. Monotonicity follows from monotonicity of intersection and closure. Since F is closed, $T_a(F) \subseteq F$. Also $A_a \cap F \subseteq A_a \cap T_a(F)$, whereas $T_a(F) \subseteq F$ gives the reverse containment after taking the corresponding closures. Hence $T_a(T_a(F)) = T_a(F)$. Finite-union preservation follows from $\overline{B \cup C} = \overline{B} \cup \overline{C}$.

Idempotence removes repeated adjacent operators. Deleting an operator from a composition can only enlarge its value: the deleted operator is contractive and all operators outside it are monotone. Iterating deletions proves the second assertion of (46.3), and hence the downset property. $\square$

Remark 46.3 (Why reduction is essential). If a occurs as a color, then $F_{a^k} = F_a \neq \emptyset$ for every k . Counting unreduced word lengths would therefore make every rank infinite. The reduction removes this artificial repetition, while retaining meaningful patterns such as aba .

Lemma 46.4 (Reduction and subsequences). *For a reduced word v and an arbitrary word w ,*

$$v \preceq w \iff v \preceq \text{red}(w).$$

Proof. One implication follows because $\text{red}(w) \preceq w$. For the other, the positions used by an embedding of a reduced word cannot use two letters from a single constant run: between two equal letters of v a different letter must occur. Map each used run to its single representative. $\square$

47 Closure-word duality and short failure certificates

The topological step is finite distributivity; the combinatorial step is a precise relation between subsequence failure and coordinatewise avoidance. We prove the latter first.

Lemma 47.1 (Diagonal avoidance). *Let $w, b \in \Sigma^m$, with $m \geq 1$, and assume $b_i \neq w_i$ for every i . Then $\text{red}(b) \not\preceq w$. Conversely, if a nonempty reduced word v is not a subsequence of w , there exists such a b with $\text{red}(b) \preceq v$ and $|\text{red}(b)| \leq m$.*

Proof. For the first direction put $v = \text{red}(b)$, and let j_i be the index of the constant run of b containing position i . Let p_i be the length of the prefix of v matched greedily while scanning $w_1 \cdots w_i$. We claim $p_i < j_i$. At the first position, $w_1 \neq b_1 = v_1$, so $p_1 = 0$. If a new run starts at i , then $j_i = j_{i-1} + 1$ and $p_i \leq p_{i-1} + 1 < j_i$. If the run does not change, reaching j_i at step i would require $w_i = v_{j_i} = b_i$, which is excluded. This proves the claim. At $i = m$, $p_m < |v|$, so greedy matching fails, equivalently $v \not\preceq w$.

For the converse, greedily match the given v in w , and let p_i be the matched prefix length *after* scanning position i . Since matching never finishes, set $b_i = v_{p_i+1}$. If there was no advance at position i , the definition of greedy matching gives $b_i \neq w_i$. If there was an advance, then $w_i = v_{p_i}$ and $b_i = v_{p_i+1} \neq v_{p_i}$ because v is reduced. The indices $p_i + 1$ are nondecreasing and increase only by one. Thus $\text{red}(b)$ is a contiguous segment of v , in particular a subsequence, and has at most m letters. $\square$

Theorem 47.2 (Universal closure-word duality). *For every topological space and finite coloring as above, and every word w of length $m \geq 1$, the following are equivalent:*

$$w \in \mathcal{W}(X, c); \tag{47.1}$$

$$v \preceq w \text{ for every } v \in \mathcal{S}(X, c); \tag{47.2}$$

$$v \preceq w \text{ for every } v \in \mathcal{S}(X, c) \text{ with } |v| \leq m. \tag{47.3}$$

In particular, each rejected m -letter word has a rejected constraint $v \in \mathcal{S}(X, c)$ of length at most m .

Proof. Since the color classes form a finite partition,

$$R_a(F) = \overline{\bigcup_{b \neq a} (F \cap A_b)} = \bigcup_{b \neq a} T_b(F).$$

All T_b preserve finite unions. Therefore the final residual in (45.3) expands as

$$G_m = R_{w_1} \cdots R_{w_m}(X) = \bigcup_{\substack{b \in \Sigma^m \\ b_i \neq w_i \ (1 \leq i \leq m)}} F_b. \quad (47.4)$$

An empty index set means an empty union, including the one-color case.

If w is rejected, proposition 45.3 says $G_m \neq \emptyset$. Some term F_b in (47.4) is nonempty. By lemma 46.2, $\text{red}(b) \in \mathcal{S}(X, c)$; by lemma 47.1, it is not a subsequence of w , and its length is at most m . Thus (47.3) implies (47.1).

Conversely, suppose some $v \in \mathcal{S}(X, c)$ is not a subsequence of w . The converse of lemma 47.1 gives an indexed word b in (47.4) with $\text{red}(b) \preceq v$. Then

$$\emptyset \neq F_v \subseteq F_{\text{red}(b)} = F_b \subseteq G_m.$$

So w is rejected. This proves (47.1) $\Rightarrow$ (47.2). The remaining implication (47.2) $\Rightarrow$ (47.3) is immediate. $\square$

Remark 47.3 (A nontrivial truncation). The assertion that length- m candidates need only length- $\leq m$ constraints is stronger than the observation that a finite-depth game has finitely many constraints. It applies even when the signature is unbounded and no finite strategy exists. A constraint longer than m cannot be embedded into w ; the theorem shows that its effect is already witnessed by some shorter *occurring* constraint. This is a topological consequence of the operator calculus and the avoidance lemma, not an arbitrary property of all infinite languages.

48 Finite depth, finite antichains, and sharp bounds

Theorem 48.1 (Finite rank criterion). *The following are equivalent: $D(X, c) < \infty$, $L(X, c) < \infty$, and $\rho(X, c) < \infty$. If their rank is r , the signature is finite, and its unique set $\mathcal{B}^{\text{bas}}(X, c)$ of subsequence-maximal words satisfies*

$$L(X, c) = \text{scs}(\mathcal{B}^{\text{bas}}(X, c)), \quad D(X, c) = \left\lceil \log_2 \text{scs}(\mathcal{B}^{\text{bas}}(X, c)) \right\rceil. \quad (48.1)$$

Proof. If an admissible word has length m , duality implies that every signature word has length at most m , hence the rank is finite. Conversely, a bounded language over a finite alphabet is finite. Concatenating all its words gives a common supersequence, hence an admissible word by duality. A finite downset is generated by its unique maximal antichain; its common supersequences are exactly those of that antichain. Now apply theorem 45.2. $\square$

Proposition 48.2 (Uniform size bounds). *For $q \geq 2$ and finite rank r , put*

$$N(q, r) = q \sum_{j=0}^{r-1} (q-1)^j.$$

Then

$$|\mathcal{S}| \leq N(q, r), \quad |\mathcal{B}^{\text{bas}}| \leq q(q-1)^{r-1}, \quad \sum_{v \in \mathcal{B}^{\text{bas}}} |v| \leq rq(q-1)^{r-1}. \quad (48.2)$$

In particular, at fixed q, r there are at most $2^{N(q, r)}$ different finite-depth admissibility profiles of rank at most r .

Proof. There are $q(q-1)^{k-1}$ reduced words of length k . This proves the first bound. Extend each member of the antichain to a reduced word of length r by appending letters different from the preceding letter. Choose any fixed deterministic extension rule. Two equal extensions would make one of their original words a prefix of the other, contradicting the antichain property. Thus extension is injective, proving the second bound and then the third. The final claim simply counts subsets of the finite pool of possible words; not all those subsets are signatures of rank at most r . $\square$

Lemma 48.3 (Universal reduced words; repository constant). *For $q \geq 2, r \geq 1$, the shortest common supersequence of all reduced q -letter-alphabet words of length r has length*

$$U(q, r) = (q-1)r + 1. \quad (48.3)$$

Proof. Use the prefix of length $U(q, r)$ of the periodic word $012 \cdots (q-1)012 \cdots$. A greedy first match uses at most q positions; each subsequent, different letter costs at most $q-1$ positions. The last match is therefore at most $q + (r-1)(q-1) = U(q, r)$.

For the lower bound, in any proposed universal word select a color whose first occurrence is latest. If a color is absent the word already fails; otherwise that occurrence has position at least q . Subsequently select, among the $q-1$ different next colors, one whose next occurrence is latest. If one is absent beyond the current position the word fails again; otherwise the position increases by at least $q-1$. This recursively chooses a reduced r -letter word whose greedy last match occurs no earlier than $U(q, r)$. Greedy matching is earliest possible, so the lower bound follows. $\square$

Corollary 48.4 (Rank controls queries). *For $q \geq 2$ and finite rank r ,*

$$r \leq L(X, c) \leq (q-1)r + 1. \quad (48.4)$$

The upper bound is attained by a finite union of chains, and by a compact countable metrizable colored space. For one active color, $r = L = 1$ and $D = 0$.

Proof. A longest constraint gives the lower bound. Every reduced word of length at most r extends to one of length r when $q \geq 2$, so lemma 48.3 and duality prove the upper bound. The realization results of section 50 realize the downset of all reduced words of length at most r , for which lemma 48.3 is sharp. $\square$

The constant in (48.3) is already proved in Part II of the repository report. Here it becomes a bound for the closure rank of arbitrary spaces. It should not be confused with the sharper repository bound involving the number of colors on the last nonempty derivative of a finite-height Hausdorff space.

49 Recovering the signature from admissibility

Duality says that the signature determines all admissible words. A converse is needed if the signature rank is to measure an intrinsic property of the game, rather than an artifact of its particular topological presentation. The following elementary separator provides that converse at finite depth.

Lemma 49.1 (A supersequence avoiding one forbidden word). *Let $\mathcal{B}^{\text{bas}}$ be a finite family of reduced words, and let $v = a_1 \cdots a_k$ be a nonempty reduced word. If $v \not\preceq b$ for every $b \in \mathcal{B}^{\text{bas}}$, there exists a reduced word w such that*

$$b \preceq w \ (b \in \mathcal{B}^{\text{bas}}), \quad v \not\preceq w, \quad |w| \leq \sum_{b \in \mathcal{B}^{\text{bas}}} |b| + k - 1. \quad (49.1)$$

Proof. Greedily match v in each b . Since matching fails, it yields a decomposition

$$b = u_1 a_1 u_2 a_2 \cdots u_j a_j u_{j+1}, \quad j < k,$$

where u_i contains no occurrence of a_i . Fill the remaining stages with empty words. For each stage i , concatenate the u_i blocks of all family members in a fixed order, obtaining U_i . It still contains no a_i . Set

$$w^* = U_1 a_1 U_2 a_2 \cdots U_{k-1} a_{k-1} U_k.$$

Every b embeds into w^* using its stage blocks and the displayed letters. Greedy matching of v into w^* matches exactly the displayed first $k - 1$ letters, because U_i contains no a_i , and then stops. Hence v is avoided. There are at most $\sum_b |b|$ letters in the blocks and $k - 1$ displayed letters. Reduce w^* ; by lemma 46.4 this preserves every reduced input constraint, still avoids v , and cannot increase length. $\square$

Theorem 49.2 (Observational completeness at finite depth). *If $D(X, c) < \infty$, then*

$$\mathcal{S}(X, c) = \{v \neq \varepsilon : v = \text{red}(v), \ v \preceq w \text{ for every } w \in \mathcal{W}(X, c)\}. \quad (49.2)$$

Thus two finite-depth colored spaces over the same alphabet have identical admissibility languages if and only if they have identical closure signatures.

Proof. One inclusion is duality. If a reduced v is absent from the signature, it is not a subsequence of any maximal signature word: the signature is a downset. Apply lemma 49.1 to its finite maximal basis. The resulting w contains every basis word, so is admissible by duality, but avoids v . This proves the other inclusion. The equivalence follows immediately. $\square$

Definition 49.3 (Word equivalence). Colored spaces over a fixed alphabet are *word-equivalent* if their admissibility languages agree. We use this term only for this explicitly defined relation, not for homeomorphism, continuous reducibility, or equality of a single optimal-depth number.

A scalar optimum loses information even when finite: different antichains can have the same SCS length. The entire admissibility language retains the signature. By contrast, when no finite strategy exists, the admissibility language is empty, and (49.2) must not be used. For example, an indiscrete two-color component together with an isolated third-color point and a three-color indiscrete space both have empty admissibility languages, but their closure signatures differ.

50 Finite and compact realizations: a complete classification

50.1 Finite posets and actual chains

Give a finite poset its upper-set topology. The closure of a subset is its downward closure. We orient a chain from its least to its greatest point.

Lemma 50.1 (Occurrences in finite posets). *For a finite colored poset P and word $v = v_1 \cdots v_k$,*

$$F_v = \{x : \exists x \leq x_1 \leq \cdots \leq x_k, c(x_i) = v_i \text{ for all } i\}.$$

For reduced v , its occurrence is equivalent to a strict chain whose color word is v . In particular, the signature of a chain colored by a reduced word b is exactly the set of nonempty reduced subsequences of b .

Proof. Induct on the word length, using that $T_a(F)$ is the downward closure of the a -colored points of F . Adjacent different colors exclude equality between adjacent witnesses, so a reduced word gives a strict chain. The final assertion specializes the witness description to a linearly ordered set. $\square$

Proposition 50.2 (Sums). *For any nonempty family of nonempty colored spaces over the same alphabet,*

$$\mathcal{S}\left(\bigsqcup_i X_i, \bigsqcup_i c_i\right) = \bigcup_i \mathcal{S}(X_i, c_i), \quad \mathcal{W}\left(\bigsqcup_i X_i, \bigsqcup_i c_i\right) = \bigcap_i \mathcal{W}(X_i, c_i). \quad (50.1)$$

Their signature rank is the supremum of the component ranks, in $\mathbb{N} \cup \{\infty\}$.

Proof. Closure in a topological sum is componentwise, so every F_v restricts to the corresponding F_v on each component. Nonemptiness is therefore existential over components. This proves the signature assertion; the word assertion follows from duality, or by gluing layerings for a common word. $\square$

The intersection statement is already in Part II of the repository. The signature description makes its compatibility information into a finite antichain calculation whenever the component ranks have a common finite bound.

Theorem 50.3 (Finite models). *Let D be a nonempty finite downset of nonempty reduced words, and let B be its maximal antichain. The disjoint union of the chains colored by the words $b \in B$ has signature exactly D . Every finite-depth colored space is therefore word-equivalent to a finite union of chains. At alphabet size $q \geq 2$ and rank r , it needs at most*

$$rq(q-1)^{r-1} \quad (50.2)$$

vertices by this construction.

Proof. The chain signatures are the principal downsets generated by the words of B , by lemma 50.1. Their union is D , by proposition 50.2. Apply this to the finite signature of a finite-depth game and use duality. The vertex count is $\sum_{b \in B} |b|$, bounded in proposition 48.2. $\square$

The model need not be a subspace or a continuous image of the original space. Its defining guarantee is preservation of all admissible words. This distinction is indispensable for Hausdorff source spaces, as discussed in section 52.

50.2 Compact convergent towers

Here is a self-contained version of the compact word realization already used in Part II of the repository. We include nonreduced rank words, which will be useful for the comparison example.

Let K_1 be a singleton. Recursively, form K_{k+1} from countably many pairwise disjoint clopen copies of K_k and a new point p . Neighborhoods of p contain all but finitely many entire copies; inside each copy use its own topology. For a word $b = b_1 \cdots b_k$, color the root of K_k by b_1 and each lower copy recursively by the tail $b_2 \cdots b_k$. Call this colored space K_b .

Lemma 50.4 (Topology and word profile of a tower). *The space K_k is compact, countable, metrizable and Hausdorff. It has Cantor–Bendixson height k , with a singleton $(k - 1)$ st derivative. For every nonempty rank word b , reduced or not,*

$$w \in \mathcal{W}(K_b) \iff \text{red}(b) \preceq w. \quad (50.3)$$

Proof. Countability and Hausdorff separation follow inductively. For compactness, an open cover has a member containing the root and all but finitely many copies; compactness of those finitely many remaining copies gives a finite subcover. An explicit metric can be constructed inductively with diameter at most one. Number the copies by $n \geq 1$, set the distance from their points to the root to 2^{-n} , the distance between different copies n, m to $2^{-n} + 2^{-m}$, and scale the old metric within copy n by 2^{-n} . This is a metric and induces exactly the stated topology.

Taking derivatives inside clopen copies commutes with restriction. Until their last nonempty derivatives disappear, infinitely many copies still meet every neighborhood of the new root. Induction therefore gives the root alone at derivative $k - 1$ and the empty set at derivative k .

For (50.3), induction on k suffices. The singleton case asks exactly that its color occur in w . Let the root color be $a = b_1$. If a layering assigns the root position i , then $w_i = a$ and its open suffix $\{\lambda \geq i\}$ contains an entire lower copy. The restriction to that copy is a layering over the suffix $w_i \cdots w_m$. By induction, $\text{red}(b_2 \cdots b_k)$ embeds there. If its first letter differs from a , its first match is strictly after i , so prepend the root's match at i . If its first letter equals a , no extra letter is needed for $\text{red}(b)$. This proves necessity.

For sufficiency, take an embedding of $\text{red}(b)$ into w . Give consecutive levels with the same color the same embedded position. The root has the smallest used position; within each lower copy use the inductively supplied layering. A suffix containing the root is the whole space, and every other suffix is a union of open suffixes within clopen copies. This proves admissibility. $\square$

Theorem 50.5 (Classification and compact representatives). *For a fixed finite alphabet, word-equivalence classes of nonempty finite-depth colored spaces are in bijection with nonempty finite downsets of nonempty reduced words. The signature gives the bijection. Every class has both a finite-poset representative and a compact countable metrizable representative. For a class of rank r , the latter can be chosen as a finite sum of towers of heights at most r .*

Proof. A finite-depth game has a finite downset signature by theorem 48.1. The signature distinguishes its word-equivalence class by theorem 49.2. Every finite downset is realized by theorem 50.3. For its maximal antichain B , take the finite disjoint union of the spaces K_b , $b \in B$. By lemma 50.4, its admissibility language consists exactly of common supersequences of B . It is therefore word-equivalent to the finite model. It has finite depth, so theorem 49.2 also gives its signature exactly, not merely its optimum. Compactness, countability and metrizability survive a finite disjoint union. $\square$

The single-word construction is inherited. What turns it into a universal classification is the new fact that *every* finite-depth topological game has a bounded, complete word signature.

51 Derivative bounds and optimal topological compression

For a Hausdorff space X , write $X^{(0)} = X$ and $X^{(j+1)} = (X^{(j)})'$, where the prime denotes the set of nonisolated points in the subspace. The finite Cantor–Bendixson height is the least positive k with $X^{(k)} = \emptyset$, when such a k exists. Thus a nonempty discrete space has height one.

Lemma 51.1 (One derivative per color change). *For a reduced word v of length k on a Hausdorff colored space,*

$$F_v \subseteq X^{(k-1)}. \quad (51.1)$$

Consequently, a space of finite height $h + 1$ has $\rho(X, c) \leq h + 1$ for every finite coloring.

Proof. We use the elementary fact that an isolated point of $\overline{B}$ belongs to B : an open set meeting $\overline{B}$ only in that point must meet B . For $k = 1$, (51.1) is immediate. Write $v = au$ where u starts with $b \neq a$. The set $F_u = \overline{A_b \cap F_{\text{tail}(u)}}$ is closed, and every isolated point of F_u has color b . Therefore

$$A_a \cap F_u \subseteq (F_u)'.$$

Derived sets of closed subspaces of a Hausdorff space are closed in the ambient space. Taking closures and using induction and monotonicity of formation of derived sets gives

$$F_v = T_a(F_u) \subseteq (F_u)' \subseteq (X^{(k-2)})' = X^{(k-1)}.$$

If the height is $h + 1$, words of length $h + 2$ cannot occur, nor can longer ones, by the downset property. $\square$

Theorem 51.2 (Optimal Hausdorff realization height). *Let (X, c) be any nonempty finite-depth colored space of rank r ; the original space need not be Hausdorff. Among its word-equivalent Hausdorff representatives of finite Cantor–Bendixson height, the least height is exactly r . The minimum is attained by a compact countable metrizable space.*

Proof. A word-equivalent representative has finite depth and the same signature, hence the same rank r , by theorem 49.2. If its height is k , then lemma 51.1 gives $r \leq k$. The tower union of theorem 50.5 has height equal to the largest basis-word length, which is r . This attains the lower bound. $\square$

This height is a property of the full finite-query behavior, not of the underlying uncolored space. A highly complicated or nonscattered space with a simple continuous coloring may have rank one and a discrete representative. Conversely, the same fixed space may carry word-inequivalent colorings of different signature ranks.

51.1 The exact formula when the second derivative is empty

Part III of the repository gives a graph formula under a finite-derived-set hypothesis. Duality removes that finiteness condition for each specified coloring.

Theorem 51.3 (Rank-one topology, with arbitrary derived-set cardinality). *Suppose X is Hausdorff and $X'' = \emptyset$. Let q_c be the number of colors actually used. Form the loopless directed graph G_c on those colors with arc $a \rightarrow b$ for $a \neq b$ exactly when*

$$A_a \cap \overline{A_b} \neq \emptyset. \quad (51.2)$$

Let $\tau(G_c)$ be its minimum directed feedback vertex number: the least number of vertices whose deletion leaves an acyclic digraph. Then

$$L(X, c) = q_c + \tau(G_c), \quad D(X, c) = \lceil \log_2(q_c + \tau(G_c)) \rceil. \quad (51.3)$$

No compactness, first-countability, or finiteness of X' is assumed.

Proof. By lemma 51.1, only words of length one or two can occur. Singleton words are exactly the active colors. A reduced pair ab occurs exactly when (51.2) holds, since $F_{ab} = \overline{A_a \cap \overline{A_b}}$ is nonempty exactly when its preclosure set is. By duality, L is the SCS length of these singleton and arc constraints.

Let w be any common supersequence and let R be the colors occurring at least twice. The colors outside R occur once; their positions give a linear order in which every arc between them must point forward. Thus $G_c - R$ is acyclic, $|R| \geq \tau(G_c)$, and $|w| \geq q_c + |R| \geq q_c + \tau(G_c)$.

For the reverse bound take a minimum feedback set R , list its colors in any order, and let D be a topological ordering of the remaining vertices. The word RDR contains every arc: arcs within D use that ordering, arcs starting in R use its first copy, arcs ending in R its last copy, and arcs within R use the first and last copies. Every active color occurs. The length is $q_c + |R|$, as required. Apply theorem 45.2 for depth. $\square$

The SCS–feedback–vertex argument is the same finite mechanism as in Part III; its extension to arbitrary X' is justified by the all-space signature and derivative theorems. For height at least three, the directed graph is generally insufficient: length-three and higher closure constraints carry additional information.

52 Finite-budget replicas and improved actual obstructions

The finite-rank hypothesis can be dropped when only bounded budgets are to be preserved. This also sharpens the repository’s finite-poset obstruction bound, which was $(m+1)q^m$ in Part II, Research question 3 and its preceding obstruction discussion.

Theorem 52.1 (Uniform finite-budget models). *Let (X, c) be an arbitrary nonempty colored space, with $q \geq 2$, and let $m \geq 1$. There is a finite colored poset P_m , with at most*

$$B_*(q, m) = mq(q-1)^{m-1} \quad (52.1)$$

points, such that

$$w \in \mathcal{W}(P_m) \iff w \in \mathcal{W}(X, c) \quad (1 \leq |w| \leq m). \quad (52.2)$$

There is also a compact countable metrizable representative of this bounded word behavior with height at most m . These models preserve every query budget d satisfying $2^d \leq m$.

Proof. Truncate the signature to the finite downset $D_m = \{v \in \mathcal{S}(X, c) : |v| \leq m\}$, and take its maximal antichain. Realize D_m by theorems 50.3 and 50.5. The antichain bound in proposition 48.2, with parameter m , proves (52.1); the compact model uses towers of height at most m .

For a candidate of length $k \leq m$, duality with its short-certificate clause says that admissibility on X is determined by $\mathcal{S}_{\leq k}$. Those words belong to D_m . Equivalently, w contains all of D_m exactly when it is admissible on X : the forward implication uses the short test, and the reverse follows from full duality. This proves (52.2). Finally, a depth- d game exists exactly when some admissible word has length at most 2^d , by theorem 45.2. $\square$

Corollary 52.2 (Actual finite subposet obstructions). *If a finite q -colored poset P has $L(P, c) > m \geq 1$, with $q \geq 2$, there is an induced subposet $Q \subseteq P$ of at most $mq(q-1)^{m-1}$ vertices with $L(Q, c|_Q) > m$.*

Proof. For every reduced candidate word w of length m , choose a signature word v_w of length at most m not embedded in w , using theorem 47.2. By lemma 50.1, select a chain in P colored by v_w . Let Q be the induced subposet on the union of all selected chains. There are $q(q-1)^{m-1}$ candidates, so the asserted size bound holds. Each selected chain remains a chain in Q , and therefore still excludes its candidate.

If Q admitted a word of length at most m , first reduce it and then append different successive letters until its length is m . Insertion of empty layers would preserve admissibility, producing one of the excluded reduced candidates. This contradiction proves the claim. $\square$

Remark 52.3 (Why a Hausdorff subspace version would be false). A finite subspace of a Hausdorff space is discrete. Its restricted coloring has an admissible word listing each active color once, hence at most q layers. A Hausdorff game needing more than q layers cannot have that behavior witnessed by a finite subspace with the induced topology. The finite-budget replica is an abstract colored poset, not such a subspace. Its infinite compact counterpart preserves accumulation instead.

The new bound is an upper bound, not a claim of optimality. Its improvement over the previous $(m+1)q^m$ is due to two changes: a failed length- m candidate now has a constraint of length at most m , and it is enough to enumerate reduced candidates rather than all words.

53 Products and an additive rank law

For colorings $c : X \rightarrow \Sigma$ and $e : Y \rightarrow \Gamma$, the product coloring is

$$(c \times e)(x, y) = (c(x), e(y)) \in \Sigma \times \Gamma.$$

This section concerns this specified pair coloring. It does not concern the supremum over arbitrary colorings of the product space. For a pair word v , write $\pi_X v$ and $\pi_Y v$ for its coordinate words; they need not be reduced even when v is.

Theorem 53.1 (Exact product signature and rank). *For every pair of nonempty colored topological spaces and every nonempty reduced word v over $\Sigma \times \Gamma$,*

$$v \in \mathcal{S}(X \times Y, c \times e) \iff \text{red}(\pi_X v) \in \mathcal{S}(X, c) \text{ and } \text{red}(\pi_Y v) \in \mathcal{S}(Y, e). \quad (53.1)$$

If the factor ranks are finite, then

$$\boxed{\rho(X \times Y, c \times e) = \rho(X, c) + \rho(Y, e) - 1.} \quad (53.2)$$

If at least one factor rank is infinite, the product rank is infinite.

Proof. The closure of a rectangle in a product topology is the rectangle of its closures. For closed $F \subseteq X$, $H \subseteq Y$, and colors (a, b) ,

$$T_{(a,b)}^{X \times Y}(F \times H) = \overline{(A_a \cap F) \times (e^{-1}(b) \cap H)} = T_a^X(F) \times T_b^Y(H).$$

Induction on word length gives the stronger set identity

$$F_v^{X \times Y} = F_{\pi_X v}^X \times F_{\pi_Y v}^Y. \quad (53.3)$$

Nonemptiness and lemma 46.2 prove (53.1).

Let the finite factor ranks be r, s . A reduced pair word of length k has $k - 1$ transitions, each changing at least one coordinate. The two coordinate words have respectively at most $r - 1$ and $s - 1$ changes when the pair word occurs. Therefore $k - 1 \leq (r - 1) + (s - 1)$, proving the upper bound $k \leq r + s - 1$.

Choose occurring reduced factor words $a_1 \cdots a_r$ and $b_1 \cdots b_s$. Follow a lattice path from $(1, 1)$ to (r, s) , advancing one coordinate at a time, and record the color pairs (a_i, b_j) along it. There are $r + s - 1$ recorded positions. Adjacent pairs differ because the changed factor word is reduced; the reduced coordinate projections are exactly the two chosen words. By (53.1) this pair word occurs, proving the lower bound.

If one rank is unbounded, take arbitrarily long occurring words in that factor and pair them with any one active color of the other. The same identity gives unbounded product rank. $\square$

Corollary 53.2 (Quantitative product consequences). *For finite-depth factors,*

$$\max\{L(X, c), L(Y, e)\} \leq L(X \times Y, c \times e) \leq L(X, c) L(Y, e), \quad (53.4)$$

$$\max\{D(X, c), D(Y, e)\} \leq D(X \times Y, c \times e) \leq D(X, c) + D(Y, e). \quad (53.5)$$

The least finite Hausdorff realization height of the product game's word profile is $r + s - 1$. More generally, for n finite-rank factors it is

$$1 + \sum_{i=1}^n (r_i - 1). \quad (53.6)$$

Proof. Fixing one coordinate restricts a product query tree to a factor, because slices of open sets are open. Pruning empty leaves and projecting output labels gives the lower bounds for leaves and depth. For the upper bounds, run one optimal factor tree and then, at each leaf, run the other factor tree. Coordinate inverse images of open sets are open, the depths add, and the leaf counts multiply. The height statement follows from theorems 51.2 and 53.1; iterate the rank formula for n factors. $\square$

The rank formula is exact even when the two layer optima provide only rough bounds on the product optimum. One can compute the exact product antichain from (53.1) by enumerating reduced pair words through length $r + s - 1$, testing their two projections, and retaining maximal words. This is a finite algorithm; no assertion of polynomial output size is made.

Together with proposition 50.2, these results give a concrete algebra of finite-depth behaviors. Sums take unions of downsets, equivalently intersections of admissibility languages. Independent products take the reduced-word languages whose coordinate projections belong to the factor

signatures. The relevant product operation is asynchronous interleaving of color changes, not mere concatenation of the factor words.

[Added 30 September 2026, batch 65] Part I's product law. The rank law (53.2) is the coloring analogue of Part I's lemma 9.1, which the manuscript does not cite: for Hausdorff spaces of last derivative ranks h and k , that is, of heights $h + 1$ and $k + 1$ in the convention of section 51, the product has last rank $h + k$, height $(h + 1) + (k + 1) - 1$. By theorem 51.2, (53.2) says the same of the least heights of Hausdorff representatives of the word profiles.

54 Higher-order information is indispensable

The finite-derived-set theory makes pairwise accumulation graphs natural. One might hope to extend it using derivative ranks, derivative cardinalities, and a bounded number of local word statistics. The following construction shows that progressively longer nested closures can change the exact query budget while all shorter occurrence tests remain unchanged.

Theorem 54.1 (Same space, same triple data, different query depth). *There is a compact countable metrizable Hausdorff space Z of Cantor–Bendixson height four with two three-colorings c_{hard} and c_{soft} such that:*

- (i) *Their closure signatures agree on every reduced word of length at most three.*
- (ii) *For each derivative index j and color a , the sets $Z^{(j)} \cap c_{\text{hard}}^{-1}(a)$ and $Z^{(j)} \cap c_{\text{soft}}^{-1}(a)$ have the same cardinality.*
- (iii) *Nevertheless,*

$$L(Z, c_{\text{hard}}) = 9, \quad D(Z, c_{\text{hard}}) = 4; \quad L(Z, c_{\text{soft}}) = 7, \quad D(Z, c_{\text{soft}}) = 3. \quad (54.1)$$

Proof. Let $\Sigma = \{0, 1, 2\}$. There are $3 \cdot 2^3 = 24$ reduced words $a_1 a_2 a_3 a_4$ of length four. Let Z be the disjoint union of 24 copies of K_4 , one indexed by each of these words.

On the component indexed by $a_1 a_2 a_3 a_4$, use that same rank word for c_{hard} . Its component signature is the reduced-subsequence downset of that word, by lemma 50.4 and theorem 49.2. The total signature is therefore all reduced words of length at most four.

On the same component, use the rank word $a_1 a_1 a_2 a_3$ for c_{soft} . Its reduction is $a_1 a_2 a_3$. Every reduced three-letter word appears as such a prefix, so the total soft signature is exactly all reduced words of length at most three. This proves (i). By lemma 48.3, their SCS lengths are respectively $2 \cdot 4 + 1 = 9$ and $2 \cdot 3 + 1 = 7$, giving (iii) by theorem 45.2.

Both colorings are defined on the identical underlying space. At derivative three there are 24 roots, eight of each color for each coloring. At each of derivatives zero, one and two, every color occurs countably infinitely many times in both colorings: for the hard coloring the relevant level colors range over the three possibilities as components vary; for the soft coloring even the next-to-root level already repeats each component's root color infinitely often. All spaces concerned are countable. Derivative four and later are empty. Thus the exact common table is

Derivative	Color 0	Color 1	Color 2
$Z^{(0)}, Z^{(1)}, Z^{(2)}$	$\aleph_0$	$\aleph_0$	$\aleph_0$
$Z^{(3)}$	8	8	8
$Z^{(j)}, j \geq 4$	0	0	0

This proves (ii). □

The two optimal master words may be taken as 012012012 and 0120120. These are nine and seven layers, respectively; they are not nine and seven questions. Balanced suffix search gives the four- and three-question trees. The comparison preserves Boolean occurrence/nonoccurrence of short patterns, not the actual subsets $F_v \subseteq Z$ or all possible local data. It should not be read as an assertion that every conceivable local invariant fails.

Proposition 54.2 (An unbounded hierarchy of missing information). *For every $r \geq 2$, a common compact countable metrizable space of height r has two three-colorings with signatures consisting respectively of all reduced words of lengths at most r and at most $r - 1$. Their layer counts are $2r + 1$ and $2r - 1$. For $r = 2^{d-1}$, $d \geq 2$, their query depths are respectively $d + 1$ and d , although all occurrence tests of length at most $r - 1$ agree.*

Proof. Repeat the preceding construction on copies of K_r indexed by all reduced r -letter words. Use the hard rank word $a_1 \cdots a_r$ and the soft rank word $a_1 a_1 a_2 \cdots a_{r-1}$. Their reduced component words are all length- r words and all length- $(r - 1)$ words, respectively. Apply the tower profile, sum law and universal-word formula. If $r = 2^{d-1}$, the two lengths are $2^d + 1$ and $2^d - 1$, which straddle the power-of-two query threshold. □

Thus no fixed truncation order captures all finite-depth colored spaces. The positive truncation theorem is necessarily tied to the proposed word budget, and the graph formula is confined to the appropriate derivative or signature-rank regime.

55 Algorithms, certificates, and the effective boundary

The mathematical reduction separates two tasks: discovering a topological signature, and solving the resulting finite word problem. Only the second is automatically effective from a finite list of words. This distinction prevents a hidden assumption that arbitrary topological closure or emptiness is computable.

55.1 Extracting a bounded signature

Given operations for closure, intersection with each A_a , and deciding emptiness of the resulting closed sets, one can compute all F_v for reduced words of length at most m recursively. There are at most $N(q, m) = q \sum_{j=0}^{m-1} (q - 1)^j$ such words. Store a word precisely when its computed closed set is nonempty, then remove words that are proper subsequences of other stored words. The result is the maximal antichain of the truncated signature.

This is an algorithm *relative to the stated operations*. An arbitrary topological presentation need not supply them effectively. For a finite poset, closure is downward reachability and the test is finite and explicit. For a space of known Hausdorff height $h + 1$, the derivative bound supplies the complete cutoff $m = h + 1$. For a general finite-depth space, one needs a rank bound or some independent termination argument; finite depth alone is not being offered as an effective stopping certificate from unspecified input.

55.2 Exact SCS by progress-state search

Let the finite basis be $B = \{b^{(1)}, \dots, b^{(t)}\}$, with lengths $\ell_1, \dots, \ell_t$. A state

$$s = (s_1, \dots, s_t), \quad 0 \leq s_i \leq \ell_i,$$

records how many letters of each word have been matched greedily. Appending a letter a updates all coordinates simultaneously:

$$s_i \mapsto s_i + \mathbf{1}\{s_i < \ell_i \text{ and } b_{s_i+1}^{(i)} = a\}. \quad (55.1)$$

Start at the zero state. Breadth-first search reaches the terminal state $(\ell_1, \dots, \ell_t)$ along a shortest master word. A predecessor map reconstructs it. There are at most

$$\prod_{i=1}^t (\ell_i + 1) \quad (55.2)$$

states and q outgoing transitions per state; straightforward transition calculation takes $O(t)$ time. Thus an explicit worst-case bound is $O(qt \prod_i (\ell_i + 1))$, with comparable finite storage after accounting for the stored state vectors.

For fixed alphabet size and signature rank, [proposition 48.2](#) bounds this state space solely in terms of those two parameters. This is a parameter-bound combinatorial algorithm *after* obtaining the signature; it is not a polynomial-time algorithm for arbitrary topological input. In particular, it does not contradict the established three-letter SCS hardness [\[5\]](#) or its repository applications.

55.3 From a word to actual open questions

Once an admissible word w has been found, compute its elimination chain $G_0 \supseteq \dots \supseteq G_m = \emptyset$. The proof of [proposition 45.3](#) gives explicit layers and suffixes:

$$\lambda^{-1}(i) = G_{m-i} \setminus G_{m-i+1}, \quad U_j = X \setminus G_{m-j+1}. \quad (55.3)$$

Binary search in these suffixes is a concrete optimal-depth strategy when w is shortest. The generator of the word may be untrusted: an independent checker can verify its subsequence certificates against a known complete basis, or check the elimination chain directly.

A failed proposed word has a particularly small semantic certificate: a reduced v with $|v| \leq |w|$, $F_v \neq \emptyset$, and $v \not\leq w$. The combinatorial part is easy to verify. The topological nonemptiness still requires valid evidence in the presentation at hand. On a finite poset, a colored chain supplies it. On a Hausdorff accumulation space, a list of finitely many isolated sample points generally does not.

55.4 What would be formalized

The proof naturally divides into three interfaces. First, finite words, run reduction, greedy subsequences, diagonal avoidance, and the separator construction are purely combinatorial. Second, the lattice of closed sets supplies monotone contractive idempotent operators and finite unions; this proves expansion and duality. Third, Hausdorff derivatives and compact tower constructions establish the topological realization theorems.

A Lean development can keep search untrusted and verify certificates against these interfaces. The finite-budget theorem can be stated with integer bounds $m \leq 2^d$ instead of real logarithms. The present package supplies executable finite checks but no Lean proof files and no assertion that these interfaces have been kernel-checked.

56 Independent verification and reproducibility

The program `code/04-closure-words-verify.py` uses only the Python standard library. The recorded results are in `data/04-closure-words-verification.json`. They come from an actual completed run under Python 3.13.5. Assertions are verification conditions, so the program explicitly refuses execution with optimization flag `-O`.

56.1 Exhaustive finite topologies

Every finite topology is the upper-set topology of its specialization preorder. The program enumerates all reflexive transitive relations on each labeled set of size at most four, including non- T_0 spaces, and all three-colorings. For each one it calculates open sets and closure directly from the relation.

Two independent computations are then compared. The first is minimax over all open questions, with no use of closure words or subsequences. For a nonempty current state S , it uses

$$d(S) = \begin{cases} 0, & S \text{ monochromatic,} \\ 1 + \min_U \max\{d(S \cap U), d(S \setminus U)\}, & \text{otherwise,} \end{cases}$$

where only nontrivial open splits are allowed and the minimum over no splits is infinite. It separately minimizes leaf count by adding the two branch leaf counts. The second computation builds the closure signature, extracts its antichain, and performs progress-state SCS search.

A finite preorder has a finite tree for a given coloring precisely when the color is constant on each topological indistinguishability class. If a class contains two colors, those colors give arbitrarily long alternating detected words and cannot be separated by open sets. Otherwise the coloring descends to the finite quotient poset, whose chains have at most n points. This justifies the program's finite-signature cutoff at n , with a check at $n + 1$ for the excluded case.

The enumerated counts are:

Points	Labeled topologies	Three-colorings	Finite-tree colorings
1	1	3	3
2	4	36	30
3	29	783	597
4	355	28,755	21,009
Total	389	29,577	21,639

These count labeled topologies, not homeomorphism classes. Every topology on at most four points is covered, with possible isomorphic repetitions.

For every coloring, the program also tests every candidate word of lengths one through the number of points, including nonreduced candidates. It compares greedy closed-set elimination, the full signature constraint test, and the short-certificate test. The recorded totals are

Check	Recorded count
Candidate-word admissibility comparisons	3,481,578
Rejected candidates with short failure certificates	2,847,462
Distinct finite signatures encountered	109
Constructed separating-word tests	7,296

All comparisons and assertions passed.

56.2 Product, graph, and extremal checks

For all 16 finite-depth two-colorings of labeled topologies on one or two points, the program checks all 256 ordered products. It constructs the product preorder independently and compares the actual nested closed sets with the rectangular identity (53.3). The record contains 9,856 such set-identity tests; all passed, as did the rank and product leaf/depth bounds.

It separately enumerates all 4,165 loopless digraphs on labeled vertex sets of sizes one through four. A subset-based acyclicity calculation obtains $\tau(G)$, and progress-state SCS independently obtains the optimum for singleton and arc constraints. All tests agree with $L = q + \tau(G)$. These are graph checks, not a finite sampling of arbitrary infinite rank-one spaces.

For the universal-word constant, the verified parameter ranges are $q = 2, 1 \leq r \leq 8$; $q = 3, 1 \leq r \leq 6$; and $q = 4, 1 \leq r \leq 3$. Each cyclic upper-bound word is checked against every length- r reduced word. Every reduced word one letter shorter than the claimed optimum is independently rejected by some constraint. The 24 hard and 24 soft rank-word assignments in theorem 54.1 are recorded explicitly; their reduced maximal bases have sizes 24 and 12.

56.3 Reproduction and limits of the evidence

From this report's directory, run

```
python3 code/04-closure-words-verify.py --output <scratch>/rerun.json
```

where `<scratch>` is a directory outside the report. This keeps the delivered verification record unchanged. The default `-max-n 4 -q 3` regenerates all counts quoted above. The measured complete run took 4.282 seconds in the execution environment; this is not a portable performance guarantee. The elapsed field is expected to vary between runs.

To compile the article, run `make pdf`; it invokes `pdflatex` three times, without an external bibliography processor. The article does not require executing the verifier, retrieving web content, or using an external graphics file.

The checks establish the stated finite computational comparisons only. Universal validity, compactness, Hausdorff derivative descent, and the infinite-space counterexamples rest on the written proofs. The tests do not establish historical novelty, publication acceptance, or formal verification.

[Added 30 September 2026, batch 65] Shipped names. The delivered text reads “The program `code/verify.py`”, “The recorded results are in `results/verification.json`” and “From the package root, run `python3 code/verify.py --output results/rerun.json`”; here `code/verify.py` is Part I’s verifier, so the command above uses the shipped name. Without `--output` the shipped verifier writes `results/verification.json` beside `code/`, that is, a new `results/` directory in this report; the recorded file is `data/04-closure-words-verification.json`, and the digests in `data/04-closure-words-validation.json` describe the delivered manuscript, PDF and verifier, not this merged report. The verifier writes its JSON with the platform’s line endings (CRLF on Windows); the recorded file is LF. A rerun on a copy (30 September 2026, Python 3.14.4, 29.8 seconds on a loaded machine) reproduced the recorded JSON after CRLF $\rightarrow$ LF except the fields `python` and `elapsed_seconds`. The delivered Makefile, shipped as `code/04-closure-words-Makefile`, is written for the delivery’s layout and should not be used here: run from this directory, its `verify` target would call Part I’s `code/verify.py`, which takes `--output` as a directory and would create `results/rerun.json/` holding Part I’s output files; its `pdf` target would build

this article in place, and its `clean` target would delete this report's auxiliary files. This merged report is built as in section 12; the report README gives the rerun commands.

57 Further research questions and proposed directions

The following questions are proposed continuations. Except for their explicit connections to the pinned repository, they are not being labeled as previously published open problems without a separate literature check.

[Added 30 September 2026, batch 65] Relation to the questions of Parts II and III. Seven of the nine questions below continue questions already printed in this report; they are kept, with their own discussion. Research question 20 is the remaining half of research question 1 and contains research question 12; research question 21 is Part II's Research question 3 with the improved bound of corollary 52.2; research question 22 refines Part II's Research questions 2 and 6; research question 24 generalizes research question 16; research question 26 is Part II's Research question 8; research question 27 combines Part II's Research questions 7 and 9 with research question 15; and research question 28 is Part II's Research question 10. Research questions 23 and 25 are new.

Research question 20 (Which signatures can a fixed space realize?). For a prescribed Hausdorff space X of finite height and fixed alphabet, characterize

$$\{\mathcal{S}(X, c) : c : X \rightarrow \Sigma\}.$$

In particular, find local necessary and sufficient conditions for attaining maximal layer counts at heights at least three.

The present paper classifies a coloring once its signature is known, and realizes every finite signature on *some* compact space. It does not classify all colorings on an independently fixed space. The repository's finite-derived-set capacity theorem is a model for the desired next step. Nested partition capacities or compatibility of locally attainable suffix signatures are plausible ingredients, but their exact axioms remain to be established.

Research question 21 (Optimal obstruction size). Determine the least $B(q, m)$ such that every finite colored poset failing an m -layer budget has an induced subposet obstruction of at most $B(q, m)$ vertices. How far can $mq(q-1)^{m-1}$ be reduced? Which antichains force many distinct witness chains, and when can their vertices be shared?

The proof gives explicit witnesses but makes no attempt to reuse points between them. Even understanding extremal critical families of reduced supersequence constraints would improve this bound.

Research question 22 (Minimal representatives). Given a finite reduced-word downset, determine the minimum number of vertices in a finite-poset representative, and the minimum number of top-rank points in a compact representative of optimal height.

The optimal height is settled here, but the disjoint-chain and disjoint-tower constructions need not optimize other resources. Sharing compatible branches may save many points or components without changing the signature. The exact conditions under which sharing introduces unwanted words need a separate analysis.

Research question 23 (Exact layer complexity of products). Express or efficiently bound the SCS length of the product signature from its two maximal antichains. When is the multiplicative upper bound $L_{X \times Y} \leq L_X L_Y$ attained, and when can it be substantially improved?

The rank law and full signature formula are exact, but shortest master words must synchronize many possible interleavings. A classification for small ranks or bounded antichain width would provide concrete test cases.

Research question 24 (Effective topological presentations). For which computable presentations can truncated signatures be extracted with explicit complexity bounds? Which closure and nonemptiness certificates can be checked rather than trusted?

Finite posets provide one complete answer. Rank-word towers provide another finitely presented class. Extending to symbolic descriptions of scattered spaces requires a precise input language and computational meaning of closure; otherwise an apparent algorithm can hide a noncomputable oracle.

Research question 25 (Comparison with partition hierarchies). How does finite-depth word equivalence compare with the difference and fine hierarchies of finite partitions, and with continuous reducibility of those partitions? Which implications or strict separations hold on common classes of spaces?

Selivanov's framework [9] is an appropriate starting point. The present invariant forgets the locations and multiplicities of closure witnesses, retaining only which finite patterns occur. Any proposed identification must respect that loss of information.

Research question 26 (Beyond finite worst-case depth). What additional structure on the unbounded signature, or on the sets F_v themselves, controls transfinite open-query games or pointwise finite strategies with no common finite bound?

When the signature rank is unbounded, the finite admissibility language is empty. It then cannot distinguish different transfinite behaviors. A well-founded rank on suitable closure trees might be relevant, but no ordinal logarithm of the finite layer formula is justified by this paper.

Research question 27 (Expected depth and restricted questions). For a probability measure on X , what data beyond the signature determine minimum expected query depth? Which parts of duality survive when queries must come from a fixed basis or another restricted family of open sets?

The normal-form proof uses arbitrary finite unions and intersections of available open sets. A restricted query class may fail that closure property. Expected-depth optimization also depends on the masses of admissible layers, which the Boolean occurrence signature deliberately discards.

Research question 28 (A small trusted formal core). Can the operator expansion, diagonal-avoidance lemma, and finite-budget model theorem be formalized as a compact reusable library, with the topological results supplied by separate instances?

A successful implementation should make the trusted boundary explicit: search produces finite certificates, verified word and closed-set lemmas connect them to semantics, and infinite topology is handled by actual proofs, not by finite testing alone.

58 Conclusion

The key invariant is neither a scalar depth nor a graph of pairwise accumulations. It is the downset of reduced words surviving iterated color closures. The finite-union expansion of greedy elimination makes this language dual to admissible layer words on every topological space. A second, purely combinatorial separator recovers the language from the full finite admissibility profile.

This gives an exact classification of finite-depth behavior and a bounded finite model for each finite budget, even when the original space has no finite-depth solution. In the Hausdorff setting, longest detected words measure the least possible representative height. At the first nontrivial height, the result removes a finite-derived-set restriction from the graph formula; at higher heights, explicit compact examples show why longer patterns are essential. For independent products, the same operators give an exact additive rank law.

The outstanding uniform problem is now more sharply separated from these settled statements: determine the signatures realizable by colorings of a *fixed* space. That problem, rather than re-solving the game separately for each coloring, is a concrete next target for the repository's research program.

A Why countable compact Hausdorff spaces are scattered

This appendix supplies the fact used after corollary 10.3 without requiring the full ordinal classification of countable compact spaces.

Lemma A.1. *A nonempty compact Hausdorff space with no isolated points has at least $2^{\aleph_0}$ points.*

Proof. Construct nonempty open sets U_s indexed by finite binary strings. Start with the whole space. Inside each U_s , select two distinct points; this is possible because no nonempty open set is a singleton. By regularity, choose nonempty open neighborhoods U_{s0}, U_{s1} with disjoint closures contained in U_s .

For each infinite binary string z , the compact sets $\overline{U_{z \upharpoonright n}}$ are nonempty and nested, so their intersection is nonempty. Choose one point from each intersection. Distinct binary strings split at some level and hence give points in disjoint closed sets. This produces an injection of $2^{\mathbb{N}}$ into the space. $\square$

Proposition A.2. *Every countable compact Hausdorff space is scattered.*

Proof. If a space is not scattered, some nonempty subspace B has no isolated points. Its closure F has no isolated points either: an isolated point of F would, by density, belong to B and be isolated in B . In a compact Hausdorff space, F is compact Hausdorff. The preceding lemma would make F uncountable, a contradiction. $\square$

This proof also shows why no additional set-theoretic hypothesis such as the continuum hypothesis is needed. Uncountability of $2^{\mathbb{N}}$ is sufficient.

B A proof audit and implementation checklist

The following points isolate possible sources of off-by-one or topology errors.

Zero questions. A constant target has $L = 1$, not $L = 0$, on a nonempty space. Thus its depth is $\lceil \log_2 1 \rceil = 0$. The empty space is assigned $\text{sm} = 0$ separately.

Derivative rank. The integer h is the index of the *last nonempty* derivative, not the height. The height is $h + 1$. A convergent sequence has $h = 1$ and height 2.

First closure. The general estimate is $F_n \subseteq X^{(n-1)}$ for $n \geq 1$, not $F_n \subseteq X^{(n)}$. The extra first step is exactly what creates the two oriented lengths $h + 1$ and $h + 2$ for a parity target.

Hausdorff separation. The lower bound with two terminal points uses disjoint open neighborhoods. The finite-height theorem is therefore stated for Hausdorff spaces. Its proof is not silently applied to arbitrary T_0 posets.

Arbitrary answers. The binary-search proof wins on every answer sequence, including one whose state becomes empty. In the ω -length certification strategy, an all-no certification path has empty final intersection. There is no hidden assumption that the Hider fixes a point beforehand.

Infinite sums. In the oriented finite-profile formula, an unbounded supremum of integers is recorded as ∞ , meaning “no finite decomposition”. The separate certification argument is needed to obtain the actual ordinal value ω .

Independent checking. The minimax implementation enumerates all open queries. It does not import the closed-layer answer as its return value. The mathematical proof and the finite computational check are deliberately distinct.

C Certificate and reproduction conventions (Part II)

In `data/02-finite-output-strategy_certificate.json` (Part I’s eight-point certificate is `data/strategy_certificate.json`), bit i of a mask denotes vertex i , with numbering starting at zero. The predecessor mask of a vertex contains *strict* predecessors. The supplied coloring consists of the chains 01, 02, 10, 12, 20, 21 in that order. The 1-based rank vector is

$$(1, 2, 1, 3, 2, 4, 2, 3, 3, 4, 3, 5).$$

At an internal tree node, query the upper set encoded by `open_mask`; follow `yes` or `no`. A leaf contains the reported color and its layer number. The word is 01201, with five layers.

The verifier requires Python 3.10 or later and no third-party packages. Do not use Python’s `-O` option, which removes assertions. The default run is the recorded through-five-point run with through-four-point fixed-word checks. A smaller independent smoke test is

```
python3 code/02-finite-output-verify.py --max-n 3 --brute-n 3 --output <scratch>
```

The six-point option is substantially more expensive because both the poset count and the number of colorings grow quickly. Its availability is not a claim that a six-point run was performed for this article.

Build the PDF with two runs of

```
pdflatex -interaction=nonstopmode -halt-on-error article.tex
```

The bibliography is embedded below; no BibTeX run, remote assets, or external figures are required. The archive also includes a Makefile and a source-status record. Font files are not distributed.

[Added 29 September 2026, batch 52] Shipped names. The smoke-test command above replaces the delivered `python3 code/verify.py --max-n 3 --brute-n 3 --output /tmp/open-query-smoke: code/verify.py` is Part I’s verifier here, and `<scratch>` is any directory outside the report. The build paragraph describes the delivered standalone article. This merged report needs three `pdflatex` runs, as in section 12. The delivered Makefile and source-status record are shipped as `code/02-finite-output-Makefile` and `02-finite-output-RESEARCH_STATUS.md`, with the source audit `02-finite-output-SOURCES.md`. That Makefile is written for the delivery’s layout; run from this directory it would call Part I’s verifier and build this article in place, so it should not be used here.

D Certificate format and reproducibility (Part III)

The archive contains (with the names shipped in this report; see the note below):

File	Purpose
<code>article.tex</code> , <code>article.pdf</code>	This merged report and its compiled PDF
<code>code/03-filter-profiles-verify.py</code>	Independent finite word, graph, and capacity checks
<code>data/03-filter-profiles-verification.json</code>	Recorded verification summary
<code>data/03-filter-profiles-verification.log</code>	Console output of the executed run
<code>data/03-filter-profiles-capacity_checks.json</code>	The 98 capacity-case outcomes
<code>data/03-filter-profiles-strategy_certificates.json</code>	Optimal words and balanced query trees
<code>data/03-filter-profiles-extremal_thresholds.csv</code>	Small exhaustive graph thresholds and equality counts
<code>data/03-filter-profiles-query_jump_thresholds.csv</code>	Exact component thresholds for 2–64 colors
<code>03-filter-profiles-SOURCES.md</code>	Pinned source audit and novelty boundary
<code>code/03-filter-profiles-Makefile</code>	The delivered build and verification targets (not for use here)

The strategy certificates use zero-based word positions. A node `ask_suffix_from_position: j` asks membership in the union of positions $j, j + 1, \dots$. The no branch goes to earlier positions and the yes branch to later positions. Leaf records give a position and its label. The interpretation as open subsets is supplied by theorem 33.1; the JSON is not an encoding of an entire infinite ultrafilter.

The verifier uses Python’s arbitrary-precision integers and assertions; run it without the `-O` flag. It has no third-party dependencies. The PDF can be rebuilt with two passes of `pdflatex`; the internal bibliography needs no separate BibTeX step.

[Added 29 September 2026, batch 56] **Shipped names.** The delivered table listed `article.tex` and `article.pdf` (the manuscript’s own source and PDF), `code/verify.py`, six files under `results/`, `SOURCES.md`, and `README.md` with `Makefile`. The manuscript, its PDF, its delivery `README` and its checksum ledger `SHA256SUMS` are not shipped; they survive in the arrival commit `e47ed8547`. `article.tex` and `article.pdf` here are the merged report, which needs three `pdflatex` runs, as in section 12. The delivered `Makefile` is shipped as `code/03-filter-profiles-Makefile`; it is written for the delivery’s layout. Run from this directory, its `verify` target would call Part I’s verifier `code/verify.py`, which without an output option replaces Part I’s recorded files in `data/`, and would write its console output to a new `results/verification.log`; its `pdf` target would build this article in place with two passes, and its `clean` target would delete this report’s auxiliary files. It should not be used here; the report `README` gives the rerun commands.

E Proof dependency and claim audit (Part IV)

The following table distinguishes new arguments in this manuscript from previously available ingredients. “New here” describes the contribution relative to the inspected baseline, not a certified historical priority claim.

Statement	Dependency	Status and boundary
Tree-layer equivalence and greedy elimination	Elementary topology and binary trees	Repository baseline; fully reproved in section 45.
Closure-word duality, including length- m counterwords	Finite-union operator expansion; diagonal avoidance	New here; arbitrary topology and finite alphabet.
Finite-rank/SCS reduction	Duality and finiteness of bounded word sets	New all-space reduction; finite-poset SCS interpretation was already in the repository.
Recovery from admissibility	Greedy forbidden-word separator	New here; requires nonempty finite admissibility language.
Compact single-word towers	Recursive compactness and suffix neighborhoods	Repository construction; reproved and extended to nonreduced rank words.
Universal finite-depth classification	Duality, recovery, finite and compact word models	New here; no claim of homeomorphism or continuous equivalence.
Optimal Hausdorff height	Derivative descent and recovery; tower realization	New here; height is exactly maximum detected reduced-word length.
Graph formula for $X'' = \emptyset$	Derivative bound; SCS of arcs	Removes the finite-derived-set restriction for fixed colorings. The finite graph optimization is inherited.
Finite-budget replicas and induced subposet bound	Short counterwords; reduced candidate enumeration	New here; actual induced-subspace conclusion only for finite posets.
Product rank formula	Rectangular closure identity	New here; applies to independent pair colorings, not arbitrary colorings of a product.
Matched-space hierarchy examples	Tower rank words and universal reduced-word constant	New examples; shorter pattern occurrence data agree, not all local structures.
Universal reduced-word constant	Cyclic word and greedy adversary	Repository baseline; not a new extremal-word claim.
Computational checks	Exhaustive finite enumeration and standard-library code	Executed finite tests; no proof-assistant or infinite-space verification claim.

No theorem of this manuscript assumes an unproved conjecture. The SCS hardness result is cited only to delimit algorithmic claims; the main proofs do not depend on it. The descriptive-set-theoretic source provides context and an explicit further-comparison question, not an imported theorem used without proof. All nonfinite conclusions are supported by the arguments in the text.

F Small worked certificates (Part IV)

F.1 A failed word and the short obstruction it generates

Take a three-color signature containing all reduced words of length at most three, and propose $w = 0120$. The coordinatewise-avoiding word $b = 1201$ satisfies $b_i \neq w_i$, but is longer than any signature word here and need not occur. Instead use $v = 210$, which does occur and is not embedded in w . Greedy matching of v in the four positions of w matches prefix lengths 0, 0, 1, 1. The converse construction of lemma 47.1 yields

$$b = (2, 2, 1, 1), \quad \text{red}(b) = 21.$$

This is an occurring obstruction of length two: $21 \not\leq 0120$. It contributes a nonempty term $F_{2211} = F_{21}$ to the residual expansion. The example illustrates that the certificate obtained by the theorem can be shorter than the originally noticed failed constraint.

F.2 The need to preserve all admissible words

Consider a chain with color word 01 and another with color word 10. Both have two layers and one query, but their signatures differ. The first admits 01 and the second does not. Their union has maximal antichain $\{01, 10\}$, shortest master word 010 or 101, and two queries. Thus equal component optima do not imply compatibility of optimum words. This binary phenomenon is already in the repository; the signature theorem explains it within the same language as arbitrary spaces.

F.3 A product-rank witness

Take occurring factor words 010 and 12. An asynchronous path gives the pair word

$$(0, 1), (1, 1), (0, 1), (0, 2).$$

Its coordinate reductions are 010 and 12, so it occurs in the product. It has length $3 + 2 - 1 = 4$. The corresponding closed set is exactly $F_{010}^X \times F_{12}^Y$, by (53.3). No relation between particular witness points in the two factors is required.

References

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- September 2026, for the game context and the topological-sum question. Part III's manuscript cites it under the shorter title, as arXiv:2510.05754v3, revised 29 May 2026, for the broader cut-and-choose setting only. Part IV's manuscript cites it under the same title (v3, 29 May 2026; first submitted 7 October 2025; doi:10.48550/arXiv.2510.05754) for the transfinite cut-and-choose framework and its point-separating and set-membership invariants, from the abstract metadata and the HTML. [Added 30 September 2026, batch 65: both titles belong to v3; the abstract page carries the shorter one, the v3 HTML full text the longer.]
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 - [5] Aurélie Lagoutte and Sébastien Tavenas. *The complexity of Shortest Common Supersequence for inputs with no identical consecutive letters*. arXiv:1309.0422v2, 2015. In particular, Corollary 6 and the discussion of typed fusions in the introduction. <https://arxiv.org/abs/1309.0422>. Part III's manuscript cites the arXiv record (2013) as background for supersequences only; its hardness input is [8]. Part IV's manuscript cites v2 (8 January 2015; first submitted 2 September 2013) for the hardness of shortest common supersequence for reduced words over three letters, as related work and complexity context, not as a dependency of its proofs.
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 - [9] Victor Selivanov. *Towards a Descriptive Theory of cb_0 -Spaces*. arXiv:1406.3942, 2014. <https://arxiv.org/abs/1406.3942>. Part IV's manuscript (16 June 2014) consulted the indexed arXiv abstract for the scope of finite-partition, difference and fine hierarchies; it asserts no detailed equivalence with its word equivalence.